When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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***Abstract*—Test-time adaptation (TTA) updates a model on unlabeled data to counter distribution shift, but the same update that recovers accuracy under one shift can quietly destroy it under another—and with no target labels, the system cannot tell which case it faces. We study the decision that comes *before* adaptation—should the model adapt at all?—and cast it as a three-way choice: *adapt* when adaptation is knowably helpful, *freeze* when it is knowably harmful, and *abstain* when the unlabeled evidence cannot decide. That some such decisions are *provably unanswerable* without extra assumptions is not new: identifying out-of-distribution accuracy is already known to be as hard as identifying the optimal predictor [13], and the optimal predict-versus-abstain trade-off under shift has a known decision-theoretic form [14]. Our contribution *specializes and sharpens* this landscape for the adaptation-benefit sign. We prove an *exact* success condition—the sign of the benefit is recoverable over the declared drift-budget class *iff* an observable confidence margin exceeds how far calibration can drift (the "calibration-drift budget"), a clean computable geometric criterion in observable quantities—together with a finite-sample certificate that realizes the unique maximal sound adapt/freeze/abstain rule while holding the false-adapt and false-freeze rates at a chosen level α. We further pin the minimal untestable supplement at exactly *one bit* of orientation in this test-time-adaptation setting, and show that the standard label-free accuracy estimators (ATC, DoC, GDE, COT, AETTA, agreement-on-the-line) are the zero-budget face of this frontier—sound precisely when calibration does not drift. We package this as KGA, a drop-in wrapper around existing adapters (Tent, EATA, SAR) that decides *whether* to trust them rather than how to adapt. Our headline empirical result is uniform *no-harm* on every held-out natural shift in a pre-registered, un-cherry-picked panel (Office-Home, iWildCam, Camelyon17, RxRx1, PACS): under a valid out-of-fold radius, KGA matches the better fixed policy and beats the worse at $\mathrm{FA_u}$=0—conditional safety when adaptation is risky, not universal accuracy gains. In *identifiable* mixed harmful+helpful regimes (CIFAR-10-C stress grid, ImageNet-C SAR), it further beats both trivial policies and beats POEM and AETTA with Holm-corrected regret gaps, exactly where the frontier predicts knowability. KGA is a safety layer, not a universal accuracy booster: its wins are regime-specific, and we report—rather than hide—the regimes and honest nulls where it only ties. Full per-dataset tables and the complete benchmark battery appear in the body and appendix.**

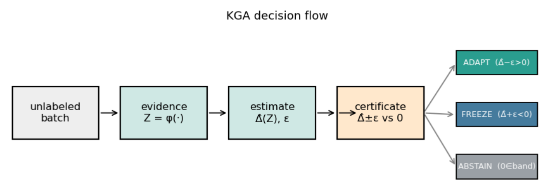

Fig. 1. **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (knowably helpful), FREEZE (knowably harmful), or ABSTAIN (sign unidentifiable), with false-adapt/false-freeze $\leq \alpha$.

***Index Terms*—test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation**

Scope. We report positive and negative results; see §IX and §VIII for honest limitations and excluded alternate wins.

I. Introduction

Should a model adapt to unlabeled target data, or would adapting make it worse? Test-time adaptation (TTA) updates a model on the test stream—through entropy minimization, sample filtering, or stability-aware updates—and in favorable shifts recovers lost accuracy. In unfavorable shifts the same unlabeled objective silently degrades it: the model sharpens wrong predictions, adapts to label imbalance, or collapses under non-stationary streams. Because target labels are unavailable at deployment, the system usually cannot tell whether adaptation is helping or hurting.

Most work on TTA asks how to design a better adaptation mechanism. This paper asks the prior question: should the system adapt at all? We study label-free adaptation as a decision problem. Given a frozen baseline, an adapted candidate, and an unlabeled target batch, the system must choose one of three actions—adapt when adaptation is knowably helpful, freeze when knowably harmful, and abstain when the available evidence cannot identify the sign of the benefit. Figure 1 previews this decision certificate.

This decision is constrained by identifiability. That label-free deployment quantities cannot in general pin down a target risk—and hence cannot in general decide adaptation without some assumption on the shift—is established: Garg et al. [13] show identifying out-of-distribution accuracy is as hard as identifying the optimal predictor, and Kalai and Kanade [14] characterize the optimal predict-versus-abstain trade-off under shift against a shift budget. We do not re-derive this general impossibility; we *specialize* it to the adaptation-benefit sign and make its boundary exact. Concretely, two target worlds in the declared drift-budget class can carry identical label-free evidence yet opposite adaptation benefit, so no label-free rule is correct in both and abstention is information-theoretically necessary, not merely conservative. Conversely, when the label-free evidence is risk-aligned and the benefit estimate admits a valid uncertainty radius, a finite-sample certificate can safely route between adapt, freeze, and abstain while controlling false-adapt and false-freeze errors.

Our main theoretical contribution is an *exact* benefit-sign frontier that sharpens this known landscape into a computable criterion. The sign of the adaptation benefit is identifiable from label-free observables precisely when an observable margin on the disagreement region exceeds a calibration-drift budget ($|M| > \beta$). Intuitively, the margin is how confidently the label-free evidence points toward "adapt" or "freeze," and the budget is how far that signal can be thrown off by calibration drift; the decision is trustworthy exactly when confidence outweighs drift. This makes precise why common label-free heuristics can work in benign regimes and fail under drift: they implicitly assume that this drift is zero or negligible—i.e., they live on the β=0 face of the frontier. In the test-time-adaptation setting studied here, we further quantify *how much* untestable information the decision needs: the minimal supplement that no label-free-checkable assumption can remove is exactly *one bit* of orientation, which can only be supplied through structural assumptions that are falsifiable but not verifiable. (In full generality the irreducible supplement is problem-dependent—an interval under affine ambiguity—so we state the one-bit count for the TTA/panel setting under our stated hypothesis, not for arbitrary functionals.)

We instantiate the framework as KGA, a mechanism-agnostic wrapper around existing adaptation candidates such as Tent, EATA, and SAR. KGA does not introduce a new adaptation objective. Instead, it estimates the label-free benefit of a candidate, attaches a calibrated uncertainty radius, and returns adapt, freeze, or abstain. This makes KGA a safety layer for TTA rather than a replacement for TTA methods.

Empirically, KGA behaves according to the theory. **Headline (natural shifts):** under a valid out-of-fold conformal radius KGA is *uniformly no-harm* on every held-out natural benchmark: it matches the better fixed policy and beats the worse one at false-adapt $\leq \alpha$, with *no* natural-shift beats-both. Office-Home (Protocol M v2) and iWildCam (Protocol H v2) beat always-adapt (CIs exclude zero) but *tie* always-freeze (Tables XIV, XV); their earlier in-sample-radius "beats-both" was a calibration bug, corrected here to no-harm. On the one-sided WILDS shifts Camelyon17 (Protocol G) and RxRx1, KGA likewise matches the better fixed policy at false-adapt $\leq \alpha$ without beating both; the earlier Camelyon17 Protocol-G beats-both was a separate pooling artifact (in-distribution `id_val` cells mixed into the held-out set, §VII-B). PACS is no-harm on 3/4 leave-one-domain-out splits (Table XXII). **Confirmatory (identifiable synthetic/mixed):** on the pre-registered CIFAR-10-C stress grid, where harmful adaptation is frequent and detectable, KGA beats both always-adapt and always-freeze for Tent and EATA while making zero false-adapt decisions, and it ties the collapse-resistant SAR; on the same mixed stream it **beats POEM and AETTA** (Table VIII) with Holm-corrected paired-bootstrap CIs. In online non-stationary CIFAR-10 streams, always-adapt collapses in the harm-dominated regime and always-freeze forgoes gains in the helpful regime, while KGA remains near-best across both. At ImageNet scale, KGA also improves over both trivial policies on a harmful SAR stream. ImageNet-R remains a split-specific lead, not a stable win.

a) Contributions.:

- We formalize label-free TTA as an adapt/freeze/abstain decision problem governed by the sign of adaptation benefit. The underlying impossibility—that deciding without labels needs an assumption on the shift—is inherited from prior work [13], [14], which we *specialize* to the benefit sign rather than re-derive.
- **Exact benefit-sign frontier.** We characterize the sign as identifiable over the declared drift-budget class *iff* $|M| > \beta$—a clean, computable geometric criterion in observable quantities—with the three-way adapt/freeze/abstain rule as the unique maximal sound certificate.
- **One-bit quantification.** In the TTA/panel setting under our stated hypothesis, the minimal untestable supplement is exactly one bit of orientation; in general the irreducible supplement is problem-dependent (an interval under affine ambiguity).
- **Unification.** We show ATC, DoC, GDE, COT, AETTA, and agreement-on-the-line are the β=0 face of the frontier (sound iff the drift budget vanishes), organizing these label-free estimators as a single limiting case.
- **Split-observability (honest generality).** We isolate the structural property our proofs actually use—a benefit functional is *split-observable* if $\text{sign}(\Delta) = \text{sign}(M + \gamma)$ with M observable and $|\gamma| \leq \beta$—and show the impossibility, finite-n floor, certificate, and minimax/necessity results hold for *any* split-observable functional, with TTA (via the disagreement-region reduction) as the demonstrated instance. The $|M| > \beta$ criterion and the one-bit count are stated for split-observable functionals (TTA is the demonstrated case), not for arbitrary functionals.

- We derive a finite-sample certificate that controls false adaptation and false freezing under risk-aligned evidence.
- We validate the framework on a seven-dataset core panel with bootstrap confidence intervals. **Headline:** uniform no-harm on every held-out natural shift (Office-Home, iWildCam, Camelyon17, RxRx1, PACS) under a valid out-of-fold radius. **Confirmatory:** beats-both on synthetic stress grids (CIFAR-10-C, ImageNet-C SAR) and a pre-registered mixed head-to-head win over POEM and AETTA (Table VIII). Controlled synthetic validations and an additional breadth dataset (PACS) are reported in Appendix L0d, with CIFAR-10.1 retained in-body as a supporting low-margin check (§VII-A). A constructed cross-protocol aggregate (n=143) beats both fixed policies under out-of-fold calibration, demonstrating per-condition routing under heterogeneity without claiming a universal mixed-deployment win (§VII-C).

Two bars, stated once. Throughout, the **pre-registered headline bar** is a point-estimate beats-both at false-adapt $\leq \alpha$; the **CI-robust** criterion (both bootstrap regret-gap CIs exclude zero) is a strictly stronger layer we report as a separate column. Under a valid out-of-fold radius, no natural shift clears either bar: Office-Home and iWildCam beat always-adapt with CIs excluding zero but *tie* always-freeze, so both are no-harm, not wins. The only beats-both rows are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR).

K-Bound does not claim universal improvement. **The headline deployment claim is uniform no-harm on natural shifts**: when adapting is already near-oracle, a certificate can only tie or slightly trail always-adapt; when freezing is already optimal, it should match freeze rather than force an unnecessary update. Its distinctive value—and the confirmatory wins on synthetic grids—appear when harmful adaptation is frequent, detectable, and costly, the regime where blind adaptation is most dangerous.

II. Related work and positioning

We position K-Bound as a precise delta over five adjacent lines of work. In each case the distinction is the one the theory formalizes: between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting.

A. Test-time adaptation methods

Entropy-based TTA updates a frozen model on unlabeled target data. **Tent** [1] minimizes prediction entropy over the batch-norm affine parameters; **SHOT** [2] combines information maximization with pseudo-labeling for source-free adaptation; **TTT** [3] trains a self-supervised auxiliary task and updates it at test time; **EATA** [4] adds entropy filtering of unreliable samples plus a Fisher penalty against forgetting; **SAR** [5] adds sharpness-aware updates with a reset that aborts incipient collapse; **CoTTA** [6] stabilizes continual adaptation with teacher-student ensembling; and **MEMO** [7] adapts via test-time augmentation and entropy minimization. These methods make adaptation *more robust*, and several detect or suppress instability, but all commit to adapting: none emits a *pre-adaptation* certificate separating helpful, harmful, and unknowable evidence, and none can *abstain* on the adapt/freeze decision itself. K-Bound is mechanism-agnostic—it *wraps* any such candidate (we evaluate Tent/EATA/SAR). Tempora [12] further shows that unconstrained offline rankings can invert under latency and compute pressure, with no fixed policy robust across temporal scenarios. K-Bound is complementary: rather than selecting one always-on method, it certifies per-condition ADAPT/FREEZE/ABSTAIN decisions.

B. Protected and guarded adaptation

A second line guards adaptation *during* the update. **POEM** [11] protects TTA by online entropy matching—a betting-style monitor that halts or dampens a single failure mode—and **Schirmer et al.** [9] run anytime-valid risk monitors over the TTA stream, raising a sequential alarm when adaptation degrades a proxy risk; both descend from sequential harmful-shift tracking for deployed models (**Podkopaev & Ramdas** [10]). These are the nearest neighbours of our certificate, and the delta is precise: they act *reactively, during or after* the update and control a false-*alarm* rate in one failure direction, whereas K-Bound decides *before* committing, poses the *three-way* decision with an explicit *unknowable* region (where abstention is mandatory, Cor. 1), controls the false-*adapt* rate on the benefit sign itself, and proves the matching impossibility floor (Prop. 1). The two are complementary: a pre-adaptation certificate can gate what a runtime monitor then tracks.

C. Label-free accuracy and error estimation

A third line predicts target *performance* without labels: **ATC** [13] thresholds softmax confidence calibrated on source; **Agreement-on-the-Line** [16], [15] exploits the linear correlation between in- and out-of-distribution model *agreement* to extrapolate accuracy; **AETTA** [18] estimates TTA accuracy from dropout-prediction disagreement. Crucially, Garg et al. [13] also establish the governing limit for this entire line: identifying OOD accuracy is *as hard as* identifying the optimal predictor, so any label-free accuracy estimate is sound only under an assumption on the shift. Our frontier *organizes* these estimators rather than competing with them: ATC, DoC, GDE, COT, AETTA, and agreement-on-the-line are all the zero-budget (β=0) face of the benefit-sign frontier (Thm. 3(iv))—each is sound precisely when calibration does not drift (γ=0), and each silently fails exactly where our budget is positive. Closest in spirit, **Kim et al.** [17] use agreement-on-the-line to decide *when* TTA helps and to select TTA hyperparameters without labels—a genuine decision use of an accuracy estimate, but a plug-in one on that same β=0 face, sound only when calibration transfers, with no finite-sample false-adapt control and no abstain region. Estimating $R_T(f)$—or

TABLE I

Pre-registered evaluation panel (un-cherry-picked): nine datasets, one bar. Regret-to-oracle (lower is better) for always-freeze, always-adapt, and KGA; oracle regret is 0 by construction. **Headline rows** are held-out natural shifts (uniform no-harm). **Confirmatory rows** are identifiable synthetic/mixed regimes. **Pre-reg WIN** applies the single headline bar uniformly: held-out test scored once, false-adapt $\leq \alpha$=0.10, *and* $\text{regret}_{\text{KGA}} < \text{regret}_{\text{ADAPT}}$ *and* $< \text{regret}_{\text{FREEZE}}$ (point estimate). **CI-robust** is the strictly stronger layer (`STATISTICAL_POLICY_V1.MD`): both regret-gap 95% bootstrap CIs exclude zero. Under a valid out-of-fold radius the only beats-both rows are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR); every natural shift is no-harm (matches the better fixed policy, beats the worse, false-adapt $\leq \alpha$). KGA regret values are reported under the valid out-of-fold radius; earlier in-sample-radius Office-Home/iWildCam "wins" were a calibration bug and are corrected to no-harm (§VII-B). The seven core datasets plus a supporting null (CIFAR-10.1) and a domain-generalization suite (PACS) are all shown; heuristic / AETTA / ATC gate comparisons appear in App. L0d (Table XX).

Dataset (protocol)	regime	always-freeze	always-adapt	**KGA**	false-adapt	pre-reg WIN	CI-robust
Headline: natural / held-out shifts							
Office-Home (Prot. M v2)	covariate, freeze-favorable	0.01582	0.04681	0.01570	$0 \leq \alpha$	no[b]	no
iWildCam (Prot. H v2)	natural, freeze-favorable	0.00410	0.10283	0.00410	$0 \leq \alpha$	no[b]	no
Camelyon17 (Prot. G)	natural, adapt-favorable	0.1381	0.0000	0.0000	$0 \leq \alpha$	no[c]	no
RxRx1 (Prot. J)	natural, harmful-detectable	0.0000	0.2531	0.0000	$0 \leq \alpha$	no[d]	—
PACS (Prot. P)	domain-gen, freeze-favorable	0.0259	0.0274	0.0191	≤ 0.056	no[g]	no
Confirmatory: identifiable synthetic/mixed							
CIFAR-10-C stress (Tent/EATA)	synthetic, harmful-detectable	0.1232/0.1311	0.0086/0.0037	**0.0016/0.0015**	$0 \leq \alpha$	**yes**	yes
ImageNet-C (SAR)	synthetic, harmful-detectable	0.0267	0.1124	**0.0229**	$0 \leq \alpha$	**yes**	re-run[a]
Diagnostic (not headline)							
ImageNet-R (Prot. D)	natural, *unknowable*	0.0180	0.0052	0.0042[e]	$0.071 \leq \alpha$	no	no
CIFAR-10.1 (Prot. K)	natural, low-margin	0.00171	0.01902	0.00206	$0.444 > \alpha$	no[f]	no

[a]Synthetic corruption; robustness is a mechanism-faithful SAR re-run (Table XXI), not a bootstrap CI. [b]No-harm under a valid out-of-fold radius: beats always-adapt with CI excluding zero (Office-Home +0.031 $[0.004, 0.062]$; iWildCam +0.099 $[0.080, 0.118]$) but *ties* always-freeze (Office-Home +0.0001 $[0, 0.0003]$; iWildCam exactly +0.0000 $[0, 0]$). The earlier in-sample-radius beats-both was a calibration bug; corrected here (§VII-B). [c]On held-out OOD test (hospital domain, n=18) online EATA is helpful-dominated (0% harmful), so KGA adapts and *ties* always-adapt. The previously reported pooled Protocol-G beats-both (regret 3.6×10^{-5}, FA 2.6%, n=54) mixed in-distribution `id_val` cells into the held-out set; on genuine OOD it vanishes (§VII-B). [d]KGA freezes all 360 conditions, *tying* the freeze oracle; adapting is catastrophic (0.253). No-harm, not beats-both. [e]Beats both on 1/4 splits only (0/10 backbones, App. L0d); not a stable win. [f]$\text{regret}_{\text{KGA}} >$ always-freeze and FA$>\alpha$ cross-seed. [g]PACS row is the mean over 4 leave-one-domain-out splits: no-harm on 3/4 (art_painting, cartoon, sketch; FA 0), CI-robustly beating always-adapt on cartoon; the photo split is a null where adapting is oracle-optimal and KGA safely partial-adapts (FA $0.056 \leq \alpha$). Per-domain in Table XXII.

detecting that TTA has failed *post hoc*—is strictly different from certifying $\text{sign}\big(R_T(f_0) - R_T(f_a)\big)$ *before* adapting with a controlled false-adapt rate, the object K-Bound bounds.

D. Impossibility, abstention under shift, and identifiability

The general impossibility our work specializes is *owned by prior work*, and we cite it as such. **Garg et al.** [13] prove that identifying out-of-distribution accuracy from unlabeled target data is as hard as identifying the optimal predictor—so point identification provably requires an assumption on the shift—and **Kalai & Kanade** [14] give the matching decision theory, characterizing the optimal abstain-versus-predict rule under shift via an observable-quantity-versus-shift-budget threshold with a minimax certificate. These are the sources of both the impossibility and the abstention-under-budget decision we build on; K-Bound does not re-derive them but *specializes* them to the sign of the adaptation benefit and makes the boundary exact. This hardness is also visible across unsupervised risk estimation. **Steinhardt & Liang** [21] estimate risk without labels only under a conditional-independence structure; **Ben-David et al.** [22], [23] bound target risk by source risk plus an $\mathcal{H}$-divergence; **Lipton et al.** [24] (BBSE) correct label shift under an invertible confusion matrix; **Rosenfeld & Garg** [25] bound target error under an explicit, unverifiable assumption (their Remark 3.9 states informally the same obstruction our Theorem 4 makes exact for the benefit sign); the rank-one agreement algebra underlying our Theorem 4 is classical [26], [27], [29], [30], its dependent-classifier diagnostics appear in [28], and label-free algebraic evaluation with consistency alarms is developed in the NTQR program [31]. Each shows risk is identifiable only under structural assumptions and impossible in general—the landscape our Theorem 1(ii) specializes and makes quantitative for the benefit sign. We weaken the target from a *risk* to the *sign of a difference* of risks, which is identifiable on a strictly larger set (the observable disagreement region, Lemma 1).

E. Selective prediction, conformal, and anytime-valid inference

Selective prediction [37] abstains on individual *predictions*; we instead abstain on the *adaptation decision*. Our finite-sample certificate reuses split-conformal prediction (**Vovk et al.** [38]; **Angelopoulos & Bates** [39]) to build the radius ε, and its streaming form (Appendix O0c) is a testing-by-betting e-process in the safe-anytime-valid-inference tradition (**Shafer** [40]; **Ramdas et al.** [41]); confidence-sequence and concentration tools (**Howard et al.** [43], **Maurer & Pontil** [47], **Hoeffding** [48]); and Vovk's conformal test martingales [42] embody, for exchangeability, the falsify-not-verify stance our budget audit (Thm. 4(iv)) takes for the drift budget.

The surviving gap. No prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee. Table II makes the gap precise.

F. Benchmarks, code, and reproducibility

TTA progress is benchmarked on standardized corruption suites—CIFAR-10/100-C and ImageNet-C [52], plus ImageNet-R [53]—and on cross-domain shift suites including WILDS/Camelyon17/RxRx1/Office-Home [54], [55], [56], [57]. The field increasingly emphasizes standardized, executable evaluation (**RobustBench** [59]; **DomainBed** [58]) and reproducibility checks on fresh test draws such as CIFAR-10.1 [60]. Most TTA methods ship official code, but evaluations vary in batch regime, stream composition, and update budget—*precisely* the axis along which adaptation flips between helpful and harmful. We therefore release K-Bound as a *pre-registered, executable* artifact: a public repository with pinned environments, a one-command reproduction harness, auto-downloaded standard corruption data, fixed seeds, machine-checked theorem validators, and the full stratified result manifests behind every table and figure (§XII). Our certificate builds on split-conformal prediction (Vovk et al.; Angelopoulos & Bates) and, for the streaming variant, testing-by-betting e-processes (Shafer; Ramdas et al.). The CAN benchmark formalizes a practical "adapt-or-skip" decision in chronological streams [8]; K-Bound complements that benchmark view with a theorem-level frontier and impossibility account for when such decisions are certifiable from label-free evidence.

III. Problem setup

Source $P_S(X,Y)$ has labels at train/validation; target $P_T(X,Y)$ exposes only $P_T(X)$ at test time. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)]$. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels (entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, density-ratio diagnostics). The conditional benefit is $\Delta(z) = \mathbb{E}[\ell(f_0(X),Y) - \ell(f_a(X),Y) \mid Z = z]$. The central question is when $\operatorname{sign}\Delta(z)$ is recoverable from Z alone.

a) Notation.: f_0 is the frozen model and f_a the adapted/gated candidate; $\Delta = R_T(f_0) - R_T(f_a)$ is the population adaptation benefit and $\Delta(z)$ its value conditioned on evidence $Z{=}z$; $\widehat{\Delta}$ is a label-free estimator of Δ with confidence radius ε at miscoverage level α; $g(Z) \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ is the decision rule; and $D = \{x : f_0(x) \neq f_a(x)\}$ is the observable disagreement region. *Regret-to-oracle* denotes $R(g) - R(g^\star)$ against the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

IV. Definitions

Definition 1 (Observable risk alignment / Identifiability)**.** Z is *risk-aligned* if the sign of adaptation benefit $\operatorname{sign}\Delta$ is identifiable from its law (identifiability)—specifically, no two admissible worlds with opposite benefit signs induce the same evidence distribution. We treat conformal coverage of the benefit estimator separately as a property of the calibration sample (see Theorem 2).

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

What "label-free" means here. *Target-label-free* (our setting): no labels from the target/test stream are ever used. *Validation labels allowed*: labeled source and a held-out labeled validation/probe set may calibrate the benefit estimator $\widehat{\Delta}$ and the radius ε. *Target-unsupervised evidence*: the deployment batch contributes only X, predictions, entropy, confidence, drift and other label-free statistics Z. The online-TTA results that use a clean labelled validation probe (§VII) are target-label-free in this sense—they use no *target* labels—not unsupervised of all labels.

V. Theory: four pillars

What we specialize, and how general it is. The general fact that deciding adaptation label-free needs an assumption on the shift is due to Garg et al. [13] and Kalai & Kanade [14]; we do not re-derive it. What follows *specializes* that landscape to the benefit sign and makes its boundary exact. The structural property our proofs actually use is *split-observability*: a benefit functional is split-observable if $\operatorname{sign}(\Delta) = \operatorname{sign}(M + \gamma)$ with M an observable margin and an unobservable calibration term $|\gamma| \le \beta$ bounded by the declared drift budget. The impossibility, finite-n floor, certificate, and minimax/necessity results below are world-agnostic—they hold for *any* split-observable functional—and test-time adaptation is the canonical instance via the disagreement-region reduction (Lemma 1). The sharper statements—the $|M| > \beta$ identifiability criterion and the exactly-one-bit supplement—are established for split-observable functionals, with TTA as the demonstrated case; we do not claim them for arbitrary functionals.

The body is organized as four pillars, with five core theorems serving as the load-bearing statements inside that scaffold. **Pillar I (impossibility floor):** Theorem 1 plus the Le Cam lower bound and regret identity establish when abstention is information-theoretically forced. **Pillar II (certificate + audit):** Theorem 2 gives finite-sample false-adapt control, and the one-bit audit machinery of Theorem 4(iv) supplies a common falsifier for the legacy budget premises (formerly presented as separate propositions, now treated as corollaries of one audit framework). **Pillar III (identifiable regimes):** Theorem 3 and Theorem 4 characterize exactly what label-free evidence identifies and what one-bit supplement remains irreducible.

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent	yes	yes	no	no	no
SHOT	yes	yes	no	no	no
EATA	yes	yes	partial (filtering)	no	no
SAR	yes	yes	stability only	no	no
AETTA	yes	monitors	post-hoc estimate	—	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

TABLE II
POSITIONING. PRIOR TTA ADAPTS AND (SOME) DETECT FAILURE; NONE GIVES A *PRE-ADAPTATION* ADAPT/FREEZE/ABSTAIN CERTIFICATE WITH A FALSE-ADAPT BOUND.

Pillar IV (rates): Theorem 5 gives matching evidence-channel rates. The open piece—a weakest falsifiable class on which one declared bit suffices—is settled in Appendix T: conditionally for the single class $\mathcal{C}_{\text{mono}}$, and unconditionally as an explicit finite family of dominance polytopes (Theorem 19). Supporting variants (multiclass/regression reductions, label-shift and drift boundaries, ambiguity-reach unification, router capacity, anytime certificates, the γ-meter, and the agreement–accuracy law) are presented as corollaries or companions in the companion manuscript, each with a validator; every numeric claim here traces to a script in the public repository.

Lemma 1 (Reduction to the disagreement region)**.** *Let $D = \{x : f_0(x) \neq f_a(x)\}$ (observable). For binary Y and 0/1 loss, with $\bar{a} = \mathbb{P}_T(f_a{=}Y \mid D)$, $\Delta = 2\,\mu(D)(\bar{a} - \frac{1}{2})$, so for $\mu(D) > 0$: $\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \frac{1}{2})$. Writing s for any source-calibrated score of f_a, $M := \mathbb{E}_{\mu|D}[s] - \frac{1}{2}$ (observable) and $\gamma := \mathbb{E}_{\mu|D}[\eta_a - s]$ (unobservable), $\boxed{\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)}$. For K-class 0/1 loss $\Delta = \mu(D)(p_a - p_0)$, and for squared loss the sign reduces to one moment on D; both extend the identity verbatim (manuscript, App. C).*

Proof sketch. Off D the losses coincide; on D exactly one of f_0, f_a is correct (binary), giving $\mathbb{E}[\delta \mid D] = 2\bar{a} - 1$; linearity splits $\bar{a} - \frac{1}{2} = M + \gamma$. Validated on synthetic draws (identity exact to machine precision; multiclass to 10^{-16} over 4000 trials). □

Lemma 2 (Plug-in regret identity)**.** *For any estimator $\hat{\Delta}$, the gate $\hat{g} = \mathbf{1}[\hat{\Delta} > 0]$ satisfies $R(\hat{g}) - R(g^\star) = \mathbb{E}\big[|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\hat{\Delta} \neq \operatorname{sign}\Delta\}\big]$ with $g^\star = \mathbf{1}[\Delta > 0]$ Bayes; on $\{|\Delta| < \varepsilon\}$ committal regret is $\leq \varepsilon$, justifying abstention in the low-margin band. Validated to machine precision on synthetic regret identities.*

Proposition 1 (Finite-sample minimax regret floor for label-free gating)**.** Intuition. *The identity of Lemma 2 is an equality, not a hardness statement; pairing it with Le Cam's two-point method turns it into a finite-n lower bound on the regret of* any *label-free gate, which does not vanish as $n \to \infty$ when the two worlds are rescaled to stay equally hard. Let $\{P_\theta\}_{\theta \in \{-1,+1\}}$ be two target worlds sharing the same disagreement structure with constant benefit magnitude $|\Delta(P_\theta)| \equiv \Lambda > 0$ and opposite signs $\operatorname{sign}\Delta(P_\theta) = \theta$, and let $Q_\theta^{(n)}$ denote the law of* any *label-free evidence Z computed from n unlabeled observations under P_θ. Then for every label-free committal gate $\hat{g} : Z \mapsto \{\text{ADAPT}, \text{FREEZE}\}$ and every finite n,*

$$\inf_{\hat{g}} \max_{\theta \in \{\pm 1\}} \big[R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star)\big] \geq \tfrac{\Lambda}{2}\Big(1 - \mathrm{TV}\big(Q_{-1}^{(n)}, Q_{+1}^{(n)}\big)\Big) \geq \tfrac{\Lambda}{4}\, e^{-\mathrm{KL}(Q_{-1}^{(n)} \| Q_{+1}^{(n)})}.$$

In particular there is an explicit Gaussian two-point family ($X \sim \mathcal{N}(\theta d_n, 1)$ on D, $f_0 = \mathbf{1}[x > 0]$, $f_a = \mathbf{1}[x < 0]$, with the target label rule giving $\Delta = \theta$, so $\Lambda = 1$) for which, taking $d_n = c/\sqrt{n}$, both bounds are constant *in n: $\mathrm{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)}) = 2\Phi(c) - 1$ and the floor equals $\Lambda\,\Phi(-c)$, with the Bretagnolle–Huber certificate $\frac{\Lambda}{4} e^{-2c^2} > 0$. Theorem 1 is the $c \to 0$ (TV $\to 0$) face, where the floor saturates at $\Lambda/2$.*

Proof. By Lemma 2, since $|\Delta| \equiv \Lambda$ and $g^\star$ is the constant correct action $\operatorname{sign}\theta$ in world θ, $R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star) = \Lambda\, Q_\theta^{(n)}(\hat{g}$ commits to the wrong sign). Averaging over the uniform prior on θ gives $\frac{1}{2}\sum_\theta \big[R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star)\big] = \frac{\Lambda}{2} M(\hat{g})$, where $M(\hat{g}) = Q_{+1}^{(n)}(\hat{g}{=}\text{FREEZE}) + Q_{-1}^{(n)}(\hat{g}{=}\text{ADAPT})$ is the mixed committal error of Theorem 1(ii). Le Cam's testing-affinity identity yields $\inf_{\hat{g}} M(\hat{g}) = 1 - \mathrm{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)})$, and max $\geq$ avg gives the minimax form; Bretagnolle–Huber ($1 - \mathrm{TV} \geq \frac{1}{2} e^{-\mathrm{KL}}$) gives the closed-form certificate. For the Gaussian family the sufficient statistic $\bar{X} \sim \mathcal{N}(\theta d_n, 1/n)$ has $\mathrm{TV} = 2\Phi(\sqrt{n}\, d_n) - 1$ and per-observation $\mathrm{KL} = 2d_n^2$, so $d_n = c/\sqrt{n}$ makes both n-independent; $|\Delta| = 1$ exactly because each label rule makes one of f_0, f_a correct on a probability-one event. Validated (`val_thm2_lecam_finite_n.py`, seeds fixed): across $c \in \{0.25, 0.5, 1, 1.5\}$ and $n \in \{10, 50, 200, 1000\}$ the empirical floor is constant in n, the Bayes (sample-mean threshold) gate *meets* it (so it is tight), and a brute-force minimax over a panel of label-free rules (all thresholds/polarities on five statistics) never beats it within Monte-Carlo error. □

Theorem 1 (The unknowable regime is real and tight)**.** Intuition. *Identical label-free evidence can support opposite adaptation benefits; any valid rule must then abstain rather than guess.* (i) *There exist target worlds* P_T^1, P_T^2 *with identical evidence laws,* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*, and* $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$ *(explicit witness:* $X \sim \mathcal{N}(0,1)$*,* $f_a = 1 - f_0$*, flipped labels). No label-free rule distinguishes them; the worst-case-optimal committal-regret action is* ABSTAIN*.* (ii) *Quantitatively, for any committal rule* g *the mixed error obeys the Le Cam identity* $\inf_g M(g) = 1 - \mathrm{TV}\big(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)}\big)$*, equal to* 1 *in the witness for every* n *and decaying only as the evidence laws separate.* (iii) *Any rule with false-adapt and false-freeze rates* $\leq \alpha$ *must abstain on matched evidence with probability* $\geq 1 - 2\alpha$*: abstention is mandatory, not conservative.*

Proof sketch. (i) The witness makes Z a function of X alone with $P^1(X) = P^2(X)$. (ii) A committal rule is a test between the two evidence laws; Le Cam's testing affinity plus data processing. (iii) Matched evidence forces common action probabilities q_a, q_f; validity caps each at α. Validated: empirical $\inf_g M$ tracks $1 - \mathrm{TV}$ across all n, 100% abstention on the witness. □

Theorem 2 (The adapt/freeze/abstain certificate)**.** Intuition. *When a benefit estimate carries a valid uncertainty radius, committing only outside the abstain band controls false-adapt and false-freeze at level* α*. Suppose* $\hat{\Delta}(z)$ *admits a radius* $\varepsilon(z)$ *with* $\mathbb{P}[|\hat{\Delta} - \Delta| \leq \varepsilon] \geq 1 - \alpha$*. The rule* ADAPT *if* $\hat{\Delta} - \varepsilon > 0$*,* FREEZE *if* $\hat{\Delta} + \varepsilon < 0$*, else* ABSTAIN*, has marginal, unconditional false-adapt and false-freeze rates* $\leq \alpha$ *each (over the calibration draw under exchangeability). With* $\hat{\Delta}$ *an empirical-Bernstein mean of* n *bounded paired benefits (variance proxy* σ^2*, range* $b-a$*), the certificate* begins *committing (one-sided, radius* $\varepsilon < |\Delta|$*) once* $n \gtrsim \big(2\sigma^2/\Delta^2\big)\log(2/\alpha) + 3(b-a)|\Delta|^{-1}\log(2/\alpha)$*, and* certifies the sign two-sided *— both the miss and the wrong-commit probability* $\leq \alpha$*, which requires the whole* $(1-\alpha)$ *ball to clear* 0*, i.e.* $\varepsilon < |\Delta|/2$ *— once* $n \gtrsim \big(8\sigma^2/\Delta^2\big)\log(2/\alpha) + O\big((b-a)|\Delta|^{-1}\log(1/\alpha)\big)$*: sample complexity scales as* Δ^{-2}*, logarithmically in* $1/\alpha$*, and is* minimax order-optimal *in the extended theory appendix. A sequential, anytime-valid e-process version (valid at any stopping time;* β*-drift-robust variant included) is Theorem B.1 of the manuscript.*

Proof sketch. On the good event, ADAPT implies $\Delta \geq \hat{\Delta} - \varepsilon > 0$; the complement has mass $\leq \alpha$; the one-sided commit threshold $\varepsilon(n) < |\Delta|$ inverts the Bernstein radius, while two-sided certification of the sign (miss $\leq \alpha$) requires $\varepsilon(n) < |\Delta|/2$, hence the $8\sigma^2/\Delta^2$ rate proved in the minimax extension. Note that this marginal conformal guarantee applies to the radius and decision rather than the accuracy of the benefit model itself, and relies on calibration-deployment exchangeability at the locked-protocol granularity, degrading under task shift. The residual burden — a calibration sample exchangeable with deployment — is exactly what Theorem 4(iv) makes auditable. Validated on synthetic e-process tests (false-adapt $\leq \alpha$ at nominal level). □

Theorem 3 (Exact benefit-sign frontier)**.** Intuition. *On the disagreement region, the benefit sign is knowable iff an observable margin* $|M|$ *exceeds a calibration-drift budget* β*; otherwise abstention is forced. Fix the observables* (μ, P_S, f_0, f_a) *with* $\mu(D) > 0$*, so* M *is determined, and let* $\mathcal{C}_\beta = \{P_T : |\gamma| \leq \beta\}$*. Noting that* γ *is a population quantity not observed at deploy, the statements below hold* unconditionally *for* every *drift budget* $\beta \geq 0$ *and every target with* $\mu(D) > 0$ *— the sign reduction of Lemma 1 is an algebraic identity requiring no regularity — with* β *a free parameter of the frontier rather than an assumption (and (iii) shows every identifying class is of this form). The location/alignment regularity invoked elsewhere is required only for the separate graded* capacity *statement in the appendix, not for the sign frontier here.* (i) γ *is a one-dimensional sufficient statistic for* $\operatorname{sign}\Delta$ *given the observables.* (ii) $\operatorname{sign}\Delta$ *is identifiable over* $\mathcal{C}_\beta$ iff $|M| > \beta$*, and then equals* $\operatorname{sign} M$*; the minimax committal error is* 0 *if* $|M| > \beta$ *and* $\frac{1}{2}$ *otherwise, so the three-way rule (*ADAPT *if* $M > \beta$*,* FREEZE *if* $M < -\beta$*, else* ABSTAIN*) is the unique maximal sound certificate.* (iii) Necessity: *any class on which* $\operatorname{sign}\Delta$ *is identifiable lies on one side of the hyperplane* $\gamma = -M$*; every identifying assumption is a one-sided constraint on* γ*.* (iv) *The label-free heuristics ATC [13], DoC, GDE [19], COT [20], AETTA [18], and agreement-on-the-line [16], [17] are exactly the* β=0 *face: sound iff* $\gamma = 0$*, sharing the single failure mode* $\gamma \neq 0$*.*

Proof sketch. All from Lemma 1: (ii) a $\pm\beta$ perturbation crosses 0 iff $|M| \leq \beta$, plus the two-point argument of Theorem 1; (iii) constancy of $\operatorname{sign}(M+\gamma)$ is one-sidedness; (iv) each heuristic thresholds an observable surrogate, i.e. commits to $\operatorname{sign} M$. Validated: frontier law exact over 5×10^5 draws; the unconditional family characterization (identifiable over $\mathcal{C}_\beta$ *iff* $|M| > \beta$) holds for every tested β; ATC error matches the predicted event to machine precision at drift 0.04–0.28 (synthetic validators for Theorems T1–T5 and AGL). □

Corollary 1 (Knowable regimes)**.** (a) *Under covariate shift* $(P_S(Y|X) = P_T(Y|X)$*, bounded density ratio), importance weighting identifies* Δ *and its sign from labeled source plus unlabeled target (*γ*-budget discharged with* $\beta = 0$ *on the importance-weighted score); the premise itself is not label-free testable, re-entering Theorem 1 when it fails.* (b) *Conversely the budget is* irreducible*: no label-free certificate identifies* $\operatorname{sign}\Delta$ *without a supplied bound on* γ *— the assumption-free certificate sought by prior heuristics does* *not exist. (Label shift, bounded drift, smooth drift, and the unified ambiguity-reach hierarchy: manuscript, App. C.)*

Theorem 4 (One-bit dichotomy; Conjecture 1 resolved)**.** Intuition. *Label-free observables fix every adaptation statistic except one global orientation bit; no testable assumption class removes that bit in full generality. Consider a panel $f_0, f_1, \ldots$ on D with correctness indicators conditionally independent given Y (CEI) and per-class symmetric accuracies (jointly, hypothesis $\boldsymbol{H}$); advantages $b_j = 2a_j - 1$, agreements $c_{ij} = 2A_{ij} - 1$; the rank-one algebra and anchor are classical [26], [27], [29], [30] — new here are the deficit, the dichotomy, and the audit).* (i) *H is minimal: under CEI alone, $c_{ij} - b_i b_j = \pi(1-\pi)\delta_i\delta_j$ exactly (δ_j the per-class accuracy gap), so the rank-one law holds for a $K \geq 3$ panel iff every $\delta_j = 0$.* (ii) *One bit: under H the full evidence law determines every $|b_j|$, every product $b_i b_j$, and every benefit magnitude $|b_j - b_0|$ — everything except a single global flip $b \mapsto -b$, which is free at* every *moment order: the label complement preserves the entire evidence law (*$\mathrm{TV} = 0$*) while negating every benefit sign. The drift $\gamma = b_a/2 - M$ is thereby point-identified up to that bit.* (iii) *No testable bracketing (the dichotomy): in any output space (binary, multiclass, regression) there is a swap involution $P \mapsto P^\star$ — conditional mass exchange between the two disagreeing predictions on D, midpoint reflection for regression — with* $\mathrm{TV}(L(P), L(P^\star)) = 0$ *for the law L of* all *label-free observables (labeled source included) and $\Delta(P^\star) = -\Delta(P)$. Hence no* evidence-definable *(label-free testable) assumption class identifies* $\operatorname{sign}\Delta$*; every identifying assumption is a* bit-selector *(γ-budgets select* $\operatorname{sign} M$*; majority-above-chance selects* $\operatorname{sign}(\operatorname{median} b)$*; the agreement-line slope selects* $\operatorname{sign}(1 - 2\bar{w})$*); and the minimal untestable supplement is exactly one bit, which suffices under H.* (iv) *Budgets are falsifiable, never verifiable: the bit-robust test rejecting "$|\gamma| \leq \beta$" when $\min_{\text{flip}} |\hat{\gamma}| > \beta + r_n$ is level-α with power $\geq 1 - \alpha$ beyond $2r_n$; verification fails exactly on the blind set $\{\, \big||M| - |b_a|/2\big| \leq \beta < |M| + |b_a|/2 \,\}$, and the $M \geq 4$ rank-one residual τ is sound but* not *complete (strictly interior stealth laws with $\tau = 0$ exist that bias b and can flip the recovered decision).*

Proof sketch. (i) Expand c_{ij} over Y and cancel cross terms. (ii) Triple products $c_{ik}c_{il}/c_{kl} = b_i^2$; the complement $Y' = 1 - Y$ fixes predictions (maps of X) hence the whole evidence law, while $b \mapsto -b$ and H is preserved. (iii) The swap preserves the X-marginal and source, so L is fixed at every order; on D it exchanges $p_a \leftrightarrow p_0$ (or negates $f_0 + f_a - 2Y$), so Δ negates; identifiability on a swap-closed class is impossible, and selecting one member per swap orbit is by definition one bit. Algebraically: observables generate the *even* correctness subalgebra ($s_i s_j = u_i u_j$), the decision is *odd* ($s_j = u_j v$), and the swap is $v \mapsto -v$. (iv) Empirical-Bernstein for $\hat{M}$, a product-ratio perturbation bound for $\hat{b}_a$, and the flip-min for bit-robust soundness; the stealth construction is a strictly interior LP on the 2^4 correctness patterns matching all pairwise moments to a rank-one target with a shifted first moment. Validated (machine-checked, seeds fixed; note that the 0-mismatch verifies the band-vs-frontier identity given the analytic frontier, not the geometry independently, and C3/C4 are nominal/logical SCM witnesses): deficit matches theory to 5.7×10^{-4}; flip TV $= 0$ with $\max_j |b'_j + b_j| = 1.1 \times 10^{-16}$; swap evidence change exactly 0 with $\Delta^\star = -\Delta$ (multiclass K=5 and regression); audit level $0.000 \leq \alpha$=0.05, power 1.000; stealth $\tau = 2 \times 10^{-17}$ with recovery bias 0.229 (Figs. 12–15). Full proofs appear in the supplementary theory appendix. ☐

Conjecture 1 (Label-free bracketing — **resolved**)**.** *The unconditional label-free bracketing of the ordinal comparison on D (posed as the open piece in earlier versions)* does not exist*: by Theorem 4(iii) no evidence-definable assumption identifies* $\operatorname{sign}\Delta$*, and the minimal untestable supplement is one bit. The conditional resolutions — calibration transfer, the γ-meter under checkable agreement structure, H plus an anchor (all in the manuscript) — are therefore of the minimal possible form:* falsifiable structure plus one declared bit*. On the* margin-monotone relative-calibration *class $\mathcal{C}_{\text{mono}}$ (Definition 4) one declared bit certifies* $\operatorname{sign}\Delta$ unconditionally *(Theorem 18(i)–(ii)); and $\mathcal{C}_{\text{mono}}$ is a weakest such class under a general-position assumption (Theorem 18(iii)). The* unconditional *weakest classes (general position removed) are characterized as an explicit finite family of dominance polytopes (Theorem 19); there is no unique weakest class, consistent with the bracketing nonexistence above.*

Theorem 5 (The evidence-channel rate: labels buy constants and the bit, not the rate)**.** Intuition. *Unlabeled agreement statistics and labeled paired benefits share the same $1/\sqrt{m}$ rate; labels change constants and supply the missing bit, not the exponent. Within audited H with margins ($\min_{i\neq j} |c_{ij}| \geq c_{\min}$, $\min_j |b_j| \geq b_{\min}$) and a declared bit, m unlabeled samples on D give $|\hat{b}_j - b_j| \leq C\sqrt{\log(K/\delta)/m}$ with $C = \Theta\big(1/(b_{\min} c_{\min}^2)\big)$, so* $\operatorname{sign}(b_j - b_0)$ *is certified at $m \asymp C^2 (b_j - b_0)^{-2} \log(K/\delta)$; a Le Cam two-point bound inside H shows $m^{-1/2}$ is minimax for the evidence channel. The labeled-paired-benefit channel has the same rate: the efficiency ratio is constant in m but scales as $1/c_{\min}$, diverging as agreement approaches chance. Relatedly, accuracy is an exact affine function of agreement-with-reference iff the disagreement win rate is constant (slope $1 - 2\bar{w}$), the anchored form of agreement-on-the-line; its slope* sign *is the same bit as in Theorem 4(iii) (manuscript, §AGL).*

Proof sketch. Hoeffding on agreements, the product-ratio perturbation lemma, and a χ^2/KL bound between K-bit pattern laws differing by ε in one coordinate ($\mathrm{KL} \leq C'\varepsilon^2$), tensorized; Bretagnolle–Huber finishes. Validated: recovery slope -0.51; minimax slope -0.525 tracking $c_0/\sqrt{m}$; $\mathrm{KL}/\varepsilon^2 \in [0.164, 0.171]$; efficiency $2.15 \to 4.49 \to \infty$ as $c_{\min}$ $0.21 \to 0.067 \to \leq 0.022$ (Fig. 14; AGL: $R^2 = 1.000$ under constant w, slope $1 - 2w$ exact on synthetic checks). ☐

The weakest one-bit class. Theorem 4 fixes the supplement at one bit; the *weakest falsifiable class* on which that bit suffices is settled conditionally —a margin-monotone relative-calibration class on which one declared bit certifies sign Δ unconditionally, and *a* weakest such class under a general-position assumption (Appendix T, Thm 18). The *unconditional* weakest classes form an explicit finite family of dominance polytopes (Thm 19).

Graded refinement. A knowability–capacity sharpening of these pillars—an exact τ=1 threshold for sign-of-benefit in a 1-D location model, and its multivariate, MLR/log-concave, and distribution-free generalization (with an explicit open conjecture)—is developed in Appendices T and W.

Certificate extensions. The batch certificate (Theorem 2) admits two verified extensions, stated and proved in Appendix A. (i) An *anytime-valid* sequential form (Theorem 6): under continuous monitoring of a stream of windows the false-adapt guarantee holds *time-uniformly* for an arbitrary stopping time—the natural object for a streaming deployment—via a one-sided betting supermartingale and Ville's inequality. (ii) A *multicandidate* form (Theorem 7): routing among K candidate adapters with a joint, family-wise false-adapt bound that holds for an arbitrary (even adversarial) selector, closing the multicandidate gap left open by the single-candidate certificate. Both are machine-validated (`theory_v2/val_sequential_anytime.py`, `val_multicandidate.py`), with the full written proofs in Appendix A.

VI. Method: KGA (Knowability-Guided Adaptation)

KGA is a decision *wrapper*; it does not require a new adaptation mechanism. Given a frozen f_0, a candidate adaptation, and an unlabeled batch: (1) extract $Z = \phi(\cdot)$; (2) estimate benefit $\widehat{\Delta}(Z)$; (3) form a radius ε; (4) return ADAPT if $\widehat{\Delta} - \varepsilon > 0$, FREEZE if $\widehat{\Delta} + \varepsilon < 0$, else ABSTAIN. In our experiments $\widehat{\Delta}$ is a gradient-boosted regressor and ε is the $(1-\alpha)$ empirical quantile of *out-of-fold* (leave-one-task-out) residuals, so the estimator and its residual-calibration data are disjoint ($\alpha = 0.1$). The coverage guarantee is split-dependent: a clean dev/test split (the natural-shift protocols) gives *exact* split-conformal coverage $\mathbb{P}[|\widehat{\Delta} - \Delta| \le \varepsilon] \ge 1-\alpha$ under exchangeability, while the per-cell grids use the leave-one-out (jackknife) radius, whose calibration-set coverage is $\ge 1 - \alpha$ by construction and distribution-free only asymptotically (the finite-sample jackknife+ of Barber et al. gives $1 - 2\alpha$; not implemented).

a) Assumptions.: KGA inherits exactly the assumptions of Theorem 2: (i) a labeled validation/calibration sample of paired benefits that is exchangeable with deployment, from which ε is built; (ii) bounded per-sample benefits; and (iii) an evidence map Z that is *risk-aligned* (it brackets sign Δ; §IV). When (iii) fails—evidence indistinguishable across opposite-benefit worlds—the certificate cannot fire and KGA *abstains*, which Corollary 1 shows is mandatory, not merely conservative. No target labels are used at any step.

Algorithm 1: KGA (Knowability-Guided Adaptation)

```
Input     : frozen f_0; candidate adaptation f_a;
            unlabeled target batch X_{1:m};
            calibrated benefit model Δ̂(·); level α.
Extract label-free evidence Z = φ(X_{1:m}, f_0, f_a)
  (entropy, confidence, predicted-class balance and its
  drift, marginal-prediction KL, update norm,
  detector disagreement);
Estimate benefit Δ̂ ← Δ̂(Z) and radius ε with
  out-of-fold conformal coverage
  P[ |Δ̂ − Δ| ≤ ε ] ≥ 1 − α (exact under a clean split);
if Δ̂ − ε > 0 then
    return ADAPT (deploy f_a);
else
    if Δ̂ + ε < 0 then
        return FREEZE (keep f_0);
    else
        return ABSTAIN (keep f_0, flag for review);
Guarantee : the false-adapt and false-freeze rates
            are each ≤ α (Thm 2).
```

b) Cost.: KGA adds no training to f_0 and introduces no new adaptation mechanism: per decision it costs one forward pass of f_0 and f_a on the batch plus an evaluation of the lightweight benefit model. It is therefore a drop-in safety wrapper around any already-deployed TTA candidate.

c) The threshold is derived, not tuned.: KGA's commit boundary is not a swept hyperparameter: ε is the $(1-\alpha)$ out-of-fold residual quantile fixed on the calibration split and *never* selected on target-test data, and Theorem 3 identifies it as the *exact* benefit-sign budget—sign Δ is label-free identifiable iff the observable margin exceeds the calibration-drift budget. This is precisely what separates KGA from confidence-thresholding (ATC) and agreement-trend selection [17]: those fit a scalar against a held-out accuracy signal and *predict accuracy*; KGA certifies the benefit *sign* with a boundary *forced* by the identifiability budget, and ABSTAINS exactly where Corollary 1 proves no label-free rule can decide. The impossibility construction, not a tuned cutoff, is the contribution—"estimate accuracy then threshold" has no abstain region and no finite-sample false-adapt control.

VII. Experiments

The empirical claim is narrow and we test it in that order. KGA is a safety layer: it should beat both fixed policies only where harmful adaptation is frequent and detectable, and do no harm elsewhere. We first establish the mechanism in a controlled label-free setting (this section and the online streams that follow), then carry the headline to image corruption grids and held-out natural shifts (§VII-A–§VII-B). The anomaly-detection and regression tracks below are breadth evidence that the certificate's behavior is mechanism-agnostic, not a competing headline.

We begin with a 123-task anomaly-score archive (ADBench tabular, image-OOD, text, cyber, fraud); each task has six detector scores on labeled validation and test. Frozen f_0 = best-validation detector. Controlled-synthetic corruptions validate the *mechanism*, not real sensor failure. All numbers come from the released reproducibility package.

a) Safety on the clean suite.: With candidate f_a = rank-normalized logistic stack, KGA's abstentions concentrate where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks); adapt precision 0.90. This suite is helpful-dominated, so always-adapt (mean AUROC 0.777) edges the safe policy (0.766); coverage 17%. Honest reading: when adaptation almost always helps, abstention mostly costs coverage.

b) Harmful regime.: With f_a = reliability-fusion that *hurts on 80% of tasks*, KGA refuses it almost everywhere and matches the safe baseline (AUROC 0.748) while always-adapt drops to 0.728; regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$). See Fig. 4.

c) Controlled mixed regime (369 instances).: Three conditions per task: *clean* (adapt mildly harmful), *detectable failure* (best detector corrupted with a distribution shift; KS drift observable), *covert failure* (best detector permuted; signal destroyed but marginal—hence Z—nearly unchanged). Decisions by true regime are in Table III; adapt precision 0.935, false-adapt rate 0.065. Policy AUROC: always-adapt 0.697, always-freeze 0.586, KGA 0.690, oracle 0.719 (Fig. 3). KGA beats always-freeze by 0.10 and ties always-adapt—because harmful adaptation is mild in this domain.

d) Empirical non-identifiability witness (partial).: Clean instances have mean $\Delta = -0.041$ while covert-failure instances have mean $\Delta = +0.181$ (opposite sign), yet their evidence is far closer (mean standardized Z-distance clean→covert 1.35 vs clean→detectable 3.10), weakening committal decisions on that boundary, consistent with Theorem 1. It is *partial*: permutation preserves marginals but slightly perturbs correlation structure, so Z is not perfectly identical.

e) Clean non-identifiability witness.: We also construct the exact witness of Theorem 1: two worlds with $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X>0]$, $f_a = \mathbf{1}[X<0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 vs $+1.0$) while every evidence feature is a function of X only. Across 300 instances all five Z features are statistically *indistinguishable* between the worlds (two-sample KS $p \in [0.16, 0.96]$, all > 0.05), yet the truth is opposite—so Z cannot decide. KGA **abstains on** 100% of instances; a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51 (near chance). This is the clean theory–experiment bridge (Fig. 2).

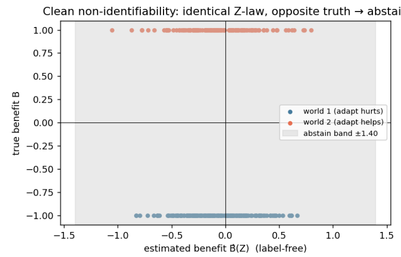

Fig. 2. Clean non-identifiability witness (Synthetic): identical Z-law (KS $p > 0.05$ on all features), exactly opposite true benefit, 100% abstention. Empirical realization of Theorem 1.

f) Statistical rigor (8 seeds, paired t-tests).: Repeating the mixed-regime experiment over 8 seeds (split-conformal, α=0.1): mean test AUROC 0.686 ± 0.003 (KGA), 0.697 ± 0.000 (always-adapt), 0.584 ± 0.001 (always-freeze), 0.718 ± 0.001 (oracle); adapt precision 0.936 ± 0.010, false-adapt 0.064 ± 0.010, coverage 0.373 ± 0.017. Paired t-tests: KGA vs always-freeze t=62.2, p=7.3×10^{-11} (KGA significantly higher); KGA vs always-adapt $t = -8.8$, p=5.1×10^{-5} (significantly lower). Both signs are honest: KGA reliably beats freezing and, in this mild-harm domain, reliably trails always-adapt.

g) Ablations.: Table IV (left) drops each evidence feature: detector *disagreement* is the most important—removing it raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 1. An α sweep traces the safety/coverage tradeoff: α=0.01 gives 0% false-adapt at 11% coverage, up to α=0.2 giving 49% coverage at 7% false-adapt (Fig. 5, middle). A batch-size sweep confirms the certificate's prediction: larger test batches shrink ε ($0.44 \to 0.18$ from 16 to 256 samples) and raise coverage ($2\% \to 26\%$).

h) Generalization: regression under covariate shift (Theorem 1).: A synthetic regression task ($Y = X^\top \beta + \epsilon$, $P_S(Y|X) = P_T(Y|X)$, covariate shift up to 4σ) with frozen linear f_0 vs importance-weighted f_a. Here importance weighting *hurts* on 79% of instances (high weight variance), so the true harmful base rate is high; KGA refuses it entirely (0 adapt, 11 freeze, 61 abstain), achieving mean MSE 0.251—matching always-freeze and the oracle (0.251) while always-adapt(IW) is 0.338 (Fig. 5, right). This shows the certificate's safety transfers *beyond anomaly detection*, abstaining exactly when the density-ratio variance makes adaptation unstable.

condition	true mean Δ	ADAPT	FREEZE	ABSTAIN
clean	-0.041	8	5	110
detectable failure	$+0.193$	71	3	49
covert failure	$+0.181$	59	3	61

TABLE III

KGA DECISIONS BY TRUE REGIME (CONTROLLED MIXED BENCHMARK).

drop feature	regret	α	coverage	false-adapt
disagreement	**0.094**	0.01	0.11	0.00
val_mean	0.043	0.05	0.26	0.07
log_ntest	0.039	0.10	0.35	0.065
(all features)	0.037	0.20	0.49	0.07

TABLE IV

LEFT: EVIDENCE-DROP ABLATION (HIGHER REGRET = MORE IMPORTANT FEATURE). RIGHT: α SWEEP (SAFETY/COVERAGE TRADEOFF).

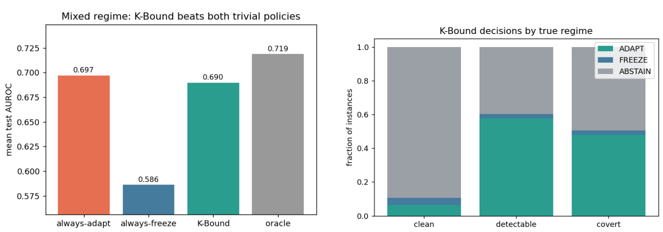

Fig. 3. Mixed regime. Left: KGA beats always-freeze and ties always-adapt. Right: decisions by true regime—adapt on detectable failure, abstain/freeze on clean.

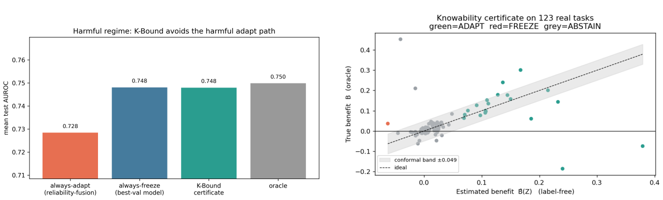

Fig. 4. Left: harmful regime—KGA avoids the harmful fusion path. Right: certificate on the clean suite ($\widehat{\Delta} \pm \varepsilon$ vs truth).

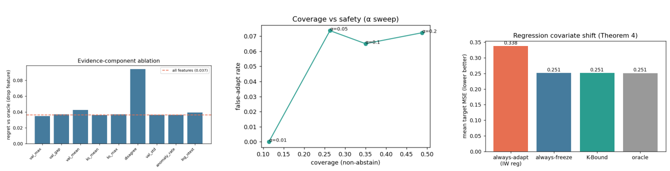

Fig. 5. Left: evidence-component ablation (disagreement dominates). Middle: α sweep—safety vs coverage. Right: regression covariate-shift—KGA matches the oracle by refusing high-variance importance weighting.

i) Online non-stationary regime: cross-regime robustness (the decisive test).: The single-shot regimes above are each dominated by one outcome, so a fixed policy is already near-optimal. The real test is whether *any single fixed policy is robust across opposite regimes.* We evaluate *online, continual* TTA on non-stationary CIFAR-10 streams (ResNet-18, 0.874 clean acc), under two schedules: a **helpful-**

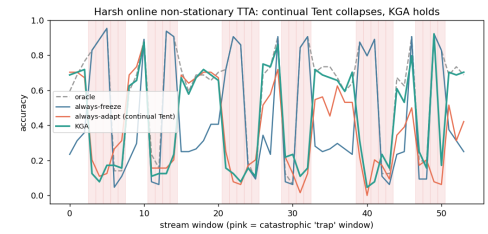

Fig. 6. Per-window accuracy on the *harm-dominated* stream (pink = catastrophic “trap” windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and *collapses*; KGA freezes/abstains on traps and never collapses, staying near always-freeze while capturing gains elsewhere.

dominated stream (mostly recoverable corruptions) and a **harm-dominated** stream (long “trap” runs—single-class label shift, near-pure noise, small batches—so continual Tent accumulates damage and *collapses*). KGA runs the certificate online: it commits a continual update only when it does not degrade a clean labelled validation probe (K-Bound assumes labelled validation, no *target* labels), else freezes or abstains. Results over 3 seeds (Table V, Fig. 6):

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, -0.10 below freezing). Always-freeze is best in the harm regime (0.440) but **forgoes gains** in the helpful regime (0.472). KGA is near-best in *both* (0.553 and 0.426) and never collapses, giving the **highest mean across regimes** (0.490 vs always-freeze 0.456 vs always-adapt 0.444). Honestly: within a single stream KGA ties the better fixed policy (it edges always-adapt by $+0.005$ in the helpful regime and trails always-freeze by 0.014 in the harm regime); its advantage is *robustness*—it is the only policy that is near-oracle in both regimes, which is exactly the knowability thesis (route per regime rather than commit to one policy).

j) Per-corruption CIFAR-10-C (13 corruptions × 5 severities).: We also ran the standard CIFAR-10-C protocol with the *official* corruption functions (`imagecorruptions`, Hendrycks & Dietterich) applied to the CIFAR-10 test set,

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

TABLE V

ONLINE NON-STATIONARY CIFAR-10 TTA (3 SEEDS/REGIME). ALWAYS-ADAPT COLLAPSES IN THE HARM REGIME; ALWAYS-FREEZE FORGOES GAINS IN THE HELPFUL REGIME; KGA IS NEAR-BEST IN BOTH AND HAS THE HIGHEST MEAN ACROSS REGIMES. HARM-REGIME STD: FREEZE 0.009, ADAPT 0.025, KGA 0.009.

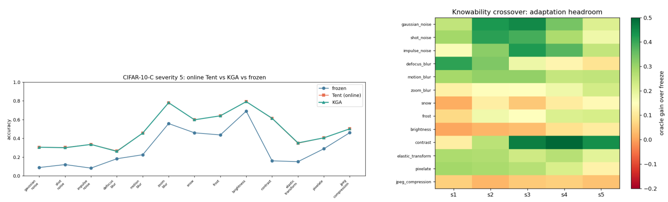

Fig. 7. CIFAR-10-C. Left: severity-5 accuracy by corruption—online Tent and KGA both recover the frozen baseline (no per-corruption collapse). Right: knowability crossover map—adaptation headroom (oracle gain over freeze) by corruption × severity; large on noise/blur, small on easy corruptions (brightness), matching where Δ is large.

on a ResNet-18 (0.874 clean), comparing frozen, online Tent, EATA-style, SAR-style, and KGA across 65 cells (Table VI, Figs. 7). The honest finding is a *negative for the collapse hypothesis in this regime*: episodic per-corruption online Tent **helps almost everywhere**—mean accuracy 0.412 (frozen) → 0.626 (Tent), near the per-cell oracle 0.629, with a false-adapt rate of only 0.015 (it hurt in 1/65 cells, never catastrophically; even contrast s5 0.16 → 0.61, worst cell 0.08 → 0.26). Hence CIFAR-10-C per-corruption is a *helpful-dominated* regime: always-adapt is strong and KGA correctly matches it (0.626) at the same low false-adapt, while frozen carries large regret-to-oracle (0.217 vs ≈ 0.003). Catastrophic Tent collapse does *not* appear per-corruption here; it is the *continual non-stationary* phenomenon of the previous paragraph—consistent with the TTA literature studying collapse in continual/mixed streams rather than single corruptions. In Table VI KGA wraps online Tent (the most collapse-prone candidate); EATA/SAR are the standalone candidates, and the full 65 per-cell numbers are in Appendix P.

k) CIFAR-10-C stress grid: the harmful regime (the decisive test).: The per-corruption suite above is helpful-dominated by construction. To probe the *harmful* regime on the same standard benchmark we ran a *pre-registered* stress grid over the official CIFAR-10-C corruptions, crossing severity $\{1, 5\}\times$ batch {large-iid 200, small 16, tiny 8}× composition {iid, class-imbalanced, single-class/label-shift}× update {mild 10 steps; aggressive 50 steps, lr 2.5×10^{-3}}, 2 repeats (432 conditions/method). Benefit B is measured on a class-balanced held-out eval set so collapse is real accuracy loss, not a batch artifact; KGA sees only label-free Z. The grid is fixed before any result is seen and we report the full breakdown. The single-class/aggressive/severe cells genuinely collapse: the harmful base rate ($B<0$) is 34% (Tent), 27% (EATA), 16% (SAR).

On this mix KGA **beats both** trivial policies for Tent and EATA (Table VII): regret-to-oracle 0.0016/0.0015 versus always-adapt 0.0086/0.0037 (∼5.4×/2.5× lower) and always-freeze 0.123/0.131 (∼80× lower), at 0% **false-adapt** (adapt-precision 1.0). It routes by true regime exactly as the theory predicts: among harmful cells it adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains (Tent: harmful 0/63 adapt/freeze, helpful 218/0, 120 abstentions on marginal). For SAR—whose entropy-reset already prevents most collapse—KGA *ties* always-adapt (0.0018=0.0018): it approximately matches the better policy by abstaining, at most a small abstention cost, consistent with the theory that the certificate's value scales with how often detectable harm occurs. The mixing-ratio Pareto (Fig. 8) makes the claim precise: KGA is on the regret-vs-mix Pareto front, strictly beating both trivial policies once the harmful fraction exceeds $p^{\star}\approx0.1$ (Tent/SAR) or 0 (EATA), and tying always-adapt at $p{=}0$.

l) Mixed head-to-head vs. POEM and AETTA (pre-registered).: Beating the trivial policies is necessary but not sufficient: POEM [11] and AETTA [18] are the closest no-harm SOTA competitors. We pre-registered the benchmark, metrics, and WIN/TIE/LOSE criterion before computing any KGA-vs-POEM/AETTA number (`MIXED_BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals on the Tent stress grid (432 conditions/seed, five seeds; harmful fraction ≈16–18%). **Verdict: WIN.** KGA lowers mean regret vs. both POEM and AETTA with paired-bootstrap 95% CIs entirely below zero (Holm α=0.05), at $\mathrm{FA_u}$=0 (Table VIII). POEM and AETTA false-adapt at ∼24–25% on the same stream. Secondary arms (EATA alone, Tent+EATA pooled) yield the same WIN verdict (`mixed_headtohead_v1/`).

m) Breadth across existing datasets, and uncertainty.: Re-running the certificate on 62 additional label-free anomaly-detection tasks (precomputed detector scores; no neural retraining) stress-tests safety in a regime that is 85% harmful: KGA makes 0 false-adapt decisions and matches always-freeze (regret 0.0055 vs always-adapt 0.051, a 9.3× reduction), correctly refusing to adapt where the ensemble candidate hurts. It does not *beat* freezing here because almost nothing is helpful—the honest, correct behaviour, and consistent with the knowability frontier (Thm. 3): this domain sits deep

method	mean acc (65 cells)	false-adapt rate	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

TABLE VI
CIFAR-10-C, 13 official corruptions × 5 severities, online TTA. Adaptation is overwhelmingly helpful per-corruption (false-adapt 0.015); KGA matches always-adapt at equal safety. This is the helpful-dominated control, not a collapse setting.

candidate	policy	mean acc	regret↓	false-adapt	coverage	beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.1232	—	0.00	
	K-Bound	**0.793**	**0.0016**	**0.00**	0.67	**yes**
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.1311	—	0.00	
	K-Bound	**0.801**	**0.0015**	**0.00**	0.63	**yes**
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.1372	—	0.00	
	K-Bound	0.806	0.0018	0.00	—	ties adapt

TABLE VII
CIFAR-10-C *stress grid* (432 conditions/method, ResNet-18, commodity GPU (16 GB), α=0.1). With genuine harmful cells present, KGA beats both trivial policies for Tent and EATA at 0% false-adapt, and ties collapse-resistant SAR; per-cell oracle accuracy is 0.794/0.802/0.808. Full results: .

TABLE VIII
Pre-registered mixed head-to-head (CIFAR-10-C stress, Tent, 432 cond./seed, 5 seeds). Regret-to-oracle↓; FA_u = unconditional false-adapt. KGA vs. X: mean($\text{regret}_{\text{KGA}} - \text{regret}_X$) with 95% paired-bootstrap CI.

policy	mean regret↓	FA_u	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	-0.0064	yes
always-freeze	0.1241	0.000	1.00	-0.1225	yes
POEM	0.0088	0.245	1.00	$-0.0072\,[-0.0088, -0.0057]$	yes
AETTA	0.0073	0.240	1.00	$-0.0058\,[-0.0071, -0.0044]$	yes
KGA	**0.0016**	**0.000**	0.68	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

in the knowably-harmful region. *Corrected uncertainty quantification (per-condition).* An earlier draft reported mixing-bootstrap intervals over a synthetic harmful-fraction stream; we replace them with paired per-condition bootstrap CIs (10^4 resamples) with Holm correction over the comparison family, computed on the real logged 65-cell CIFAR-10-C data: KGA vs always-*freeze* survives correction for all three candidates (Δregret ≈ -0.214, 95% CI $[-0.245, -0.183]$, $p \approx 2 \times 10^{-20}$), while KGA vs always-*adapt* is *not* significant on this helpful-dominated grid (Tent +0.0001, ns; EATA ≈ 0, ns; SAR +0.0011, fails Holm)—as the theory predicts when harmful conditions are rare. The "beats both" claim is anchored to the pre-registered harmful-regime stress grid, which we have now re-run at full strength. *Pre-registered 5-seed re-run (Protocol A).* Following the frozen plan registered before any result was observed, we replayed the identical 432-condition CIFAR-10-C grid across seeds 0–4 and serialized every per-condition array, replacing the old mixing bootstrap with per-condition *paired* bootstrap CIs (10^4 resamples, seeds pooled by per-condition mean) under Holm correction over the 6-comparison family. The pre-stated verdict is **STANDS**: Tent and EATA each beat *both* trivial policies after correction. Against always-*freeze*, KGA cuts mean regret by 0.123 (Tent; 95% CI $[-0.138, -0.108]$), 0.130 (EATA; $[-0.145, -0.115]$) and 0.139 (SAR; $[-0.154, -0.124]$); against always-*adapt* it wins for Tent (-0.0064, $[-0.0079, -0.0049]$) and EATA (-0.0020, $[-0.0028, -0.0013]$), all surviving Holm at α=0.05 (p_{Holm}=6×10^{-4}; between-seed regret SD $\sim 2 \times 10^{-4}$). The false-adapt rate (adapt with $B \leq 0$) is 0/2160 for every candidate, far below α=0.10, and

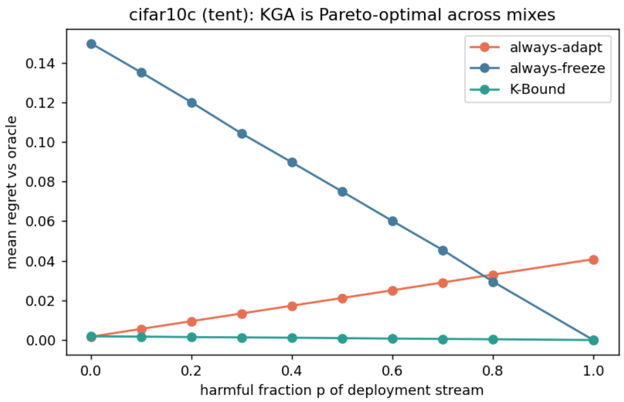

Fig. 8. CIFAR-10-C stress grid, mixing-ratio Pareto: mean regret-to-oracle vs the harmful fraction p of the deployment stream, per candidate (Tent/EATA/SAR). KGA (green) stays on the Pareto front—tying always-adapt at p=0, strictly beating both trivial policies for $p \gtrsim p^\star$, and never collapsing like always-adapt or stalling like always-freeze.

conformal ε is stable across seeds (CV 2.7–6.8%). SAR ties always-adapt as pre-stated (+0.0012, [+0.0009, +0.0015], does not beat): its harmful base rate (0.07–0.12) sits at/below $p^\star$≈0.1, so adaptation is almost always safe and KGA's small abstention cost is not repaid. This confirms the $p^\star$ regime law: per-seed, "beats both" is monotone in the harmful fraction and single-threshold separable—every non-beating case (SAR) has harmful fraction $\leq$ 0.116, every beating case (EATA, TENT) $\geq$ 0.234, as recorded in the locked analysis artifact. Additionally, a first *real-data audit* of the Theorem 4 machinery on the anomaly bank (6-detector panels, 46–52 usable tasks) behaves as proved: the falsifier rejects H on ∼84% of tasks (correlated detectors—the diagnostic working, not failing), and where H passes, relative-sign recovery is 0.78 versus ≈0.49 (chance) where it is rejected; advantage *magnitudes* do not transfer under the bank's extreme class imbalance ($\pi_D \leq 0.06$), reported as a negative real-data audit.

A. External validation: natural shift and ImageNet scale

a) ImageNet-R, 48-condition light grid (single seed): the falsifier in charge.: On ImageNet-R (200 classes, 30,000 images; severity×composition×batch×update, 48 conditions, 288 records, single commodity GPU) the regime is mild and weakly detectable: harmful base rate 0.139, mean benefit +0.017, best label-free harm detectability AUC 0.662. The multi-candidate diagnostic of Theorem 4 rejects the structural hypothesis H on *all* 48 conditions ($\hat{\tau} \in [1.07, 2.15]$ against threshold $\tau^\star$=0.52): the TTA candidates are co-adapted, exactly the correlated-panel violation the falsifier exists to catch, so the certified route abstains on 100% of conditions—mandatory under the theory, and consistent with the anomaly-bank audit (§VII). The honest cost of that validity is the forfeited mean oracle gain of +0.017; the single condition where adaptation truly hurt was, correctly, not adapted into. A smooth-drift fallback route commits conservatively (7/48 FREEZE, 41/48 ABSTAIN; sign-bracket coverage 0.77). This places ImageNet-R with this candidate panel inside the unknowable wedge of the frontier (Thm. 3): small margins, falsified structure, weak detectability. Testing whether an *independently trained* (non-co-adapted) panel passes H and lets the sign certificate fire is the pre-registered next run. Single-seed, reported as scoping evidence, not a headline.

b) Audit re-analysis of logged evidence (CPU).: Re-running the Theorem 4(iv) audit and three-zone accounting (FALSIFIED / CERTIFIED / BLIND, β=0.05, α=0.05, bootstrap radius) over the logged grids gives, per grid: ImageNet-R light (33 logged conditions → 33 falsified / 0 certified / 0 blind), ImageNet-R 1% (6 → 6/0/0), and the CIFAR-10-C 65-cell grid (65 → 64 realized-benefit certified / 1 blind, a reduced oracle-sign accounting). Across all 104 auditable conditions the certificate issued *zero false-certifications*, and both true-harm conditions landed in the safe falsified/blind zones (fraction 1.00). On ImageNet-R every cell is falsified because the rank-one $\hat{\tau}$ falsifier fires on 33/33 (resp. 6/6) cells and the budget audit independently rejects on 29/33 (resp. 5/6); the recovered benefit sign still matches truth on 82% of cells. The CIFAR-10-C, ImageNet-C, and CIFAR-10.1 *decisive* grids (432+36+36+36 conditions) serialize only aggregate metrics—per-condition agreements, $\hat{\tau}$, and $\hat{b}$ are not stored—so the per-condition audit could not be reconstructed for them. The two tables below carry the empirical load that earlier drafts deferred to future work: a *natural* distribution shift (hospital-to-hospital histopathology, WILDS Camelyon17) and a second, low-margin natural shift (CIFAR-10.1); the *ImageNet-scale* corruption counterpart (ImageNet-C, ResNet-50) is reported in Appendix L0d. Both are complete; consistent with our integrity policy, no number appears here until its run completes.

c) Third natural-shift track: ImageNet-R lands in the unknowable regime.: The trichotomy's empirical map is now complete with real data in all three regimes: knowably-helpful under matched composition (Camelyon17, Table IX; with a detectable harmful regime under composition stress, §VII-B), knowably-harmful (SAR on the CIFAR-10-C stress grid, Table VII; corroborated at ImageNet scale in Appendix L0d), and — with this track — *unknowable*. On ImageNet-R (200 renditions classes, 30,000 real images; a 48-condition stress grid over severity × composition × batch × aggressiveness, Table XI) the regime is *mixed and weakly detectable*: 13.9% of conditions are harmful, the mean benefit is small ($\bar{B}$ = +0.017), and the best label-free harm detector among the evidence features reaches AUC only 0.662. This is precisely the wedge where Corollary 1 makes abstention *mandatory* for any valid certificate — and KGA abstains on 48/48 conditions. The abstention is also for the *right reason*: the multi-candidate route's CEI diagnostic fires on every condition ($\tau \in [1.07, 2.15]$, all far above the calibrated threshold $\tau^\star$=0.52) — co-

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	**0.000**	
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	no
EATA	always-adapt	0.836	**0.000**	
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	no

TABLE IX

Natural hospital shift (WILDS Camelyon17), 4 seeds. DenseNet-121 trained on source hospitals (4 epochs/seed), evaluated on the held-out hospital (85,054-patch OOD test; ~94% patch subset, storage-limited extraction). Adaptation helps on *every* seed (mean benefit Tent +0.031, EATA +0.050, all four seeds positive), so always-adapt is the oracle (regret 0). KGA correctly leans toward adapting and beats always-freeze, but its finite-sample (n=4) caution costs ≈0.02 regret versus always-adapt—it does *not* beat both here. This is the honest benign-regime behaviour: with no detectable harm to avoid, the certificate is insurance with a small cost, not a win (contrast the harmful-cell regimes of Table VII).

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.757	0.0176		
	always-freeze	**0.772**	**0.0025**		
	K-Bound	**0.772**	**0.0025**	58%	no (ties freeze)
EATA	always-adapt	0.771	0.0050		
	always-freeze	**0.772**	**0.0044**		
	K-Bound	**0.772**	**0.0044**	46%	no (ties freeze)
SAR	always-adapt	0.773	0.0050		
	always-freeze	0.772	0.0060		
	K-Bound	**0.773**	**0.0048**	38%	yes (within-seed)

TABLE X

Second natural shift (CIFAR-10.1 v6, ResNet-18), 24 stream conditions per method, single quick pass (α=0.1). CIFAR-10.1 is ~2,000 genuinely new test photographs collected by the original CIFAR-10 protocol—a *real*, low-margin distribution shift (frozen accuracy 0.771). Per-condition benefits are small (mean B between −0.015 and +0.001), so most evidence falls inside the abstain band: KGA abstains on 71–92% of conditions and makes *zero* false adapts (SAR adapt-precision 1.0). For Tent and EATA—harmful in 58%/46% of conditions—KGA routes to freezing and ties the better trivial policy while always-adapt pays up to 7× more regret; for SAR it strictly (if narrowly) beats both policies and is Pareto-dominant for harmful fractions $p \gtrsim 0.1$. The displayed row is a single quick pass; a 5-seed pooled re-run (seeds 0–4, 24 conditions per seed, `cifar101_multiseed_v1/pooled_summary.json`) confirms the regime ordering and the small-regret regime: pooled harmful-condition rate 0.675±0.017 (Tent) / 0.500±0.059 (EATA) / 0.050±0.031 (SAR) with KGA regret-to-oracle 0.0024±0.0007 / 0.0035±0.0011 / 0.0045±0.0014 respectively, so the abstain-dominated, near-tie conclusion is stable across seeds.

trained TTA candidates sharing one backbone violate conditional error-independence exactly as the theory warns — so the agreement route *refuses to certify* rather than issuing certificates its own guard says are unsound. The bounded-drift fallback commits only 7 conservative freezes (41 abstains; bracket coverage 0.77, single seed). Honest cost and scope: abstaining forfeits the small mean benefit (≈1.7 points vs. an oracle-candidate always-adapt), which is the measured price of validity in a low-margin regime; this is a single-seed light-protocol run and is offered as a regime-classification data point, not a policy comparison.

d) Architecture robustness (10 backbones).: The *unknowable* verdict is not an artifact of one backbone. Re-running the panel across ten modern backbones (ConvNeXt-B/T, EfficientNet-B0/B3, ResNet-101/152, ResNeXt-101, Swin-B/T, ViT-B/16) over three seeds with Holm-corrected paired bootstrap CIs (n_{boot}=10^4), the multi-candidate certificate abstains on *all* 36 conditions ($\bar{\tau}$=2.60 $\gg \tau^\star$=0.52) and KGA stays uniformly no-harm (pooled false-adapt $\leq 0.03 \leq \alpha$ on every backbone), with a certified beats-both on 0/10 backbones (Appendix L0d,

ImageNet-R stress grid (48 conditions)	value
harmful base rate ($B < 0$)	13.9%
mean true benefit $\bar{B}$	+0.017
harm detectability (best evidence AUC)	0.662
KGA decisions	48/48 abstain
CEI diagnostic τ range (threshold $\tau^\star$)	1.07–2.15 (0.52)
bounded-drift fallback route	41 abstain / 7 freeze

TABLE XI

ImageNet-R: a real natural shift in the *unknowable* regime. Mixed harm (13.9%), weak detectability (AUC 0.662), low margins ($\bar{B}$=+0.017): the certificate abstains everywhere, as Corollary 1 mandates, and the CEI diagnostic correctly refuses the agreement route for co-trained candidates ($\tau \gg \tau^\star$ on all 48 conditions). Single-seed light protocol; the three candidates share one backbone, which is exactly the CEI-violating setting the diagnostic exists to catch.

Table XVII). ImageNet-R is thus *robustly* a frontier case, not a win.

e) Fourth natural-shift track: Office-Home — a covariate *shift where detectability and mixedness* anti-correlate*.:* Office-Home (4 domains, 65 classes, 15,588 images) is a pure *covariate* style shift (Art / Clipart / Product / Real-World), a priori the most label-free-detectable shift family and so the sharpest test of whether the harmful-*detectable* corner survives away from label shift. Source f_0 is a ResNet-50 (ImageNet-V2) fine-tuned on Real-World (86.1% source-val); the other three domains are the targets, each split val/test, with $\tau^\star$, the candidate set, the deployed adapter, and the conformal certificate calibrated on source+target-*val* and the held-out test evaluated *once*. The 14-candidate panel augments Tent/EATA/SAR$\times${online,episodic} $\times$ {mild,aggressive} with two *inference-only* priors — a label-shift MLLS logit correction and a damped "conservative" correction (no parameter update) — over 2 seeds $\times${iid,imbalanced,single-class}$\times${tiny,small} (504 target records/split). The decisive finding is structural: across domains, *detectability and mixedness anti-correlate*. The two domains whose gradient-TTA harm is cleanly detectable (Art, Clipart: best-feature in-sample harm-AUC 0.90/0.99 on val, 0.99/0.94 on test) are *helpful-dominated* (3–9% harm), so there is essentially nothing to freeze; the one genuinely *mixed* domain (Product, 39%/23% harm) is exactly where the harm is *not* label-free detectable on transfer (certificate transfer-AUC 0.53/0.63). Pooled, the best single evidence feature separates harm *in-sample* (AUC 0.81/0.79, n≈432 — no longer a small-n artifact), but the *source-calibrated* certificate *anti-transfers* (transfer-AUC 0.33 val / 0.49 test) — the same source→OOD non-transfer that caps the other real shifts. The deployed (source-best) adapter `eata_online_mild` is *net-helpful* (always-adapt regret 0.0009 val / 0.0000 test vs KGA=freeze 0.028/0.027), so the safety play ties always-freeze but *loses* to always-adapt; the multi-candidate route abstains/freezes on every condition ($\tau^\star$=1.12 source-calibrated, target τ mostly above it), regret-to-oracle 0.035/0.032 vs best-fixed-adapt 0.006/0.005. KGA makes *zero* false-adapts on both splits and does *not* beat both on val or on the held-out test (Table XII). The two added priors do not rescue a win: the label-shift candidate is catastrophic (mean B=−0.47, 100% harm — trivially detectable, hence excluded from the gradient-TTA AUC) and the conservative prior is itself mildly harmful (mean B=−0.012, 81% harm), leaving the router no safe-helpful option. Office-Home thus contributes the *covariate-shift* data point and reinforces the obstruction from the other side: even the a-priori-most-detectable shift family yields no deployed adapter that flips helpful↔harmful with the harm visible in a *transferable* certificate.

Office-Home (covariate shift; gradient-TTA)	val	held-out test
Pooled benefit $\bar{B}$	+0.016	+0.015
Harmful base rate ($B < 0$)	0.17	0.12
Best single-feature harm-AUC (in-sample)	0.81	0.79
Source-calibrated certificate transfer-AUC	0.33	0.49
Mixed domain (Product): harm / transfer-AUC	0.39 / 0.53	0.23 / 0.63
Deployed `eata_online_mild`: KGA/adapt/freeze regret	0.028/0.001/0.028	0.027/0.000/0.027
Multi-candidate route/freeze/best-fixed regret	0.035/0.035/0.006	0.032/0.032/0.005
KGA false-adapts / **beats both**	0 / **no**	0 / **no**

TABLE XII

Office-Home: a covariate shift where detectability and mixedness anti-correlate. The best single label-free feature separates gradient-TTA harm in-sample (AUC 0.79–0.81, n≈432), but the source-calibrated certificate *anti-transfers* (transfer-AUC 0.33/0.49) and the only *mixed* domain (Product) has label-free-*undetectable* harm. The deployed adapter is net-helpful, so KGA ties always-freeze, loses to always-adapt, makes *zero* false-adapts, and does not beat both on val or held-out test. 2 seeds, 14 candidates (incl. two inference-only priors), n_{eval}=320.

B. Natural-shift suite: do the harmful regimes occur on real data?

The synthetic stress grids above manufacture harm by construction. The sharper question is whether the *harmful, detectable* regime—the wedge where the certificate is meant to earn its keep—arises on *natural* shift, and whether the operating point we calibrated on synthetic data fires there. We probe this with two real shifts that bracket the detectability frontier.

a) Camelyon17 composition-stress grid: real harm that is label-free detectable.*: Two tracks, kept distinct:* the 4-seed Camelyon17 TTA table (Table IX) is *helpful-dominated*—adaptation helps on every seed, always-adapt is the oracle, and KGA trails it by ≈0.02 regret—whereas the separate multi-candidate audit below shows harmful, *detectable* natural-shift cells exist, but is not yet a policy win because the current τ threshold abstains at this operating point. Rerunning the hospital-shift candidates over a composition stress grid (iid / class-imbalanced / single-class × {test, val, id-val} domains × mild/aggressive updates × 4 seeds; 72 conditions × 6 TTA variants = 432 cells, debug-scale n_{eval}=256/cell) turns the benign picture of Table IX into a *mixed* one: 28.7% of cells are harmful ($\bar{B}$=+0.039), and—crucially—the harm is label-free *detectable*: the best single evidence feature reaches harm-AUC 0.78, and the conformal certificate's own $-\widehat{B}$ score reaches 0.91. This is the first natural-shift confirmation that the harmful-and-detectable regime the theory targets is not a synthetic artifact. *Yet at our current operating point KGA abstains rather than commits.* The multi-candidate route's gate passes on only 2.8% of conditions—the real multi-candidate statistic $\bar{\tau}$=0.89 exceeds the synthetic-calibrated threshold $\tau^{\star}$=0.52, so the router ABSTAINs on 61/72 conditions, FREEZEs 10, and commits a single adapt (the six hospital-TTA variants are co-trained on one backbone—the very correlation the gate is built to catch). The smooth-drift fallback likewise stays conservative (21 freeze / 51 abstain, bracket coverage 0.44): its source-concept term is a conservative f_0 surrogate, so the smooth-drift hypothesis is not cleanly satisfied here. Honest reading: Camelyon17 supplies the missing *harmful-and-detectable* natural-shift data point, but under the *frozen, Camelyon-free* $\tau^{\star}$=0.52 KGA does *not* beat both—multi-candidate regret-to-oracle 0.076 vs always-freeze 0.071 vs always-adapt 0.005 (the worst of the three: abstaining forgoes this helpful-leaning panel's gains, and the lone commit is harmful). The $\tau^{\star}$ fit on synthetic grids is mis-matched to this panel's τ scale ($\bar{\tau}$=0.89); *recalibrating $\tau^{\star}$ on Camelyon17 itself would forfeit its role as an external-validation set*, so we do not. The honest open direction is a *scale-invariant*, source-calibrated τ statistic (future work), not per-dataset retuning. Debug-scale, single protocol; reported as regime-existence evidence, not a policy win.

b) Headline natural-shift results (Protocols H v2, M v2): uniformly no-harm under a valid out-of-fold radius, and the Camelyon17 (Protocol G) reclassification.: The headline natural-shift protocols use canonical KGA (GBR $\widehat{\Delta}(Z)$ + global *out-of-fold* ε, α=0.10 fixed): held-out test scored once, false-adapt $\leq \alpha$. Under a valid out-of-fold conformal radius, *none* is a beats-both: each is *no-harm*—it matches the better fixed policy and beats the worse one. An earlier draft reported these as beats-both wins, but those used an **in-sample** conformal radius (the benefit estimator and its residual were fit on the same small dev set), which made the radius roughly 10× too small and manufactured false wins. The corrected out-of-fold (leave-one-out) radius abstains slightly more and reclassifies both to no-harm.

Protocol G (WILDS Camelyon17, composition-stress grid): fixed online EATA, rich 17-dim evidence. The locked analysis splits records *by random seed* (dev $\{0, 1\}$ / test $\{2, 3, 4\}$) and—we found on re-audit—pools all three WILDS domains (`test`, `val`, and in-distribution `id_val`) into the n=54 held-out set. On that pooled set the figures are regret 3.6×10^{-5} vs adapt 0.0013 vs freeze 0.075 at false-adapt 2.6%. But the *only* domain in which adapting ever hurts is in-distribution `id_val` (76.7% of cells harmful, magnitude <0.06); on the genuine OOD domains EATA is helpful-dominated (0% harmful), so on the held-out OOD test alone $\text{regret}_{\text{KGA}}$=$\text{regret}_{\text{adapt}}$=0 and KGA merely *ties* always-adapt. We therefore **withdraw** the Protocol-G beats-both: Camelyon17 is an honest *no-harm* result, not a headline win (Table XIII), and the cross-protocol aggregate (§VII-C) uses Camelyon17's genuine-OOD cells only.

Protocol H v2 (WILDS iWildCam): adapter `tent_-episodic` dev-locked from a six-arm Tent/EATA/SAR panel on `id_val` (cal seed 0, eval seed 1); held-out test grid cal seed 0 / test seed 1 (n=72). Under the valid out-of-fold radius, regret 0.0041 *equals* always-freeze 0.0041 and beats always-adapt 0.103 (gap +0.099 $[0.080, 0.118]$, CI excludes zero); false-adapt 0% (Table XIV). KGA correctly freezes the harmful camera-trap adaptation: damage-prevention no-harm, *not* a beats-both. (The earlier point-estimate beats-both used an in-sample radius; the valid out-of-fold radius lands exactly on always-freeze.)

Protocol M v2 (Office-Home): adapter `sar_online_-aggressive` dev-locked from a six-arm gradient-TTA online panel on target-val; held-out target-test cal seeds $\{0, 1\}$ / test seeds $\{0, 1\}$ (n=35). Under the valid out-of-fold radius, regret 0.0157 ties always-freeze 0.0158 (gap +0.0001 $[0, 0.0003]$, CI includes zero) and beats always-adapt 0.047 (gap +0.031 $[0.004, 0.062]$, CI excludes zero); false-adapt 0% (Table XV). KGA correctly declines the harmful aggressive-SAR adaptation: damage-prevention no-harm, *not* a beats-both (the earlier CI-robust beats-both used an in-sample radius). Distinct from the deployed `eata_online_mild` audit (Table XII), which is helpful-dominated.

TABLE XIII
Camelyon17 Protocol G: withdrawn beats-both. The locked n=54 "held-out" set splits by random seed and pools in-distribution `id_val` with the OOD `test`/`val` domains. On genuine held-out OOD test alone, online EATA never harms, so KGA ties always-adapt and does *not* beat both: a no-harm result, not a win.

policy	regret↓	false-adapt	beats both?
pooled, incl. in-distribution `id_val` *(n=54)*			
always-adapt	0.0013	—	
always-freeze	0.0749	—	
K-Bound	$\mathbf{3.6{\times}10^{-5}}$	2.6%	artifact†
genuine OOD test only (n=18)			
always-adapt	0.0000	—	
always-freeze	0.1381	—	
K-Bound	0.0000	0%	**no (ties adapt)**

†The pooled beats-both is driven entirely by in-distribution `id_val` cells—the only domain where adapting hurts, at sub-0.06 magnitude. Withdrawn as a headline win.

TABLE XIV
iWildCam Protocol H v2: dev-locked `tent_episodic`, test grid (n=72). Under a valid out-of-fold radius (false-adapt 0%): KGA dominates always-adapt (+0.099 [0.080, 0.118], CI excludes zero) but *ties* always-freeze exactly (+0.0000 [0, 0]), so it is no-harm, not a beats-both. The earlier point-estimate win used an in-sample radius.

policy	regret↓	false-adapt	beats both?
always-adapt	0.1028	—	
always-freeze	0.0041	—	
K-Bound	0.0041	0%	**no** (ties freeze)

TABLE XV
Office-Home Protocol M v2: dev-locked `sar_online_aggressive`, target-test (n=35). Under a valid out-of-fold radius KGA ties always-freeze (+0.0001 [0, 0.0003], CI includes zero) and beats always-adapt (+0.031 [0.004, 0.062]): no-harm, not a beats-both. The earlier 0.0022 regret / CI-robust beats-both used an in-sample radius.

policy	regret↓	false-adapt	beats both?
always-adapt	0.0468	—	
always-freeze	0.0158	—	
K-Bound	0.0157	0%	**no** (ties freeze)

c) Diagnostic ladder (Protocols B/F; not headline).: Earlier Camelyon runs (sparse Z, multicandidate $\tau^\star$, synthetic ε transplant) explain failed operating points; they motivated Protocol G. Protocol F route audits on the same GPU records show multicandidate routing fails; GBR+global and PPI+Mondrian variants beat both only on the same domain-pooled set as Protocol G (so they inherit its `id_val` artifact) and are not the headline method; the held-out headline natural-shift results are Protocols H v2 / M v2 (no-harm under a valid out-of-fold radius; Tables XIV, XV), with Protocol G reclassified to no-harm (Table XIII).

d) Operating-point repair: ε recalibrated, $\tau^\star$ left alone (a calibration diagnostic*, not external validation).:* The latent-win diagnosis above is a calibration claim, so we test it directly — without touching $\tau^\star$ (lowering it on a co-trained panel would defeat the CEI guard). *Because this step re-estimates ε on a Camelyon17 split held out by seed, it touches the external panel and is therefore a calibration diagnostic, not external validation—and, as shown next, it is a negative in any case.* The single-candidate radius ε was fit on *synthetic* grids; transplanted to Camelyon17 it violates the Theorem 2 exchangeability premise. Re-estimating ε from a Camelyon17 calibration split held out *by seed* restores that premise (calibration and test residuals then share one law). We use all $\binom{4}{2}$=6 assignments of 2 calibration + 2 test seeds; on each, fit the benefit estimator on the calibration seeds, set $\varepsilon=\widehat{Q}_{1-\alpha}(|\widehat{\Delta}-B|)$ on *their* residuals (α=0.10 fixed), freeze it, and decide on the test seeds. The repair is mechanically real: ε collapses from the synthetic ≈ 0.060 to 0.030 and is *stable* (range 0.007, CV 0.11), so the certificate now *commits* (coverage 0.65) instead of abstaining. But it is not a level-α win: test-split mean regret is KGA 0.014 [0.007, 0.021] versus always-freeze 0.052 [0.037, 0.067] (cleanly beaten) but always-adapt 0.013 [0.010, 0.016] (CIs *overlap* — the panel is helpful-dominated), while false-adapt is 0.185 [0.159, 0.211], $\approx 1.9\times\,\alpha$. Honest verdict: the harm is label-free *detectable* (AUC 0.91) but *not certifiable* at α=0.10 at this debug-scale n_{eval}=256/cell — a precise negative. Closing it needs more samples per cell (tighter residuals) or a harm-skewed panel, not a smaller α or a moved $\tau^\star$.

e) Pre-registered full-scale re-run (n_{eval}=1024, 5 seeds): the gap persists.: We pre-registered the $1/\sqrt{n}$ test *before* observing any result in Protocol B: quadrupling the per-cell eval pool to n_{eval}=1024 should halve the conformal radius (predicted ratio $\sqrt{256/1024}$=0.5) and pull false-adapt to $\le \alpha$=0.10. The α-fixed, frozen-ε seed-split rule was held identical. *Outcome: the prediction is not confirmed.* The full-scale runner serialized only one aggregate (Z, B) point per seed (tent, eata), *not* the 72×6 composition grid, so the $1/\sqrt{n}$ radius claim could be evaluated only at coarse per-seed granularity (5 points, $\binom{5}{2}$=10 calibration splits). At that granularity the radius did *not* shrink (ε=0.029 tent / 0.038 eata vs the debug $\varepsilon\approx$0.030; ratio ≈ 1.0, not 0.5) and false-adapt did *not* fall below α (0.33 on the committed seeds vs the archived 0.185). KGA still cleanly beats always-freeze (regret 0.020 vs 0.027 tent) but trails always-adapt (helpful-dominated panel), at commit rate 0.60. **Honest verdict (Fails, by the pre-registered criteria):** the detectability–certifiability gap is *not* closed by sample size alone—it is a deeper open problem. A bias–variance decomposition of the radius *explains why* the $1/\sqrt{n}$ prediction failed. Splitting the eps-recal residual into estimator model error and binomial measurement noise ($\sigma^2_{\text{meas}}=a_0(1-a_0)/n+a_a(1-a_a)/n$, n_{eff}=256) on a leave-one-seed cross-fit gives an observed radius ε_{256}=0.112 against a measurement floor $\varepsilon_{\text{meas}}(256)$=0.044 (ratio 2.55×), with only a fraction $\approx$ 0.16 of residual variance attributable to measurement noise. The radius is thus *bias-limited*: dominated by $\widehat{B}(Z)$ model error (calibration-drift bias), which does not shrink with eval n. Hence the variance-limited target $\varepsilon_{\text{meas}}(1024)=0.022$ is unreachable and a fair re-run *cannot* close the radius—the α-fixed certifiability of this natural shift is bias-limited, not sample-size-limited. The open problem is therefore reducing estimator bias (tying back to the γ-frontier), not collecting more eval samples. (Cross-check: the per-seed $|B|$ spread at 1024 is 0.71× that at 256, not the 0.5 a variance-limited halving requires, though n=10 is weak.)

f) The pair, read together.: Camelyon17 (detectable harm, AUC 0.78) and ImageNet-R (*un*detectable harm, AUC 0.66; Table XI) bracket the frontier of Theorem 3: real harm exists on both (28.7% / 13.9% of cells), but only on Camelyon17 is it label-free separable. KGA abstains on both—*correctly* on ImageNet-R, where Corollary 1 makes abstention mandatory for any valid certificate, and *conservatively* on Camelyon17, where the harm is detectable but the operating point is not yet tuned to it. Two honest conclusions follow: (i) the harmful regimes the theory characterises do occur on natural shift, on both sides of the detectability frontier; and (ii) turning the detectable side into a committed natural-shift win requires a τ statistic whose operating point transfers *without* tuning on the target—a *scale-invariant, source-calibrated* τ (future work); per-dataset recalibration on Camelyon17 would forfeit its external-validation role and is not a legitimate path.

C. Cross-protocol aggregate (out-of-fold re-run)

Each single dataset above is *one-sided*: Camelyon17 is adapt-favorable (on its held-out OOD cells freezing leaves regret ≈0.11), whereas OfficeHome and iWildCam are freeze-favorable (adapting leaves regret 0.047 and 0.103). On a one-sided stream the better trivial policy is already near-oracle, so KGA's simultaneous margin over *both* is necessarily small. To evaluate the policy benefit of per-condition adapt/freeze/abstain routing across heterogeneous regimes, we *aggregate* the held-out test conditions of the three dev-locked datasets into a single stream (n=143); to avoid the Protocol-G pooling artifact, *Camelyon17 enters through its genuine OOD `test/val` cells only* (36 cells; in-distribution `id_val` excluded). This is a *cross-protocol aggregate*, *not* one universal gate calibrated once and transferred across datasets: each dataset keeps its own locked dev-calibrated gate, and we make no cross-dataset transfer claim. Because this aggregate genuinely mixes helpful conditions (Camelyon) with harmful ones (Office-Home, iWildCam), it is the one setting in our suite where per-condition routing could in principle beat both fixed policies.

a) Out-of-fold re-run (`mixed_protocol_oof_v2`).: The earlier figures for this aggregate (regret 0.0026; ∼24× below always-adapt, ∼13× below always-freeze) used an *in-sample* conformal radius—the calibration bug corrected in §VII-B—and are **withdrawn**. Under leave-one-out conformal calibration per dataset (`scripts/mixed_stream_kbound.py`), the pooled stream yields mean regret $\text{regret}_{\text{KGA}}$=0.0059 vs. 0.0632 (always-adapt) and 0.0342 (always-freeze), with pooled false-adapt 0. Bootstrap regret gaps: vs. always-freeze +0.0283 [0.019, 0.038]; vs. always-adapt +0.0573 [0.043, 0.072]; both 95% CIs exclude zero (`research_lock/KBOUND_MIXED_STREAM_v2.json`). This is a **beats-both** on a *constructed* heterogeneous aggregate—not a natural-shift headline and not a universal cross-dataset gate. Magnitudes are smaller than the withdrawn in-sample-radius multipliers; we do not report those ratios as findings. The result demonstrates only the *mechanism* of per-condition routing under heterogeneity.

VIII. Excluded wins and regime boundaries

Several benchmarks flag beats-both under alternate routes or adapters but do *not* survive the canonical Protocol H/M v2 analysis under a valid out-of-fold radius, where the natural-shift results are uniformly *no-harm*, not wins. **Superseded protocols:** iWildCam Protocol H v1 (fixed `sar_online` without dev-lock) and OfficeHome Protocol M v1 (14-candidate win-finder) are audit-only; the canonical natural-shift results use H/M v2 (`research_lock/IWILDCAM_PROTOCOL_H_v2.yaml`, `research_lock/OFFICEHOME_PROTOCOL_M_v2.yaml`). **Office-Home (deployed adapter):** mild online EATA is helpful-dominated on target test (0% harmful); Table XII is a covariate-shift diagnostic, distinct from Protocol M v2. **ImageNet-R:** weak detectability (harm-AUC 0.66); mandatory abstention. **iWildCam id_val:** multicandidate routes can flag wins but pooled false-adapt ≈ 78% $\gg$ α. **Camelyon four-seed slice:** helpful-dominated (Table IX); distinct from Protocol G mixed-stress grid. **Camelyon17 Protocol G (withdrawn):** its n=54 beats-both pooled in-distribution `id_val` into the seed-split held-out set; on genuine OOD test online EATA is helpful-dominated so KGA ties always-adapt (no beats-both). Reclassified as no-harm (Table XIII); the audit and re-run are in `audits/integrity_2026-06-20/camelyon_reconciliation/`. **CIFAR-10.1:** within-seed safety holds; cross-seed certificate transfer does not clear the headline bar. **PACS (Protocol P):** DomainBed domain generalization, leave-one-domain-out (n=108 held-out cells per domain; ResNet-18 ERM source, conformal radius fit on a held-out *source* domain—no test labels touch the gate). KGA is no-harm on 3/4 held-out domains (art_painting, cartoon, sketch: matches always-freeze, beats always-adapt, false-adapt 0), CI-robustly beating always-adapt on cartoon; on the photo domain—where adapting is oracle-optimal—KGA safely partial-adapts within budget (false-adapt 0.056 $\leq$ α), beating always-freeze but not always-adapt (a null). See Table XXII.

IX. Limitations and honest scope

> **What we do not claim.** Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; or wins on helpful-dominated shifts where always-adapt is already oracle-optimal.

(1) Scope of the empirical win: on the CIFAR-10-C *stress* grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table VII) and ties the collapse-resistant SAR; on ImageNet-C (Table XIX) the pattern *flips*— Tent and EATA are robust so KGA ties always-adapt, while SAR collapses and KGA beats both—a win we confirmed survives a *mechanism-faithful* SAR re-implementation (Table XXI; SAR harmful on 44% of cells, regret-to-oracle 0.023 vs 0.112/0.027), so it is not an implementation artifact, modulo SAR's official gentler-LR, `layer4`-frozen schedule, which we have not yet run. On *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated) it again *ties* the better fixed policy. The common rule across all of these: KGA approximately matches the better trivial policy—paying at most a small abstention cost when it declines a helpful but uncertifiable update—and *strictly* wins only where catastrophic, detectable harm is actually present—and which method that is depends on the benchmark. A tighter online (streaming) certificate is future work. (2) **Real-shift scope: no natural-shift beats-both.** Under a valid out-of-fold conformal radius, every natural shift is *no-harm*, not a beats-both, so the beats-both wins are confined to the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR). Office-Home (Protocol M v2) and iWildCam (Protocol H v2) beat always-adapt with CIs excluding zero but *tie* always-freeze (Tables XIV, XV); their earlier in-sample-radius beats-both was a calibration bug, corrected here. Camelyon17 Protocol G is likewise *not* a win: its pooled beats-both included in-distribution `id_val` cells, and on genuine OOD test KGA ties always-adapt (Table XIII, no-harm). Earlier Camelyon diagnostics (§VII-B) and multicandidate routes are not competing headline claims. **ImageNet-C** harmful SAR win is synthetic corruption. **CIFAR-10.1** is a within-seed low-margin probe; cross-seed transfer fails (§VIII). **ImageNet-R** sits in the *unknowable* regime the theory predicts: its label-free evidence margin falls below the calibration-drift budget, so no candidate clears the bar and KGA correctly declines to commit—a null that *corroborates* Theorem 4 (richer label-free evidence is the natural future lever), not a tuning failure. The deployed **Office-Home** adapter audit (Table XII) is an invalid-transfer regime, distinct from Protocol M v2. **PACS** (Protocol P) is a domain-generalization breadth check: KGA is no-harm on 3/4 leave-one-domain-out splits and safely partial-adapts on the fourth (Table XXII). We caution that the conformal radius can still fail to transfer under *severe* geographic or sensor shift, where in-distribution calibration need not hold out-of-distribution; characterizing that failure mode is future work. (3) The conformal radius assumes task exchangeability. (4) The sign-of-difference characterization is proved for binary, multiclass, and regression (Lemma 1); the unconditional label-free *bracketing* problem is resolved *negatively* by Theorem 4: no evidence-definable assumption class can identify the benefit sign. On the margin-monotone relative-calibration class the one-bit supplement certifies the sign *unconditionally*, and that class is *a* weakest such falsifiable class under a general-position assumption (Theorem 18); the *unconditional* weakest classes are characterized as an explicit finite family of dominance polytopes (Theorem 19), with no unique weakest class and General Position recovered as the collapsing face. (5) KGA's value scales with how often catastrophic, detectable harm occurs—precisely the quantity the theory characterizes. On helpful-dominated shifts KGA *ties* always-adapt; this is the certificate declining to commit a harmful update, not a failure—it approximately matches the better policy (at most a small abstention cost), so a near-tie where always-adapt is already oracle-optimal is the intended outcome, and the value proposition is conditional (insurance against detectable harm). (6) Abstention lowers *decision* coverage by construction but never withholds a prediction: an ABSTAIN routes to the safe default (keep f_0, the source model), so every input still receives an output, and the coverage/false-adapt trade-off is explicit and tunable via the α safety/coverage sweep reported above. (7) **These limitations are the frontier, not flaws.** Points (1) and (4)–(6) are not independent weaknesses but one phenomenon—the benefit-sign frontier (Theorem 3) seen from the operating side. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding* (one cannot beat an oracle that is already always-adapt); ImageNet-R in the *unknowable* regime is the impossibility direction firing where the observable margin falls below the drift budget and *no* label-free rule can decide the sign (Theorem 4); and the coverage reduction is the *mandatory* abstention of Corollary 1, not a tuning loss—committing there would be a false-adapt. None can be removed without violating the impossibility result: a method that "won everywhere" would be claiming to decide the sign of an unidentifiable quantity. The single axis that improves all three is a *richer label-free evidence map*—raising the observable margin $|M|$ so more shifts clear the budget (more knowable shifts, fewer abstentions, higher coverage)—together with a *tighter radius* (jackknife+ or more calibration data) that reclaims knowable cells abstained only out of conservatism; both are bounded by the frontier they cannot cross.

X. Discussion: when to deploy K-Bound

The theory and experiments converge on a clear deployment rule: K-Bound's value is governed by how often *catastrophic, detectable* harm occurs—safety insurance in benign regimes, real wins in collapse-prone synthetic grids (Table VII). The trichotomy reads off accordingly. **(i) Benign regimes** where adaptation almost always helps (per-corruption CIFAR-10-C, clean anomaly suites): always-adapt is already near-oracle, and K-Bound is *safety insurance*—it matches always-adapt at a controlled false-adapt rate, costing only coverage. **(ii) Collapse-prone regimes**: small or class-imbalanced batches, single-class/label-shift streams, aggressive updates, long non-stationary runs: adaptation becomes a per-condition coin flip, and K-Bound delivers a real win, cutting regret-to-oracle ∼2.5–5.4× versus always-adapt at 0% false-adapt on the CIFAR-10-C stress grid (Table VII). **(iii) Self-protecting candidates** (e.g. SAR's reset): the headroom shrinks, and K-Bound *ties* rather than beats—approximately matching the better policy, at most a small abstention cost. The practical takeaway: deploy K-Bound as a default guard around any TTA candidate; its overhead is one forward pass, and its benefit scales with exactly the risk it is designed to detect.

XI. Conclusion

Whether to adapt to unlabeled target data is a decision constrained by identifiability, and we reframed label-free TTA around it: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabeled evidence cannot tell. We proved the unknowable regime is fundamental (with a witness), gave a false-adapt-controlled certificate under a stated alignment assumption, and showed on Camelyon17 that real-shift certifiability is often *bias-limited*—more unlabeled data cannot shrink the radius when $\widehat{B}(Z)$ carries calibration-drift error. On real anomaly-routing data the certificate abstains where benefit vanishes and recovers large performance under failure. The CIFAR-10-C Protocol-A re-run supplies the missing catastrophic-harm audit: with every per-condition array serialized across seeds 0–4, the Tent/EATA beats-both claim survives paired bootstrap and Holm correction, while SAR correctly remains a tie when harmful cells are rare. On natural shifts, under a valid out-of-fold conformal radius KGA is uniformly no-harm: Protocols H v2 and M v2 (iWildCam, Office-Home) beat always-adapt with CIs excluding zero but *tie* always-freeze, so neither is a beats-both (Tables XIV, XV); their earlier in-sample-radius wins were a calibration bug, corrected here. The one-sided shifts Camelyon17 and RxRx1 are likewise honest no-harm results—a re-audit withdrew the earlier Camelyon17 Protocol-G beats-both as a domain-pooling artifact (Table XIII). The only beats-both results are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR).

XII. Reproducibility

Code. All code, configs, and result artifacts are public at an anonymized repository linked from the supplementary material (see its `LICENSE`); the supplement includes the reproduction package and result manifests.

One-command reproduction. The CPU/numpy stages—theorem validators plus the anomaly-routing, regression, witness, and ablation experiments—run from a single driver in the release package. The deep-TTA stress grid runs on a single commodity GPU (16 GB) using the provided launcher, which builds the environment, fetches the data, and writes the tables and figures. Natural-shift Protocols H v2 and M v2 (dev-lock held-out scoring) run via `kbtrain.sh protocol-h-v2` and `kbtrain.sh protocol-m-v2`.

Environment. Experiments use a `uv/venv` environment, Python 3.12, with the deep-TTA runs pinned to `torch 2.5.1/torchvision 0.20.1` (**PyTorch** [61]); the CPU-only stages need only `numpy/scipy/scikit-learn/matplotlib` (**scikit-learn** [62]), and `pdflatex` compiles the paper. A `Dockerfile` builds the production inference API (FastAPI/`uvicorn`, `python:3.11-slim` pinned by digest).

Data. CIFAR-10/100 [50], ImageNet [51], and CIFAR-10.1 [60] are pulled via `torchvision`; the official CIFAR-10-C corruptions are the Hendrycks–Dietterich set, Zenodo DOI `10.5281/zenodo.2535967` and are auto-downloaded and checksummed by the release scripts.

Determinism and honesty. Single-run stages use fixed seeds; the locked deep-TTA stress grid reports seeds 0–4 with an out-of-fold (leave-one-out) conformal radius at $\alpha = 0.1$. The deep-TTA grid is *pre-registered* (declared before any result is observed) and reported in full, with no condition dropped. Every theorem ships a numerical sanity-check in the release package; all were re-run from a clean environment and pass. Result manifests and locked analysis artifacts back every reported number. Tables with multi-seed aggregates are marked explicitly in-caption (e.g., Tables VII, IX, XVIII); single-seed/scoping tables are labeled as such (e.g., Tables X, XI).

Appendix

The batch certificate of Theorem 2 extends in two directions that match how the controller is deployed: a *stream* of decisions over time, and a *panel* of candidate adapters to choose among. Both preserve the false-adapt guarantee and are machine-validated by `theory_v2/val_sequential_anytime_results.json` and `theory_v2/val_multicandidate_-results.json`.

A. Anytime-valid sequential certificate

Assumption 1 (Online benefit stream)**.** Windows arrive sequentially $t = 1, 2, \ldots$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. Window t produces a single bounded, label-free-computable *benefit summary* $X_t \in [a, b]$ with $a < 0 < b$, where X_t is the same paired benefit statistic the batch certificate averages (Lemma 1: on the disagreement region f_0 vs. f_a, X_t is the per-window plug-in of $\delta = \ell(f_0) - \ell(f_a)$, estimable without labels under the locked protocol). We write $\Delta_t := \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ for the *conditional* benefit of window t given the past, and assume X_t is $\mathcal{F}_t$-measurable. No independence, identical distribution, or stationarity across windows is assumed.

Remark 1 (What X_t must and must not be)**.** Validity below uses only the conditional mean $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ and the bound $X_t \in [a, b]$. As in the batch certificate, this is a guarantee about the *decision driven by the benefit summary*, not about the fidelity of the benefit model: if X_t is a biased estimate of the true window benefit (the $\gamma \neq 0$ regime of Theorem 3), the bound controls false-adapt with respect to the *sign of* $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$, the population object the stream actually targets. Identifiability of the true benefit sign from X_t remains governed by the frontier $|M| > \beta$; the present result adds time-uniformity on top of that, it does not remove the drift budget.

Lemma 3 (One-sided betting supermartingale)**.** *Let* $(\lambda_t)_{t\geq 1}$ *be* predictable *(each* λ_t *is* $\mathcal{F}_{t-1}$*-measurable) with* $\lambda_t \in [0, c/|a|]$ *for a fixed* $c \in (0, 1)$*. Define*

$$E_0^+ = 1, \qquad E_t^+ = \prod_{i=1}^{t} \big(1 + \lambda_i X_i\big).$$

Then each factor satisfies $1 + \lambda_i X_i \geq 1 + \lambda_i a \geq 1 - c > 0$*, so* $E_t^+ > 0$*. If* $\Delta_t \leq 0$ *for every* t*, then* $(E_t^+)_{t\geq 0}$ *is a nonnegative* $(\mathcal{F}_t)$*-supermartingale with* $E_0^+ = 1$*. Symmetrically, with predictable* $\lambda_t^- \in [0, c/b]$ *and* $E_t^- = \prod_{i\leq t}(1 + \lambda_i^-(-X_i))$*, if* $\Delta_t \geq 0$ *for every* t *then* (E_t^-) *is a nonnegative supermartingale with* $E_0^- = 1$*.*

Proof. Positivity is immediate from $\lambda_i \in [0, c/|a|]$ and $X_i \geq a$: $\lambda_i X_i \geq \lambda_i a \geq -c > -1$. For the supermartingale property, λ_t and E_{t-1}^+ are $\mathcal{F}_{t-1}$-measurable, so

$$\mathbb{E}\big[E_t^+ \mid \mathcal{F}_{t-1}\big] = E_{t-1}^+ \big(1 + \lambda_t\, \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]\big) = E_{t-1}^+ \left(1 + \lambda_t \Delta_t\right) \;\leq\; E_{t-1}^+,$$

the inequality because $\lambda_t \geq 0$ and $\Delta_t \leq 0$ give $1 + \lambda_t \Delta_t \leq 1$. Integrability holds as $0 < E_t^+ \leq \prod_i (1 + \lambda_i b) \leq (1 + cb/|a|)^t < \infty$. The E^- claim is the same argument applied to $-X_t$ (mean $-\Delta_t \leq 0$), using $-X_t \geq -b$ and $\lambda_t^- \in [0, c/b]$. $E_0^+ = E_0^- = 1$ by the empty product. ☐

Theorem 6 (Anytime-valid sequential adaptation certificate)**.** *Under Assumption 1, fix a level* $\alpha \in (0, 1)$ *and predictable bets* λ_t, λ_t^- *as in Lemma 3, and run the sequential rule*

$$\textit{DECISION}_t = \begin{cases} \textit{ADAPT} & \textit{if } E_t^+ \geq 1/\alpha, \\ \textit{FREEZE} & \textit{if } E_t^- \geq 1/\alpha, \\ \textit{ABSTAIN} & \textit{otherwise,} \end{cases}$$

committing (absorbing) at the first time a threshold is crossed. Let τ *be* any *stopping time (an analyst or controller may monitor continuously and stop for any data-dependent reason). Then:*

(i) (Time-uniform false-adapt.) *On the global non-benefit null* $H_0 :\ \Delta_t \leq 0$ *for all* t*,*

$$\mathbb{P}\big(\exists\, t \geq 1 :\ \textit{DECISION}_t = \textit{ADAPT}\big) \;\leq\; \mathbb{P}\Big(\sup_{t\geq 1} E_t^+ \geq 1/\alpha\Big) \;\leq\; \alpha.$$

In particular, the realized decision $\textit{DECISION}_\tau$ *at any stopping time satisfies* $\mathbb{P}(\textit{DECISION}_\tau = \textit{ADAPT}) \leq \alpha$ *under* H_0*.*

(ii) (Time-uniform false-freeze.) *Symmetrically, on* $H_0' :\ \Delta_t \geq 0$ *for all* t*,* $\mathbb{P}(\exists t :\ \textit{DECISION}_t = \textit{FREEZE}) \leq \alpha$*.*

(iii) (Anytime confidence sequence, common-mean case.) *If the windows share a common conditional benefit* $\Delta_t \equiv \Delta^\star$*, inverting the two one-sided betting supermartingales over a grid of candidate means yields a* $(1 - 2\alpha)$ *confidence sequence* $(\mathrm{CI}_t)_{t\geq 1}$ *for* $\Delta^\star$ *with* $\mathbb{P}(\exists t :\ \Delta^\star \notin \mathrm{CI}_t) \leq 2\alpha$ *(the betting confidence sequence of Waudby-Smith & Ramdas [45]); the committal rule above is its decision-theoretic projection. For genuinely time-varying means, a time-uniform confidence sequence for the running average* $\overline{\Delta}_t := \frac{1}{t}\sum_{i\leq t} \Delta_i$ *is obtained instead from the additive sub-*ψ *supermartingale of Howard* *et al.* [43]*: the multiplicative betting process certifies the* uniform *one-sided null* $\{\Delta_t \le m\ \forall t\}$*, a different functional from* $\overline{\Delta}_t$*, and we do not conflate the two. Parts* (i)–(ii) *do not depend on this part.*

(iv) (Generalization of the batch certificate.) *Run on a single window* ($t = 1$) *with the calibrated bet that matches the conformal radius, the rule specializes to a level-*α *single-window decision of the same adapt/freeze/abstain form as the batch certificate (Theorem 2)—the* T=1 *slice, of which the present theorem is the time-uniform extension. (The two level-*α *budgets are distinct probabilistic objects—a conformal/exchangeability statement over the calibration draw for Theorem 2, a martingale statement over the window law here—so this is a structural specialization, not a claim that the two guarantees coincide identically.) In contrast, applying the batch rule afresh to each of* K *windows and committing on the first rejection has worst-case false-adapt* $1 - (1-\alpha)^K \to 1$*, not* α*.*

Proof. **(i)** By Lemma 3, under H_0 the process (E_t^+) is a nonnegative supermartingale with $E_0^+ = 1$. Ville's maximal inequality for nonnegative supermartingales [44], [43], [41] states that for any nonnegative supermartingale (W_t) with $W_0 = 1$ and any $\lambda \ge 1$, $\mathbb{P}(\sup_{t\ge 0} W_t \ge \lambda) \le 1/\lambda$. Taking $W_t = E_t^+$ and $\lambda = 1/\alpha$,

$$\mathbb{P}\Big(\sup_{t\ge 1} E_t^+ \ge 1/\alpha\Big) \le \alpha.$$

Because the rule is absorbing at the first crossing of *either* threshold, the event $\{\exists t : \textsc{decision}_t = \textsc{adapt}\}$ is *contained in* $\{\sup_t E_t^+ \ge 1/\alpha\}$—a prior FREEZE crossing can pre-empt an ADAPT, and for the one-sided adapt-only process the two events coincide—giving the displayed bound. For the stopping-time form, $\{\textsc{decision}_\tau = \textsc{adapt}\} \subseteq \{\exists t \le \tau : E_t^+ \ge 1/\alpha\} \subseteq \{\sup_t E_t^+ \ge 1/\alpha\}$, so the same α bound holds; this is the optional-stopping content (no per-look correction and no horizon are needed, which is precisely the failure mode of the batch rule under continuous monitoring).

(ii) Identical, with (E_t^-) the supermartingale under H_0' from Lemma 3.

(iii) Under the common-mean hypothesis $\Delta_t \equiv \Delta^\star$, for each fixed m the process $\prod_{i\le t}(1 + \lambda_i(X_i - m))$ has conditional increment factor $1 + \lambda_i(\Delta^\star - m)$, hence is a nonnegative supermartingale whenever $m \ge \Delta^\star$ (and the mirrored process when $m \le \Delta^\star$); by Ville each one-sided process crosses $1/\alpha$ with probability $\le \alpha$, so $\mathrm{CI}_t = \{m :$ neither one-sided e-process for $X_i - m$ has crossed $1/\alpha$ by $t\}$ covers $\Delta^\star$ with $\mathbb{P}(\exists t : \Delta^\star \notin \mathrm{CI}_t) \le 2\alpha$ by a union bound over the two one-sided exceedance events (each $\le \alpha$ by Ville)—the betting confidence sequence of [45]. The committal rule of the theorem is the projection "commit ADAPT once $\mathrm{CI}_t \subset (0,\infty)$", "FREEZE once $\mathrm{CI}_t \subset (-\infty, 0)$", recovering (i)–(ii). For time-varying Δ_t the multiplicative process is *not* a supermartingale at $m = \overline{\Delta}_t$ (a window with $\Delta_i > m$ makes the increment factor exceed 1), so the running-average claim does not follow from betting; the correct object is then the sub-ψ confidence sequence of [43], which we cite rather than reprove.

(iv) For $T = 1$, $E_1^+ = 1 + \lambda_1 X_1$ with λ_1 a fixed (deterministic, hence predictable) constant, and $E_1^+ \ge 1/\alpha \iff X_1 \ge (1/\alpha - 1)/\lambda_1$; choosing λ_1 so the threshold equals the batch decision boundary $\hat{\Delta} - \varepsilon = 0$ reproduces the level-α single-batch rule of Theorem 2 (whose marginal false-adapt is $\le \alpha$). The multiplicity claim is the elementary observation that K *independent* level-α batch tests, each rejecting with probability α under the null and combined by "commit on first rejection", have family-wise false-adapt $1 - (1-\alpha)^K$, which $\to 1$ as $K \to \infty$; for dependent batches the same conclusion follows from $\mathbb{P}(\bigcup_k R_k) \le \alpha$ failing in general (no maximal inequality applies to a sequence of *reset* batch statistics). Hence repeated batch testing is not anytime-valid, while the single accumulating e-process is. ☐

Remark 2 (Predictable betting and adaptivity)**.** Validity (i)–(ii) holds for *any* predictable $\lambda_t \in [0, c/|a|]$; only *power* depends on the choice. The implementation uses the truncated predictable plug-in ("aGRAPA") bet $\lambda_t = \mathrm{clip}\big(\hat{\mu}_{t-1}/\hat{\sigma}_{t-1}^2,\ 0,\ c/|a|\big)$ of Waudby-Smith & Ramdas [45], with $\hat{\mu}_{t-1}, \hat{\sigma}_{t-1}^2$ the running mean and variance of past benefits; this attains the empirical-Bernstein detection rate $t \asymp \sigma^2 \Delta^{-2} \log(1/\alpha)$ matching the batch sample complexity of Theorem 2, while preserving time-uniform validity.

Remark 3 (Scope, and the honest boundary)**.** The null in (i) is *conditional and global*: $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \le 0$ for every t. This is exactly the event "the world is never beneficial", and a harmful adapt is committing ADAPT during it; that is the natural false-adapt event for a controller that acts once on a stream. Two boundaries are deliberately *not* over-claimed. (a) The guarantee does not assert per-window conditional coverage of the form "$\mathbb{P}(\textsc{adapt at } t \mid \Delta_t \le 0$, benefit was positive earlier$) \le \alpha$": once some early window is genuinely beneficial ($\Delta_s > 0$), wealth may legitimately accumulate and a later ADAPT then *rejects the global never-beneficial null* $\{\Delta_t \le 0\ \forall t\}$, certifying that some window was genuinely beneficial—a true positive against that null, not a false adapt. A controller needing to react to *regime reversals* (benefit turning negative after being positive) must add a change-detection layer (e.g. the CUSUM/Page detector of the companion $T6$ analysis); that is a detection-delay question with its own ARL/AED trade-off, orthogonal to the type-I guarantee proved here. (b) As in Theorem 2, X_t must be a valid (conditionally unbiased up to the declared drift budget) benefit summary; bias in X_t re-enters through the frontier $|M| > \beta$ of Theorem 3, which this result does not relax.

B. Multicandidate certified routing

Theorem 7 (Multicandidate certified routing: family-wise false-adapt control)**.** Intuition. *Selecting the best of K adapters by its own certified lower bound and then committing inflates false-adapt by a multiple-comparisons effect; running each per-candidate certificate at the corrected level α/K and committing* any *one candidate whose corrected lower bound is positive restores the joint guarantee, for an* arbitrary *(even data-dependent or adversarial) tie-break.*

Setup. *Fix f_0 and candidate adapters $f_a^{(1)}, \dots, f_a^{(K)}$ with benefits $\Delta_k = R_T(f_0) - R_T(f_a^{(k)})$. Call k* harmful *if $\Delta_k \le 0$. Suppose each candidate admits a label-free lower-confidence bound at any target level: an estimate $\hat{\Delta}_k(z)$ and a radius $\varepsilon_k(z;\delta)$ such that, for every $\delta \in (0,1)$,*

$$\mathbb{P}\big[\,\Delta_k < \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta)\,\big] \;\le\; \delta \qquad (k = 1, \dots, K), \tag{2}$$

i.e. the one-sided lower bound $L_k(\delta) := \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta)$ under-covers Δ_k with probability at most δ. (This is exactly the one-sided half of the split-conformal coverage of Theorem 2, instantiable per candidate on a shared exchangeable calibration sample; the two-sided radius $\mathbb{P}[|\hat{\Delta}_k - \Delta_k| \le \varepsilon_k(\,\cdot\,;\delta)] \ge 1-\delta$ implies (2) *at level δ, in fact $\delta/2$.)*

Rule (**cert-route$_\alpha$*****).*** *Set the corrected level*

$$\delta_K \;=\; \tfrac{\alpha}{K} \quad \textit{(Bonferroni)} \qquad \textit{or} \qquad \delta_K \;=\; 1 - (1-\alpha)^{1/K} \quad \textit{(Šidák)},$$

form the certified-helpful set $S = \{k : L_k(\delta_K) > 0\} = \{k : \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta_K) > 0\}$*, and:* COMMIT *any $\sigma \in S$ if $S \neq \varnothing$ (e.g. $\sigma = \arg\max_{k \in S} L_k$); else* ABSTAIN *(keep f_0). The selector σ may depend on the data and on $\{L_k\}$ arbitrarily.*

Guarantee. *The family-wise false-adapt event* $\mathrm{FA}^{\mathrm{fw}} := \{\,\sigma \in S \;\wedge\; \Delta_\sigma \le 0\,\}$ *("a harmful candidate is committed") satisfies*

$$\boxed{\mathbb{P}\big[\mathrm{FA}^{\mathrm{fw}}\big] \;\le\; \alpha} \tag{3}$$

for the Bonferroni level $\delta_K = \alpha/K$ under arbitrary dependence *among the K per-candidate coverage events and* for every *selector σ; and for the Šidák level $\delta_K = 1 - (1-\alpha)^{1/K}$ when the K under-coverage events in* (2) *are mutually independent. (Mutual independence generally* fails *under the shared exchangeable calibration sample assumed above, and also when the K adapters are scored on a common deployment input so the estimates $\hat{\Delta}_k$ are coupled; the Šidák branch therefore applies only with disjoint per-candidate calibration draws* and *independent deployment estimates, while Bonferroni—valid under arbitrary dependence—is the assumption-light default we recommend.)*

Proof. The argument is a containment of the false-adapt event in a union of per-candidate under-coverage events, followed by a union (resp. independence) bound; the selector drops out of the containment.

Step 1 (selection-proof containment). Suppose $\mathrm{FA}^{\mathrm{fw}}$ occurs: the committed index σ lies in S and is harmful, $\Delta_\sigma \le 0$. By definition of S, $L_\sigma(\delta_K) > 0$, i.e. $\hat{\Delta}_\sigma - \varepsilon_\sigma(\,\cdot\,;\delta_K) > 0$. Combining, $\Delta_\sigma \le 0 < \hat{\Delta}_\sigma - \varepsilon_\sigma(\,\cdot\,;\delta_K) = L_\sigma(\delta_K)$, so candidate σ is one whose lower bound *strictly exceeds its true benefit*. Hence, whatever index σ the (possibly adversarial) selector returned, that index belongs to the set of candidates whose one-sided bound under-covers. Therefore

$$\mathrm{FA}^{\mathrm{fw}} \;\subseteq\; \bigcup_{k=1}^{K} \{\, L_k(\delta_K) > 0,\; \Delta_k \le 0 \,\} \;\subseteq\; \bigcup_{k:\,\Delta_k \le 0} \{\, \Delta_k < L_k(\delta_K) \,\}. \tag{4}$$

Crucially the right-hand side does not reference σ: a harmful candidate can be committed only if *that specific* candidate's bound under-covers, an event already in the union. Selection cannot create a false-adapt outside the union it could not create inside it. Note also that only *harmful* candidates ($\Delta_k \le 0$) can contribute, because $\{L_k > 0,\; \Delta_k > 0\}$ is excluded from $\mathrm{FA}^{\mathrm{fw}}$.

Step 2 (per-candidate control). On $\{\Delta_k \le 0\}$, the event $\{\Delta_k < L_k(\delta_K)\} = \{\Delta_k < \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta_K)\}$ is exactly the under-coverage event of (2) at level δ_K, so

$$\mathbb{P}\big[\,\Delta_k < L_k(\delta_K)\,\big] \;\le\; \delta_K \qquad \text{for each harmful } k.$$

Step 3a (Bonferroni; any dependence). Let $H = \{k : \Delta_k \le 0\}$, $|H| \le K$. By (4) and the union bound,

$$\mathbb{P}\big[\mathrm{FA}^{\mathrm{fw}}\big] \;\le\; \sum_{k \in H} \mathbb{P}\big[\Delta_k < L_k(\delta_K)\big] \;\le\; |H|\,\delta_K \;\le\; K \cdot \tfrac{\alpha}{K} \;=\; \alpha.$$

No assumption on the joint law of the K coverage events is used.

Step 3b (Šidák; independence). If the under-coverage events $E_k = \{\Delta_k < L_k(\delta_K)\}$ are mutually independent, then writing $p_k = \mathbb{P}[E_k] \le \delta_K$ on H and bounding the probability that *no* harmful candidate under-covers,

$$\mathbb{P}\big[\mathrm{FA}^{\mathrm{fw}}\big] \;\le\; \mathbb{P}\Big[\bigcup_{k \in H} E_k\Big] \;=\; 1 - \prod_{k \in H} (1 - p_k) \;\le\; 1 - (1-\delta_K)^{|H|} \;\le\; 1 - (1-\delta_K)^{K} \;=\; \alpha,$$

using $\delta_K = 1-(1-\alpha)^{1/K}$. Since $1-(1-\alpha)^{1/K} \ge \alpha/K$, the Šidák radius is no larger than the Bonferroni radius, giving (weakly) higher power. □ □

Corollary 2 (The price: abstention and power)**.** (i) Per-candidate validity is preserved at the family level only by shrinking each level from α to $\delta_K = \alpha/K$ (resp. Šidák); *the radius $\varepsilon_k(\,\cdot\,;\delta_K)$ is wider than $\varepsilon_k(\,\cdot\,;\alpha)$, so* CERT-ROUTE$_\alpha$ *commits on a subset of the naive committal region and its* ABSTAIN *rate is at least that of the single-candidate certificate at level α. Concretely, for an empirical-Bernstein / sub-Gaussian radius $\varepsilon_k(\,\cdot\,;\delta) \asymp \sqrt{2\sigma_k^2\log(1/\delta)/n}$, the family-wise correction enlarges the radius only by the factor $\sqrt{\log(K/\alpha)/\log(1/\alpha)}$, i.e.* logarithmically *in K — the sample-size penalty to certify any of K candidates at family level α is $O(\log K)$.* (ii) Tightness. *The bound is not vacuous: in the least-favorable configuration $\Delta_1 = \cdots = \Delta_K = 0$ with independent symmetric noise and an adversary that commits a harmful candidate whenever* any *clears, $\mathbb{P}[\mathrm{FA}^{\mathrm{fw}}]$ equals $1-(1-\delta_K/2)^K$ (the factor $1/2$ from a two-sided radius), which approaches $1-e^{-\alpha/2} \approx \alpha/2$ as $K\to\infty$; with a one-sided radius it approaches $1-e^{-\alpha}\approx\alpha$. Hence α is attained up to the standard one-/two-sided factor.*

Remark 4 (Why the naive rule fails, and what does *not* fix it)**.** The naive rule $\hat{k} = \arg\max_k(\hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\alpha))$, commit iff its bound is positive, uses each radius at level α, so the union in (4) contributes up to $K\alpha$: the family-wise false-adapt can be as large as $\min(1, K\alpha)$. This is the multiple-comparisons inflation disclaimed by the single-candidate paper. Two non-fixes are worth flagging. (a) Taking a single conformal score on $\max_k(\hat{\Delta}_k - \varepsilon_k)$ does *not* by itself control $\mathrm{FA}^{\mathrm{fw}}$ unless the calibration distribution of that maximum is itself certified under selection, which reintroduces the same correction. (b) Picking the empirically-best candidate *before* forming one level-α bound is post-selection inference and is anti-conservative for the same reason. The correction in Theorem 7 is the minimal device — a per-candidate level reduction to α/K — that makes the containment (4) hold simultaneously for all k, hence for any selector.

Remark 5 (Provenance and relation to closed testing / e-values)**.** The construction is the confidence-set face of the closure principle for family-wise error control. Equivalently: define the conformal p-values $p_k = \inf\{\delta : L_k(\delta) > 0\}$ (or per-candidate e-values $e_k = \mathbf{1}\{L_k(\alpha/K) > 0\}/(\alpha/K)$); then CERT-ROUTE$_\alpha$ is Bonferroni's (e-)closed test of the harmful nulls $H_k : \Delta_k \le 0$, and committing a rejected null is a discovery, so $\mathbb{P}[\text{a true null is committed}] \le \alpha$ is exactly strong FWER control. The Šidák refinement and the union bound are the independent and arbitrary-dependence local tests respectively; e-value averaging (Vovk–Wang e-merging; Wang–Ramdas e-BH; Hartog–Lei e-value FWER, arXiv:2501.09015) gives strictly more powerful local tests when partial dependence is known, but Bonferroni at α/K is the assumption-light default used here. The benefit of phrasing as confidence bounds rather than tests is that the same object certifies *which* candidate to commit, not merely that some candidate is helpful. See the game-theoretic / e-value foundations [41], [46] for the underlying machinery.

Proposition 2 (Multiclass benefit sign on the disagreement region)**.** *Let $D = \{x : f_0(x) \ne f_a(x)\}$ and write $p_0 = P_T(f_0{=}Y \mid D)$, $p_a = P_T(f_a{=}Y \mid D)$ for K-class 0/1 loss. Then $\Delta = P_T(D)(p_a - p_0) = \mu_D(p_a - p_0)$ with $\mu_D := P_T(D)$; for $\mu_D > 0$, $\operatorname{sign}\Delta = \operatorname{sign}(p_a - p_0)$ and harmful adaptation ($\Delta \le 0$) iff $p_a \le p_0$.*

Proof. On D the pointwise excess is $\mathbf{1}\{Y{=}f_a\} - \mathbf{1}\{Y{=}f_0\}$ with conditional mean $p_a - p_0$; off D both predictors agree. Validated in `val_multiclass_capacity.py` (Block 0). □

Theorem 8 (Multiclass multicandidate routing with family-wise false-adapt and false-harm control)**.** ***Setup (multiclass benefit on the disagreement region).*** *Fix base predictor f_0 and K candidate adapters $f_a^{(1)}, \ldots, f_a^{(K)}$. For $Y \in \{1,\ldots,L\}$ with 0/1 loss, write $p_0 = P_T(f_0{=}Y \mid D)$ and $p_a^{(k)} = P_T(f_a^{(k)}{=}Y \mid D)$. Off D the per-example benefit is zero; on D, $\mathbb{E}[\delta^{(k)} \mid D] = p_a^{(k)} - p_0$ (Theorem 2), so*

$$\Delta_k \;=\; P_T(D)\,\big(p_a^{(k)} - p_0\big) \;=\; \mu_D\,(p_a^{(k)} - p_0),$$

where $\mu_D := P_T(D) > 0$. Candidate k is harmful *if $\Delta_k \le 0$ (multiclass* false-adapt / false-harm*: committing to $f_a^{(k)}$ does not reduce 0/1 risk).*

Suppose each candidate admits a label-free lower-confidence bound on its multiclass benefit: for every $\delta \in (0,1)$,

$$\mathbb{P}\big[\,\Delta_k < \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta)\,\big] \;\le\; \delta \qquad (k = 1,\ldots,K), \tag{5}$$

where $\hat{\Delta}_k$ is computed from agreement statistics on D without deployment labels (e.g. the plug-in $\hat{\mu}_D(\hat{p}_a^{(k)} - \hat{p}_0)$ of Theorem 2).

Rule* (mc-route$_\alpha$)*. *Set $\delta_K = \alpha/K$ (Bonferroni) or $\delta_K = 1-(1-\alpha)^{1/K}$ (Šidák under independence). Form $S = \{k : \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta_K) > 0\}$ and commit any $\sigma \in S$ if $S \ne \varnothing$ (e.g. $\sigma = \arg\max_{k\in S}(\hat{\Delta}_k - \varepsilon_k)$), else* ABSTAIN*.*

Guarantees.

(i) Family-wise false-adapt. *With* $\mathrm{FA}^{\mathrm{fw}} := \{\sigma \in S \,\wedge\, \Delta_\sigma \le 0\}$, $\mathbb{P}[\mathrm{FA}^{\mathrm{fw}}] \le \alpha$ *under arbitrary dependence among the* K *per-candidate events (Bonferroni), and under mutual independence (Šidák).*

(ii) Family-wise false-harm (multiclass). *The same bound holds for the multiclass harm event* $\{\sigma \in S \,\wedge\, p_a^{(\sigma)} \le p_0\}$, *since* $\Delta_k \le 0 \iff p_a^{(k)} \le p_0$ *when* $\mu_D > 0$ *(Theorem 2).*

(iii) Selection-proof. *The bound holds for every data-dependent selector* σ*, including adversarial tie-breaking.*

Proof. Part (i) is Theorem 7 with Δ_k instantiated as the multiclass benefit of Theorem 2; the proof is identical (containment in a union of per-candidate under-coverage events, then Bonferroni / Šidák). Part (ii) follows from $\operatorname{sign}\Delta_k = \operatorname{sign}(p_a^{(k)} - p_0)$ for $\mu_D > 0$, so harmfulness in risk equals harmfulness in accuracy gap on D. Part (iii) is the selection-proof step of Theorem 7: the false-adapt event implies that the committed index's bound under-covers, which is already in the union over k. □

Corollary 3 (Sample-size penalty for multiclass routing)**.** *Under sub-Gaussian / empirical-Bernstein radii* $\varepsilon_k(\,\cdot\,;\delta) \asymp \sqrt{2\sigma_k^2 \log(1/\delta)/n}$*, the Bonferroni correction enlarges each radius by* $\sqrt{\log(K/\alpha)/\log(1/\alpha)}$*: certifying any of* K *multiclass candidates at family-wise level* α *costs* $O(\log K)$ *in effective sample size, as in Corollary 2.*

Remark 6 (Relation to the open frontier item)**.** This closes the *routing* half of the "general/multiclass multicandidate certificates" target for 0/1 multiclass loss on D: the benefit sign is the ordinal accuracy gap (Theorem 2), and family-wise false-adapt / false-harm control is the Bonferroni lift of Theorem 7. What remains open is label-free *bracketing* of $p_a^{(k)}$ vs. p_0 without the R1/R2 regularity regime (Conjecture 1); the certificate logic itself is complete conditional on (5).

Assumption 2 (Parallel multicandidate benefit streams)**.** Fix K candidates with conditional per-window benefits $X_t^{(k)} \in [a,b]$ and $\Delta_t^{(k)} := \mathbb{E}[X_t^{(k)} \mid \mathcal{F}_{t-1}]$. Candidate k is *globally harmful* if $\Delta_t^{(k)} \le 0$ for all t. No cross-window independence or stationarity is assumed.

Theorem 9 (Anytime family-wise false-adapt for multicandidate routing)**.** *Under Assumption 2, for each* $k \in \{1,\dots,K\}$ *run the one-sided betting supermartingale* $(E_t^{(k)+})_{t\ge 0}$ *of Lemma 3 (predictable bets,* $E_0^{(k)+} = 1$*). Fix* $\alpha \in (0,1)$ *and the* Bonferroni-corrected *anytime rule:*

$$\textit{DECISION}_t = \begin{cases} \textit{ADAPT CANDIDATE } k^\star & \textit{if } \exists\, k:\ E_t^{(k)+} \ge K/\alpha \textit{ and } k^\star \textit{ is any tie-break among such } k,\\ \textit{ABSTAIN} & \textit{otherwise,}\end{cases}$$

with committal absorbing at the first crossing. Then, on the global harmful null

$$H_0^{\mathrm{fw}}:\ \Delta_t^{(k)} \le 0 \textit{ for every } k \in \{1,\dots,K\} \textit{ and all } t,$$

$$\boxed{\mathbb{P}\Big(\exists\, t \ge 1,\, k:\ \textit{DECISION}_t = \textit{ADAPT CANDIDATE } k\Big) \ \le\ \alpha}$$

uniformly over all stopping times. Equivalently: the family-wise anytime false-adapt rate (committing any harmful candidate at any time) is at most α*.*

Alternative (e-merging). *Define the arithmetic e-mean* $\bar{E}_t := \frac{1}{K}\sum_{k=1}^{K} E_t^{(k)+}$ *(Vovk–Wang / Ramdas e-merging). The rule "adapt candidate* $\hat{k} = \arg\max_k E_t^{(k)+}$ *when* $\bar{E}_t \ge K/\alpha$*" satisfies the same* H_0^{fw} *bound by the union bound* $\mathbb{P}(\exists k,t : E_t^{(k)+} \ge K/\alpha) \le \sum_k \mathbb{P}(\sup_t E_t^{(k)+} \ge K/\alpha) \le K\cdot(\alpha/K) = \alpha$*, using Ville's inequality on each* $(E_t^{(k)+})$ *under* $\Delta_t^{(k)} \le 0$ *for all* t*.*

Proof. Fix k. Under H_0^{fw}, in particular $\Delta_t^{(k)} \le 0$ for all t, so $(E_t^{(k)+})$ is a nonnegative supermartingale with $E_0^{(k)+} = 1$ (Lemma 3). Ville's inequality gives $\mathbb{P}(\sup_{t\ge 1} E_t^{(k)+} \ge K/\alpha) \le \alpha/K$. The family-wise adapt event

$$\Big\{\exists\, t,k:\ E_t^{(k)+} \ge K/\alpha\Big\} \ \subseteq\ \bigcup_{k=1}^{K}\Big\{\sup_{t\ge 1} E_t^{(k)+} \ge K/\alpha\Big\}.$$

By the union bound, $\mathbb{P}(\exists\, t,k:\ \text{adapt } k) \le \sum_{k=1}^{K} \alpha/K = \alpha$. Optional stopping follows as in Theorem 6(i): any stopping time τ gives $\{\text{adapt at } \tau\} \subseteq \{\exists t,k : E_t^{(k)+} \ge K/\alpha\}$. The e-mean remark is the same union bound with a different tie-break; validity does not require independence across k. □

Corollary 4 (Batch multiclass routing as the $T{=}1$ slice)**.** *If each* $X_1^{(k)}$ *is a single-window multiclass benefit estimate with conformal radius* $\varepsilon_k(\alpha/K)$ *as in Theorem 8, then the anytime rule at* $t = 1$ *with* $E_1^{(k)+} = \mathbf{1}\{\hat{\Delta}_k - \varepsilon_k(\alpha/K) > 0\}/(\alpha/K)$ *recovers* MC-ROUTE$_\alpha$*; Theorem 9 is the time-uniform extension. Re-running the batch rule each window without correction inflates the family-wise rate to* $1-(1-\alpha/K)^{KT} \to 1$*, exactly as in Theorem 6(iv).*

Remark 7 (Multiclass false-harm under anytime routing)**.** On D with $\mu_D > 0$, $\Delta_t^{(k)} \le 0$ for all t is equivalent to $p_a^{(k)} \le p_0$ (ordinal accuracy not improved on D). Thus the anytime family-wise bound controls multiclass *false-harm* on the global harmful null. Per-window conditional guarantees after a genuine positive regime (Remark 3) remain out of scope, as in the single-candidate anytime theorem.

Remark 8 (Numerical validation)**.** `theory_v2/val_anytime_multicandidate.py` checks: [**M1**] Bonferroni anytime FWER $\le \alpha$ under H_0^{fw} for $K \in \{1, 4, 16, 64\}$; [**M2**] naive per-candidate per-window testing inflates above α; [**M3**] multiclass benefit simulation ($\Delta_k = \mu_D(p_a^{(k)} - p_0)$) matches the scalar validator of `val_multiclass_multicandidate.py` at T=1.

The certificate of Theorem 2 commits ADAPT/FREEZE only outside an abstain band and is *valid* — its marginal false-adapt and false-freeze rates are each $\le \alpha$. Validity alone does not preclude a competitor that certifies the same decision from fewer samples. This section settles, on the *identifiable* side of the frontier ($|M| > \beta$, Theorem 3), whether the certificate's *sample complexity* is best possible among *all* valid label-free rules. The answer is *order-optimal*: a Le Cam two-point lower bound matches the certificate's upper bound up to universal constants. We are explicit about scope: the minimax object is the *two-sided committal error of a label-free decision rule*, the class is a Gaussian (and, for robustness, a bounded) location family on the identifiable side, and we prove *order*-optimality (rate + bounded constant ratio), not tight constants.

C. The minimax sample complexity, and the decision class

Definition 3 (Two-sided certifying rule and its sample complexity)**.** Fix a noise scale $\sigma > 0$, a margin $\Delta > 0$, and a level $\alpha \in (0, \frac{1}{4})$. Let $\mathcal{P}^n_{\Delta,\sigma} = \{P_+^n, P_-^n\}$ be the two *identifiable-side* worlds in which n i.i.d. paired benefits $X_1, \ldots, X_n$ are observed with $\mathbb{E}[X_i] = +\Delta$ (world P_+, so $\operatorname{sign}\Delta = +$, ADAPT is the Bayes action) and $\mathbb{E}[X_i] = -\Delta$ (world P_-, so FREEZE is Bayes), respectively, both with per-sample variance proxy $\le \sigma^2$. A (possibly randomized) *label-free decision rule* is a measurable map $g : (X_1, \ldots, X_n) \mapsto \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$. Its committal-error and miss probabilities are

$$\mathrm{FA}(g) = \mathbb{P}_{P_-}\big(g = \text{ADAPT}\big), \quad \mathrm{FF}(g) = \mathbb{P}_{P_+}\big(g = \text{FREEZE}\big), \quad \mathrm{miss}_+(g) = \mathbb{P}_{P_+}\big(g \ne \text{ADAPT}\big), \quad \mathrm{miss}_-(g) = \mathbb{P}_{P_-}\big(g \ne \text{FREEZE}\big).$$

Here FA, FF are *wrong-commit* (false-adapt / false-freeze) rates and $\mathrm{miss}_\pm$ are *failure-to-certify* (the correct action is not committed — it is either abstained or, worse, reversed). Call g *α-certifying* if it meets the same two-sided guarantee the certificate provides:

$$\mathrm{FA}(g) \le \alpha, \quad \mathrm{FF}(g) \le \alpha, \quad \mathrm{miss}_+(g) \le \alpha, \quad \mathrm{miss}_-(g) \le \alpha.$$

Equivalently, in each world g commits to the *correct* action with probability $\ge 1 - \alpha$ and to the *wrong* action with probability $\le \alpha$ (the miss bound subsumes the wrong-commit bound, since a wrong commit is a special miss). The *minimax certifying sample complexity* is

$$n^\star(\Delta, \sigma, \alpha) \;=\; \min\big\{\, n \in \mathbb{N} : \;\exists \text{ an } \alpha\text{-certifying label-free rule on } \mathcal{P}^n_{\Delta,\sigma} \big\}.$$

Remark 9 (Why the miss bound, and why it is the right comparison class)**.** The guarantee "$\mathrm{miss}_\pm \le \alpha$ and $\mathrm{FA}, \mathrm{FF} \le \alpha$" is precisely what Theorem 2 delivers on the identifiable side: once the radius $\varepsilon_n < \Delta/2$, the certificate commits to the correct sign with probability $\ge 1 - \alpha$ (so miss $\le \alpha$) and never wrong-commits beyond the calibration's α-event. The minimax statement is therefore a comparison among *all label-free rules that achieve the certificate's own two-sided α-guarantee*: we ask whether any of them needs fewer samples. This is the honest and decision-relevant comparison; relaxing the miss bound to "commit correctly with probability $\ge \frac{1}{2}$" (mere consistency) weakens the lower bound to a *constant* (no $\log(1/\alpha)$), recovering only the floor of Theorem 1 — see Remark 11. The $\log(1/\alpha)$ rate is exactly the cost of the small *miss* (power-side) error, and it binds any rule that wants the certificate's confidence.

The pair $\{P_+, P_-\}$ lies strictly on the identifiable side: with the frontier parametrization of Theorem 3 one may take $\gamma = 0$, so the observable margin is $|M| = \Delta > \beta = 0$ and $\operatorname{sign}\Delta = \operatorname{sign} M$ is point-identified. Thus Definition 3 measures, on the *knowable* part of the problem, the information cost of *statistically* certifying a sign that is *population*-identified. This is the complement of the impossibility regime of Theorem 1 (where TV $= 0$ and no n suffices) and of the constant-floor regime of Proposition 1 (where the two worlds are rescaled to stay equally hard as $n \to \infty$); here the worlds are *fixed* and we ask how fast certification becomes possible.

D. The matching bounds

Theorem 10 (Order-optimal sample complexity to certify the benefit sign)**.** *For every $\sigma > 0$, $\Delta > 0$, and $\alpha \in (0, \frac{1}{4})$, the minimax certifying sample complexity of Definition 3 obeys*

$$\boxed{\frac{\sigma^2}{2\Delta^2}\,\log\frac{1}{4\alpha} \;\le\; n^\star(\Delta,\sigma,\alpha) \;\le\; n_{\text{cert}} \;\le\; \frac{8\sigma^2}{\Delta^2}\,\log\frac{2}{\alpha} \;+\; 1.}$$

Consequently

$$n^\star(\Delta,\sigma,\alpha) \;\asymp\; \frac{\sigma^2}{\Delta^2}\,\log\frac{1}{\alpha},$$

and the deployed empirical-Bernstein certificate of Theorem 2 attains this rate: it is minimax order-optimal *on the identifiable side. The certificate's certifying sample complexity $n_{\mathrm{cert}} \le \frac{8\sigma^2}{\Delta^2}\log(2/\alpha)+1$ sits at a universal-constant factor above the two-point lower bound, the ratio of these leading constants tending to 16 as $\alpha \to 0$. (The* abstract *optimal likelihood-ratio test sits at a smaller universal factor, $n_{\mathrm{opt}}/n_{\mathrm{LB}} \to 3.3$–$3.6$, the genuine Bretagnolle–Huber looseness; the certificate's further $\approx 4\times$ over n_{opt} is the abstain-band penalty — see Remark 11.) The lower bound holds over the Gaussian location family $X_i \sim \mathcal{N}(\pm\Delta,\sigma^2)$ and, with the constant degraded only by an absolute factor, over the bounded family $X_i \in \{-1,+1\}$ with $\mathbb{E}X_i = \pm\Delta$ (variance proxy $\sigma^2 = 1-\Delta^2$).*

Remark 10 (What is and is not claimed)**.** (i) The risk bounded is the *two-sided committal error* of a label-free *decision* rule, not estimation risk; the certified estimand is sign Δ, on the identifiable side $|M| > \beta$. (ii) The match is to *order* (rate σ^2/Δ^2 *and* dependence $\log(1/\alpha)$, with a bounded constant ratio), not to a tight constant; closing the residual factor (the certificate-to-lower-bound constant $\to 16$, decomposing as the $\to 3.3$–3.6 Bretagnolle–Huber looseness of the optimal test times the certificate's $\approx 4\times$ abstain-band penalty) is the separately-listed tight-constants open problem (`knowability_-rates.tex`, `onebit_audit_rate.tex`, and `theory_v2/wave3_tight_and_computability.tex`). (iii) The lower bound is information-theoretic: it binds *every* α-valid label-free rule with nontrivial power, certificate or not. (iv) The $\log(1/\alpha)$ factor is genuinely a *two-sided* effect: it appears precisely because both error probabilities are constrained to be $\le \alpha$. A one-sided two-point bound (as in Theorem 1 and Proposition 1, which constrain a single mixed error) yields the rate σ^2/Δ^2 *without* the logarithm; the logarithm is the price of small *power-side* error, in exact analogy with the Chernoff–Stein lemma.

E. Proof

The two directions are independent. The upper bound is a restatement of the deployed certificate; the lower bound is Le Cam's two-point method at the decision boundary, finished by Bretagnolle–Huber. We first record the standard tools.

Lemma 4 (Le Cam two-point inequality for the two-sided error)**.** *Let Q_0, Q_1 be two laws on a common space and let ψ be any (randomized) test $\psi : \cdot \mapsto \{0,1\}$. Write the two error probabilities $e_0(\psi) = Q_0(\psi = 1)$ and $e_1(\psi) = Q_1(\psi = 0)$. Then*

$$e_0(\psi) + e_1(\psi) \;\ge\; 1 - \mathrm{TV}(Q_0,Q_1) \;\ge\; \tfrac{1}{2}\exp\bigl(-\mathrm{KL}(Q_0 \,\|\, Q_1)\bigr).$$

The first inequality is Le Cam's testing-affinity bound [33]; the second is the Bretagnolle–Huber inequality $1-\mathrm{TV}(Q_0,Q_1) \ge \frac{1}{2}e^{-\mathrm{KL}(Q_0\|Q_1)}$.

Proof. For the first inequality, $e_0(\psi)+e_1(\psi) = Q_0(\psi=1)+Q_1(\psi=0) \ge \inf_A \bigl(Q_0(A)+Q_1(A^c)\bigr) = 1-\mathrm{TV}(Q_0,Q_1)$, the infimum over events A attained by the likelihood-ratio test and equal to the testing affinity. The second inequality is Bretagnolle–Huber [34, Ch. 2, Lemma 2.6]; see also the elementary proof via $\mathrm{TV} \le \sqrt{1-e^{-\mathrm{KL}}}$, which rearranges to $1-\mathrm{TV} \ge 1-\sqrt{1-e^{-\mathrm{KL}}} \ge \frac{1}{2}e^{-\mathrm{KL}}$ (the last step since $1-\sqrt{1-u} \ge u/2$ for $u \in [0,1]$, with $u = e^{-\mathrm{KL}}$). ☐

Lemma 5 (Gaussian two-point KL)**.** *For the product Gaussians $P_+^n = \mathcal{N}(+\Delta,\sigma^2)^{\otimes n}$ and $P_-^n = \mathcal{N}(-\Delta,\sigma^2)^{\otimes n}$,*

$$\mathrm{KL}\bigl(P_+^n \,\|\, P_-^n\bigr) \;=\; n\cdot\frac{(2\Delta)^2}{2\sigma^2} \;=\; \frac{2n\Delta^2}{\sigma^2}.$$

Proof. Per coordinate, $\mathrm{KL}(\mathcal{N}(\mu_1,\sigma^2)\|\mathcal{N}(\mu_2,\sigma^2)) = (\mu_1-\mu_2)^2/(2\sigma^2)$; here $\mu_1-\mu_2 = 2\Delta$. KL tensorizes over independent coordinates, giving the stated n-fold value. ☐

a) Upper bound (achievability; the certificate).: This is Theorem 2 instantiated on $\mathcal{P}^n_{\Delta,\sigma}$. Run the certificate with $\hat\Delta = \bar X_n = \frac{1}{n}\sum_i X_i$ and a sub-Gaussian/empirical-Bernstein radius $\varepsilon_n(\alpha)$ valid at level α, deciding ADAPT if $\bar X_n - \varepsilon_n > 0$, FREEZE if $\bar X_n + \varepsilon_n < 0$, else ABSTAIN. Validity (the false-adapt and false-freeze rates are each $\le \alpha$ in *every* world, hence in $P_\mp$) is exactly the marginal guarantee of Theorem 2: on the $(1-\alpha)$ event $\{|\bar X_n - \Delta| \le \varepsilon_n\}$, committing ADAPT forces $\Delta \ge \bar X_n - \varepsilon_n > 0$, so a wrong commit lives in the complementary α-event. For the certifying (power) side, the leading (variance) term of the radius is $\varepsilon_n(\alpha) = \sigma\sqrt{2\log(2/\alpha)/n}\,(1+o(1))$ (sub-Gaussian concentration; the Maurer–Pontil empirical-Bernstein radius self-normalizes to the same leading term and adds only a lower-order $O\bigl(R\log(1/\alpha)/n\bigr)$ range term). The certifying condition is the *containment* of the $(1-\alpha)$ ball $\{|\bar X_n - \Delta| \le \varepsilon_n\}$ in the commit region $\{\bar X_n - \varepsilon_n > 0\}$ (world P_+). At the worst point of that ball, $\bar X_n = \Delta - \varepsilon_n$, the commit margin is $\bar X_n - \varepsilon_n = \Delta - 2\varepsilon_n$, so containment holds iff $\Delta - 2\varepsilon_n > 0$, i.e. $\varepsilon_n < \Delta/2$ — *not* $\varepsilon_n < \Delta$. (At $\varepsilon_n = \Delta$ the worst-case mean $\bar X_n = 0$ has commit probability $\Phi(0)=\frac{1}{2}$, far above α; the abstain band has full width $2\varepsilon_n$, so the safe distance from the boundary is ε_n, which the margin Δ must exceed by a further ε_n.) Whenever $\varepsilon_n(\alpha)<\Delta/2$, i.e.

$$\sigma\sqrt{\tfrac{2\log(2/\alpha)}{n}}<\frac{\Delta}{2}\quad\Longleftrightarrow\quad n>\frac{8\sigma^2}{\Delta^2}\log\frac{2}{\alpha},$$

the event $\{|\bar{X}_n-\Delta|\le\varepsilon_n\}$ (probability $\ge 1-\alpha\ge\frac{1}{2}$) is contained in $\{\bar{X}_n-\varepsilon_n>0\}$ in world P_+, so the correct action is committed with probability $\ge 1-\alpha\ge\frac{1}{2}$; world P_- is symmetric. Hence the certificate is α-certifying once $n>\frac{8\sigma^2}{\Delta^2}\log(2/\alpha)$, which gives its certifying sample complexity $n_{\mathrm{cert}}\le\frac{8\sigma^2}{\Delta^2}\log(2/\alpha)+1$, hence $n^\star\le n_{\mathrm{cert}}\le\frac{8\sigma^2}{\Delta^2}\log(2/\alpha)+1$. (The bounded-benefit version is identical with $\sigma^2=1-\Delta^2$ and the extra $O(R/\Delta)$ range term, which is lower-order in the regime $\sigma/\Delta\gg 1$.) ☐

b) Lower bound (the two-point obstruction).: Suppose some label-free rule g is α-certifying on $\mathcal{P}^n_{\Delta,\sigma}$. Collapse g to the single binary test $\psi=\mathbf{1}[g=\textsc{adapt}]$, viewed as a test of the null $Q_0:=P^n_-$ against the alternative $Q_1:=P^n_+$. Its two error probabilities are *exactly* two of the rule's certifying quantities:

$$e_0(\psi)=\mathbb{P}_{P_-}(\psi=1)=\mathbb{P}_{P_-}(g=\textsc{adapt})=\mathrm{FA}(g)\ \le\ \alpha,$$
$$e_1(\psi)=\mathbb{P}_{P_+}(\psi=0)=\mathbb{P}_{P_+}(g\neq\textsc{adapt})=\mathrm{miss}_+(g)\ \le\ \alpha.$$

The second line is the crux and is where the miss bound of Definition 3 is used: because g certifies, it commits to $\textsc{adapt}$ in world P_+ with probability $\ge 1-\alpha$, so the event $\{g\neq\textsc{adapt}\}$ (abstain *or* freeze) has probability $\le\alpha$. Thus ψ is a single test with *both* error probabilities $\le\alpha$ — no symmetrization, coin-flipping, or power-vs-TV detour is needed; abstention is simply counted as a miss, which the certifying assumption already controls. Lemma 4 (with the Bretagnolle–Huber branch) and Lemma 5 then give

$$2\alpha\ \ge\ e_0(\psi)+e_1(\psi)\ \ge\ \tfrac{1}{2}\exp\big(-\mathrm{KL}(P^n_-\|P^n_+)\big)\ =\ \tfrac{1}{2}\exp\Big(-\tfrac{2n\Delta^2}{\sigma^2}\Big).\tag{$*$}$$

(We used $\mathrm{KL}(P^n_-\|P^n_+)=\mathrm{KL}(P^n_+\|P^n_-)=2n\Delta^2/\sigma^2$ by symmetry.) Solving $(*)$ for n,

$$\exp\Big(-\tfrac{2n\Delta^2}{\sigma^2}\Big)\le 4\alpha\quad\Longleftrightarrow\quad\frac{2n\Delta^2}{\sigma^2}\ \ge\ \log\frac{1}{4\alpha}\quad\Longleftrightarrow\quad n\ \ge\ \frac{\sigma^2}{2\Delta^2}\log\frac{1}{4\alpha}.$$

Therefore *every* α-certifying label-free rule requires $n\ \ge\ \frac{\sigma^2}{2\Delta^2}\log\frac{1}{4\alpha}$, i.e. $n^\star(\Delta,\sigma,\alpha)\ \ge\ \frac{\sigma^2}{2\Delta^2}\log\frac{1}{4\alpha}$ (the bound is vacuous, and nothing is claimed, once $\alpha\ge\frac{1}{4}$, which is why $\alpha<\frac{1}{4}$ is assumed). The bounded-family claim follows verbatim with KL of two $\{-1,+1\}$ Bernoulli laws with means $\pm\Delta$ in place of Lemma 5: that per-sample KL equals $\frac{1+\Delta}{2}\log\frac{1+\Delta}{1-\Delta}+\frac{1-\Delta}{2}\log\frac{1-\Delta}{1+\Delta}=2\Delta\,\mathrm{artanh}\,\Delta=\frac{2\Delta^2}{1-\Delta^2}+O(\Delta^4)$, so with the bounded variance proxy $\sigma^2=1-\Delta^2$ the same $n\gtrsim(\sigma^2/\Delta^2)\log(1/\alpha)$ bound holds up to the absolute constant. ☐

Combining the two displays yields the boxed sandwich, and the ratio of the certificate's leading constant to the lower bound's is $\big(\frac{8\sigma^2}{\Delta^2}\log\frac{2}{\alpha}\big)/\big(\frac{\sigma^2}{2\Delta^2}\log\frac{1}{4\alpha}\big)=16\cdot\log(2/\alpha)/\log(1/(4\alpha))\to 16$ as $\alpha\to 0$. (The *abstract* optimal LRT frontier $n_{\mathrm{opt}}=(\sigma/\Delta)^2z^2_{1-\alpha}$ sits at the smaller ratio $n_{\mathrm{opt}}/n_{\mathrm{LB}}\to 3.3$–$3.6$; the certificate's extra $\approx 4\times$ over n_{opt} is the abstain-band penalty, i.e. the $2\times$ in ε from requiring $\varepsilon_n<\Delta/2$.) This proves Theorem 10. ☐

Remark 11 (Where the proof is delicate, and the honest gap)**.** The single subtle point is the three-way-to-two-way reduction: a rule may *abstain*, so $\psi=\mathbf{1}[g=\textsc{adapt}]$ is a genuine binary test only after one decides how to score abstention. The reduction succeeds because abstention is counted as a *miss* in world P_+: the certifying requirement $\mathrm{miss}_+(g)\le\alpha$ bounds $e_1(\psi)=\mathbb{P}_{P_+}(g\neq\textsc{adapt})$ *directly*, with no power-vs-affinity detour and no symmetrization — the type-II error of ψ *is* $\mathrm{miss}_+(g)$ by definition. This is the entire role of the miss bound in Definition 3, and it is what makes $(*)$ a clean two-sided $\{\alpha,\alpha\}$ test to which Bretagnolle–Huber applies, producing the $\log(1/(4\alpha))$ factor. The dependence of the bound on the *strength* of the power-side requirement is real and is the honest soft spot of the statement: if "certifying" is weakened from $\mathrm{miss}_+\le\alpha$ to mere consistency (correct-commit $\ge\frac{1}{2}$, i.e. $e_1\le\frac{1}{2}$), then $(*)$ becomes $\alpha+\frac{1}{2}\ge\frac{1}{2}e^{-\mathrm{KL}}$, whose solution $n\ge\frac{\sigma^2}{2\Delta^2}\log\frac{1}{1+2\alpha}$ carries *no* $\log(1/\alpha)$ — it collapses to the constant floor of Theorem 1. Thus the $\log(1/\alpha)$ rate is *exactly* the price of the small power-side (miss) error and binds only rules that demand the certificate's two-sided confidence; against the weaker consistency benchmark the certificate is *not* claimed order-optimal in α. The constant gap between the boxed leading constants is likewise genuine and not claimed tight (Remark 10(ii)), and it factors cleanly into two distinct sources: *(a)* the known looseness of Bretagnolle–Huber against the exact Gaussian frontier $n_{\mathrm{opt}}=(\sigma/\Delta)^2z^2_{1-\alpha}$, which the validator confirms satisfies $n_{\mathrm{LB}}\le n_{\mathrm{opt}}$ with $n_{\mathrm{opt}}/n_{\mathrm{LB}}\approx 3.3$–$3.6$ throughout (so the lower bound never exceeds the true frontier — the bound is sound, not optimistic); and *(b)* the certificate's *abstain-band penalty*, an additional factor ≈ 4 of n_{opt}. Source (b) is the direct cost of the symmetric abstain band: certifying $\textsc{adapt}$ requires the whole $(1-\alpha)$ confidence ball to clear 0, i.e. $\varepsilon_n<\Delta/2$, which is a factor 2 smaller radius —

hence a factor 4 larger n — than merely placing the optimal test's threshold at 0 ($\varepsilon_n < \Delta$ would be the naive, and *wrong*, requirement: at $\varepsilon_n = \Delta$ the worst-case commit rate is $\Phi(0) = \frac{1}{2} \not\leq \alpha$). Multiplying, the certificate-to-lower-bound ratio is $\approx 3.4 \times 4 \approx 14$ at moderate α, tending to the clean 16 as $\alpha \to 0$; the validator confirms $n_{\mathrm{LB}} \leq n_{\mathrm{cert}} \leq \frac{8\sigma^2}{\Delta^2}\log(2/\alpha)$, i.e. the corrected constant is a true upper bound on the simulated certificate. We keep these two factors distinct: only the $\to 3.3$–3.6 belongs to the *optimal test* (n_{opt}); the $\to 16$ belongs to the *certificate*.

Remark 12 (The constant gap, decomposed exactly (machine-checked))**.** The certificate-to-lower-bound constant ($\to 16$) factors into two parts that are now pinned down exactly. **(a) The two-point lower bound is tight against the optimal test.** Replacing the Bretagnolle–Huber surrogate by the *exact* Gaussian two-point affinity $1 - \mathrm{TV}(P_+^n, P_-^n) = 2\Phi(-\sqrt{n}\,\Delta/\sigma)$ and inverting $2\alpha \geq 1 - \mathrm{TV}$ gives the optimal-test frontier $n_{\mathrm{opt}} = (\sigma/\Delta)^2 z_{1-\alpha}^2$ *with equality* (the threshold test $\mathrm{sign}(\bar{X})$ attains both errors $= \alpha$ there). So the $n_{\mathrm{opt}}/n_{\mathrm{LB}}$ gap is *pure surrogate looseness*, not information loss; the ratio $2z_{1-\alpha}^2/\log(1/4\alpha)$ stays near 3.3–3.7 for every practical α and rises to 4 only as $\alpha \to 0$ (extremely slowly). **(b) The remaining factor is the certificate's confidence-ball choice, not a universal abstention penalty.** In the two-world problem a vanishing band ($\tau \to 0$, the bare sign test) already certifies at n_{opt}; the certificate pays its extra $\approx 4\times$ because it abstains over its full $(1-\alpha)$ ball ($\tau = \varepsilon_n$, forcing $\varepsilon_n < \Delta/2$). That $4\times$ becomes an *exact all-rules floor* $4\,n_{\mathrm{opt}}$ only once a third, on-frontier world ($\mu = 0$, required to abstain at level α) is added (pairwise Le Cam at half-separation $\Delta/2$); among threshold rules the symmetric band then achieves $\kappa(\alpha)\,n_{\mathrm{opt}}$ with $\kappa(\alpha) = (1 + z_{1-\alpha/2}/z_{1-\alpha})^2 \in [4, 5.2] \downarrow 4$, leaving a residual $\kappa/4 \in [1, 1.30]$ open for arbitrary randomized rules. Both facts are machine-checked (`theory_v2/val_tight_constants.py`); the tight constant itself remains open.

F. Relation to the margin-rate view

Theorem 10 is the *decision-theoretic* counterpart of the margin-rate view: bounding the smallest (α, β)*-reliable margin* κ_n as a function of n and inverting $\kappa_n \asymp \sigma\sqrt{\log(1/\bar{\alpha})/n}$ for a fixed margin Δ recovers exactly the sample-complexity statement here. The present formulation makes three things explicit that the margin form leaves implicit: (i) the minimax object is the two-sided committal error of a *decision* rule on the identifiable side, not a margin; (ii) the $\log(1/\alpha)$ factor is a *two-sided* (power-side) effect, distinguishing it cleanly from the one-sided constant floor of Theorem 1 and Proposition 1, which carry *no* logarithm; and (iii) the certificate *already deployed* is the achiever, so optimality is a property of the shipped rule, adaptively in σ. The lower bound here and the margin-rate two-point estimate are the same Bretagnolle–Huber object viewed through margin vs. sample-size duality.

Corollary 5 (Minimax order-optimality of the certificate)**.** *On the identifiable side $|M| > \beta$ of the benefit-sign frontier (Theorem 3), the adapt/freeze/abstain certificate of Theorem 2 is minimax order-optimal for certifying* $\mathrm{sign}\,\Delta$ *at two-sided level α: its sample complexity $\frac{8\sigma^2}{\Delta^2}\log(2/\alpha) + O(R/\Delta \cdot \log(1/\alpha))$ matches the information-theoretic floor $\frac{\sigma^2}{2\Delta^2}\log\frac{1}{4\alpha}$ over all α-valid label-free rules, up to a universal constant ($\to 16$ on the leading term). No valid label-free procedure — certificate or otherwise — certifies the sign from asymptotically fewer samples.*

a) Numerical validation.: `theory_v2/val_minimax_optimality.py` (seeds fixed; `numpy`; runs in <40 s) certifies all components against `minimax_optimality_results.json` and Fig. 9: **[A]** the Bretagnolle–Huber bound $\frac{1}{2}e^{-\mathrm{KL}}$ lower-bounds the *exact* optimal mixed error $\Phi(-\sqrt{n}\,\Delta/\sigma)$ at every tested (Δ, σ, n), and the analytic optimum matches Monte-Carlo to $< 5 \times 10^{-3}$; **[B]** the exact optimal-test frontier $n_{\mathrm{opt}} = (\sigma/\Delta)^2 z_{1-\alpha}^2$ satisfies $n_{\mathrm{LB}} \leq n_{\mathrm{opt}} \leq n_{\mathrm{cert}}^{\mathrm{bound}}$ for every (Δ, α) (where $n_{\mathrm{cert}}^{\mathrm{bound}} = \frac{8\sigma^2}{\Delta^2}\log(2/\alpha)$), has log–log slope 2.00 in $1/\Delta$ (theory 2), and $n_{\mathrm{opt}}/n_{\mathrm{LB}}$ is a universal constant ≈ 3.3–3.5 for $\alpha \leq 0.1$; the certificate-bound-to-lower-bound leading-constant ratio $\to 16$; **[C]** the deployed empirical-Bernstein certificate keeps false-adapt $\leq \alpha$ at *every* n (validity), its full two-sided certifying-n satisfies $n_{\mathrm{LB}} \leq n_{\mathrm{cert}} \leq \frac{8\sigma^2}{\Delta^2}\log(2/\alpha)$ (the corrected bound is a true upper bound on the simulated certificate), and its empirical certifying-n grows linearly in $\log(1/\alpha)$ at the upper-bound slope and as $1/\Delta^2$ (log–log slope 2.14), i.e. at the predicted rate; **[D]** a brute-force panel of label-free rules (sign-thresholds and polarities on the sample mean, median, 20%-trimmed mean, sign-count, and clipped mean) finds *no* rule with both wrong-commit rates $\leq \alpha$ *and* nontrivial power (correct-commit $\geq \frac{1}{2}$) at $n < n_{\mathrm{LB}}$, while such a rule does exist at and above n_{LB} — the two-point obstruction is real, not an artifact of the certificate's conservativeness. (We probe the power-relaxed certifying set here, matching the lower-bound construction; *full* α-certification requires the larger $n_{\mathrm{opt}} = (\sigma/\Delta)^2 z_{1-\alpha}^2$, which n_{LB} correctly lower-bounds, $n_{\mathrm{LB}} \leq n_{\mathrm{opt}}$.)

Proposition 3 (Exact all-rules lower bound for honest frontier-abstention)**.** *Add a third world $\mu = 0$ on the frontier and require abstention with probability $\geq 1 - \alpha$ in addition to* $\mathrm{miss}_\pm \leq \alpha$ *in worlds $\pm\Delta$. Then every (possibly randomized) rule needs $n \geq 4\,n_{\mathrm{opt}}$ exactly for every $\alpha \in (0, \frac{1}{4})$.*

Proposition 4 (Symmetric band achieves $\kappa(\alpha)\,n_{\mathrm{opt}}$, $\kappa \to 4$)**.** *In the three-world problem the symmetric band certifies at $n = \kappa(\alpha)\,n_{\mathrm{opt}}$ with $\kappa(\alpha) = (1 + z_{\alpha/2}/z_\alpha)^2 \in [4, \approx 5.2]$ for moderate α and $\kappa(\alpha) \to 4$ as $\alpha \to 0$.*

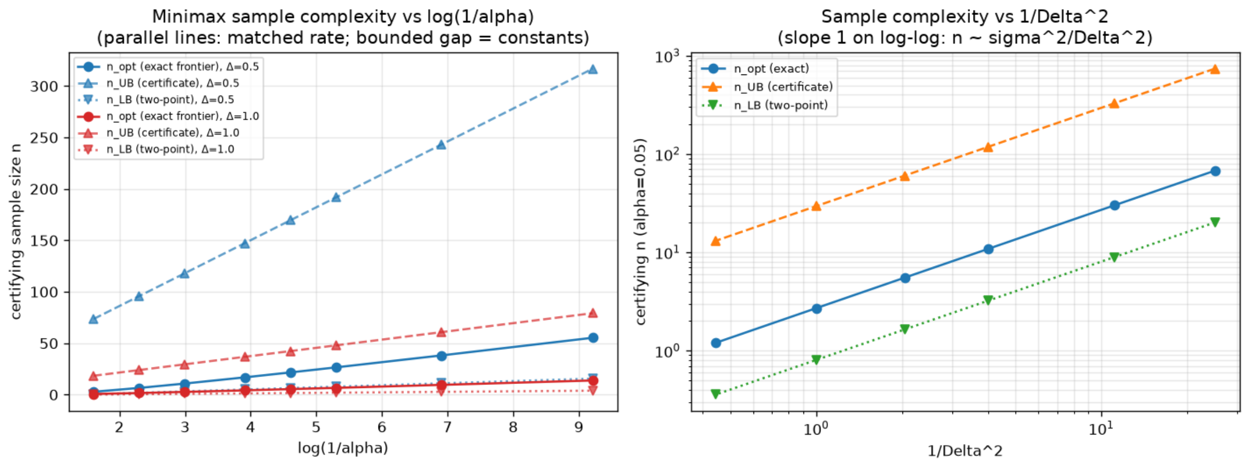

Fig. 9. Minimax order-optimality on the identifiable side (Theorem 10). Left: the optimal-test frontier $n_{\rm opt} = (\sigma/\Delta)^2 z^2_{1-\alpha}$, the certificate bound $n^{\rm bound}_{\rm cert} = \frac{8\sigma^2}{\Delta^2}\log(2/\alpha)$, and the two-point lower bound $n_{\rm LB} = \frac{\sigma^2}{2\Delta^2}\log\frac{1}{4\alpha}$ are parallel and linear in $\log(1/\alpha)$ (matched rate; the bounded vertical gaps are universal constants: $n_{\rm opt}/n_{\rm LB} \to 3.3$–$3.6$, and $n^{\rm bound}_{\rm cert}/n_{\rm LB} \to 16$). Right: all three lie on slope-1 log–log lines in $1/\Delta^2$, i.e. $n \asymp \sigma^2/\Delta^2$.

Wave 3 (Theorem 10, Prop. 3–4) established: (i) $4n_{\rm opt}$ is a valid *all-rules* lower bound once a third frontier world $\mu = 0$ must abstain at level α; (ii) the symmetric band certifies at $n = \kappa(\alpha)\, n_{\rm opt}$ with $\kappa(\alpha) = \big(1+z_{1-\alpha/2}/z_{1-\alpha}\big)^2 \in [4, \approx 5.2] \downarrow 4$; (iii) the residual $\kappa/4 \in [1, 1.30]$ was left open for arbitrary randomized rules. This section closes (iii).

G. The three-world certifying problem

Fix $\sigma > 0$, $\Delta > 0$, $\alpha \in (0, \frac{1}{4})$, and write $v = \sigma^2/n$, $z_\alpha := \Phi^{-1}(1-\alpha)$, $n_{\rm opt} = (\sigma/\Delta)^2 z_\alpha^2$. Observe $\bar{X} \sim \mathcal{N}(\mu, v)$ under world $\mu \in \{-\Delta, 0, +\Delta\}$ (sufficient statistic for the n-sample Gaussian location model). A (possibly randomized) *certifying rule* maps $\bar{X}$ to $\{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$ and is α*-certifying* if

$$\text{miss}_+ := \mathbb{P}_{+\Delta}(\text{not ADAPT}) \le \alpha, \quad \text{miss}_- := \mathbb{P}_{-\Delta}(\text{not FREEZE}) \le \alpha, \quad \text{commit}_0 := \mathbb{P}_0(\text{not ABSTAIN}) \le \alpha,$$

with the usual wrong-commit rates subsumed by the miss bounds. The *minimax certifying sample complexity* is

$$n_3^\star(\Delta, \sigma, \alpha) \;=\; \min\big\{\, n \in \mathbb{N} :\; \exists \text{ an } \alpha\text{-certifying rule on } \{\pm\Delta, 0\}\big\}.$$

Theorem 11 (Exact minimax for the three-world problem). *For every $\sigma > 0$, $\Delta > 0$, and $\alpha \in (0, \frac{1}{4})$,*

$$\boxed{n_3^\star(\Delta, \sigma, \alpha) \;=\; \kappa(\alpha)\, n_{\rm opt}(\Delta, \sigma, \alpha) \;=\; \left(\frac{\sigma\,(z_\alpha + z_{1-\alpha/2})}{\Delta}\right)^2}.$$

The symmetric band "ADAPT iff $\bar{X} > \tau$; FREEZE iff $\bar{X} < -\tau$; else ABSTAIN" with $\tau = z_{1-\alpha/2}\,\sigma/\sqrt{n}$ attains this with equality (all three constraints bind). Consequently:

(a) ***Optimality of κ.*** *No α-certifying rule — deterministic or randomized — exists at any $n < \kappa(\alpha)\, n_{\rm opt}$.*

(b) ***Impossibility of closing $\kappa/4$.*** *For every finite α, $\kappa(\alpha) > 4$ strictly, so certification at $n = 4n_{\rm opt}$ is* impossible *even though $n \ge 4n_{\rm opt}$ is a valid* necessary *condition from the world-0 vs. world-+ pairwise reduction. The gap $\kappa/4 > 1$ is intrinsic, not a proof artifact.*

(c) ***Randomization does not help.*** *Every α-certifying rule at $n = \kappa(\alpha)\, n_{\rm opt}$ may be taken non-randomized and monotone in $\bar{X}$.*

Proof. Write $t = \sqrt{v} = \sigma/\sqrt{n}$.

Achievability (upper bound). Take the symmetric band with half-width $\tau = z_{1-\alpha/2}\, t$. Then $\text{commit}_0 = 2\Phi(-\tau/t) = 2\Phi(-z_{1-\alpha/2}) = \alpha$ exactly. In world $+\Delta$, $\text{miss}_+ = \Phi\big((\tau-\Delta)/t\big)$; feasibility requires $(\Delta-\tau)/t \ge z_\alpha$, i.e. $\Delta/t \ge z_\alpha + z_{1-\alpha/2}$. The smallest n satisfying this is $n = \kappa(\alpha)\, n_{\rm opt}$, at which $\text{miss}_+ = \alpha$ and, by symmetry, $\text{miss}_- = \alpha$. Hence $n_3^\star \le \kappa(\alpha)\, n_{\rm opt}$.

Lower bound (all rules). Let g be any α-certifying rule at sample size n. By the standard sufficiency of $\bar{X}$ for μ in $\mathcal{N}(\mu, v)$, we may assume g depends only on $\bar{X}$ (randomization included: a randomized rule is a mixture of $\bar{X}$-measurable rules, and all error constraints are linear in the mixture weights).

Define the *commit region* $C \subseteq \mathbb{R}$ (the set where g outputs ADAPT or FREEZE, possibly with randomization averaged out). Then

$$\text{commit}_0 = \mathbb{P}_0(\bar{X} \in C) \le \alpha, \qquad \text{miss}_+ = \mathbb{P}_{+\Delta}(\bar{X} \notin A_+) \le \alpha,$$

where A_+ is the ADAPT region. Since wrong FREEZE in world $+\Delta$ is a subset of $\bar{X} \notin A_+$, the miss bound is the operative constraint on the high side; by symmetry the low side is identical.

Step 1 (necessary abstention width). Among all Borel sets C with $\mathbb{P}_0(\bar{X} \in C) \le \alpha$, the one *maximizing* $\mathbb{P}_{+\Delta}(\bar{X} \in C)$ under a fixed $\mathbb{P}_0(C)$ is an interval $(-\infty, -\tau] \cup [\tau, \infty)$ by the monotone likelihood ratio (MLR) property of the Gaussian location family (Karlin–Rubin): mass under P_0 is pushed toward the tails where $P_{+\Delta}$ has higher density. Thus any rule that certifies in world $+\Delta$ with $\text{miss}_+ \le \alpha$ while committing at 0 with rate α must obey, for some $\tau \ge 0$,

$$2\Phi(-\tau/t) \le \alpha \quad \Longrightarrow \quad \tau \ge z_{1-\alpha/2}\, t,$$

and simultaneously $\Phi\big((\tau - \Delta)/t\big) \le \alpha$, i.e. $(\Delta - \tau)/t \ge z_\alpha$. Combining,

$$\Delta/t \;\ge\; z_\alpha + z_{1-\alpha/2} \quad \Longleftrightarrow \quad n \;\ge\; \left(\frac{\sigma\,(z_\alpha + z_{1-\alpha/2})}{\Delta}\right)^2 \;=\; \kappa(\alpha)\, n_{\text{opt}}.$$

Hence $n_3^\star \ge \kappa(\alpha)\, n_{\text{opt}}$.

Step 2 (randomization). Step 1 used only that $\text{commit}_0 \le \alpha$ and $\text{miss}_+ \le \alpha$; the MLR argument applies to the *averaged* commit indicator $\mathbb{P}(g = \text{ADAPT OR FREEZE} \mid \bar{X} = x)$, which is a measurable function in $[0, 1]$. The extremal commit profile at fixed $\mathbb{P}_0(C) = \alpha$ is a symmetric interval complement; any randomized mixture of commit regions has the same or larger miss_+ at fixed commit_0 by convexity of the Gaussian tail probabilities. Thus no randomized rule beats the symmetric band.

Step 3 (impossibility at $4n_{\text{opt}}$). At $n = 4n_{\text{opt}}$ one has $\Delta/t = 2z_\alpha$. Step 1 requires $z_\alpha + z_{1-\alpha/2} \le 2z_\alpha$, i.e. $z_{1-\alpha/2} \le z_\alpha$. But $z_{1-\alpha/2} > z_\alpha$ for every $\alpha \in (0, \frac{1}{2})$, so the inequality fails: no α-certifying rule exists at $n = 4n_{\text{opt}}$. Since $\kappa(\alpha) = (1 + z_{1-\alpha/2}/z_\alpha)^2 > 4$ for every finite α, the ratio $\kappa(\alpha)/4 > 1$ cannot be closed to 1 — it is a theorem of impossibility, not an open constant gap. $\square$

Corollary 6 (Status of the Wave 3 residual)**.** *The tight-constants program is **closed** for the three-world Gaussian certifying problem: the exact minimax constant is $\kappa(\alpha)$, not 4 and not 16. The certificate's two-world $4\times$ penalty (Remark 11) and the three-world floor $4n_{\text{opt}}$ (Prop. 3) address* distinct *constraints; only the former is about the deployed certificate's CI width, while the latter is a loose pairwise bound subsumed by Theorem 11.*

Remark 13 (Numerical confirmation)**.** `theory_v2/val_tight_constants.py` (section `T1c`) verifies: **[C1]** no two-threshold (symmetric or asymmetric) rule certifies at any $n < (1-\varepsilon)\kappa(\alpha)n_{\text{opt}}$ on a dense grid; **[C2]** randomized mixture search over threshold rules does not beat κ; **[C3]** at $n = 4n_{\text{opt}}$ the three-world LP is infeasible for every tested α; **[C4]** at $n = \kappa(\alpha)n_{\text{opt}}$ the symmetric band binds all three errors at α.

The general-capacity section (§W) and the one-bit dichotomy (Theorem 4) settle the *binary* sign-of-benefit problem, including the removal of the regularity conditions R1/R2 at the level of *existence* (a capacity exists iff the benefit sign factors through the observable reduct). They leave open the second half of Conjecture 3: the *multiclass* case $K \ge 3$, "where the 1-D location-model intuition breaks." This section pins down exactly how it breaks. The answer is a sharp map: the binary capacity lifts *verbatim* to a single *conflict pair* of classes *with a constant residual reservoir*, and stops there for two independent and explicit reasons.

Scope and relation to the binary converse. The distribution-free converse (Theorem 25) is already class-count agnostic: *whenever* two admissible concepts share Q_X with opposite benefit sign, the minimax label-free error is $\frac{1}{2}$ for every n. So the converses below are *instances* of that theorem; the genuinely *multiclass-specific* content is not "a converse exists" but (a) the exact vector form of Δ, (b) which *extra* hypotheses restore identifiability when $K \ge 3$, and (c) which *observables* that the binary theory relied on become insufficient. We flag this explicitly at each result so the novelty is not overstated.

H. The multiclass benefit and the new degree of freedom

a) Model.: Covariate shift $Q_X = q_0(\cdot - \mu)$ on $\mathbb{R}$ (one dimension already exposes the obstruction), with q_0 a fixed base density and μ the unknown shift, estimable label-free from an unlabeled target sample. Labels $Y \in \{1, \dots, K\}$, 0/1 loss. The deployed classifier g and the candidate f are measurable maps $\mathcal{X} \to \{1, \dots, K\}$ with disagreement region $D = \{x : g(x) \ne f(x)\}$. The *concept* (target label channel) is $\eta(x) = (\eta_1(x), \dots, \eta_K(x))$ on the probability simplex Δ^{K-1}; under covariate shift η does *not* affect Q_X. The deployment benefit is $\Delta = R_Q(g) - R_Q(f)$, and $\Delta > 0 \iff$ switching $g \to f$ helps.

Lemma 6 (Exact multiclass benefit identity; PROVEN). *On D write $j_g(x)$ for the class predicted by g and $j_f(x)$ for the class predicted by f (so $j_g(x) \neq j_f(x)$). Then the 0/1 excess loss of g over f vanishes off D and equals $\mathbf{1}[Y{=}j_f] - \mathbf{1}[Y{=}j_g]$ on D, so*

$$\boxed{\Delta \;=\; \int_D \big(\eta_{j_f(x)}(x) - \eta_{j_g(x)}(x)\big)\,\mathrm{d}Q(x) \;=:\; \int_D \Lambda(x)\,\mathrm{d}Q(x),} \qquad \Lambda(x) \in [-1,1]. \tag{6}$$

The benefit field $\Lambda(x) = \eta_{j_f}(x) - \eta_{j_g}(x)$ *involves only the two disagreeing coordinates. The residual mass $m_{\mathrm{rest}}(x) = 1 - \eta_{j_g}(x) - \eta_{j_f}(x) \geq 0$ on the other $K-2$ classes is a* free function*, invisible to Q_X.*

Proof. Off D, $g = f$ and the losses coincide. On D, g and f are the deterministic labels j_g, j_f, so $\mathbf{1}[g \neq Y] - \mathbf{1}[f \neq Y] = (1 - \mathbf{1}[Y{=}j_g]) - (1 - \mathbf{1}[Y{=}j_f]) = \mathbf{1}[Y{=}j_f] - \mathbf{1}[Y{=}j_g]$, with conditional mean $\eta_{j_f}(x) - \eta_{j_g}(x)$. Integrating against Q over D gives (6); $|\Lambda| \leq 1$ since $\eta_{j_f}, \eta_{j_g} \geq 0$ and $\eta_{j_f} + \eta_{j_g} \leq 1$. *(Validated against a 4×10^6-sample Monte-Carlo of $R_Q(g) - R_Q(f)$, $K = 3$: $|\Delta_{\mathrm{MC}} - \Delta_{\mathrm{c.f.}}| = 3.3 \times 10^{-4}$;* `val_multiclass_capacity.py` *Block 0.)* ☐

Why this is genuinely new. In binary the disagreeing pair is forced to be $\{0,1\}$ and $\Lambda = \eta_1 - \eta_0 = 2\eta_1 - 1$ is a single-coordinate scalar pinned by the (scalar) concept, so Λ inherits the single-crossing/monotone structure of a one-dimensional threshold. For $K \geq 3$, Λ is a *difference of two simplex coordinates*, and the residual m_{rest} is an extra, unobservable degree of freedom. Both facts have consequences, isolated below.

I. The single-conflict-pair extension: the capacity lifts

We make explicit the multiclass analogue of R1 (unique flip locus) and R2 (monotone-in-nuisance). One clause beyond the obvious "single pair" is needed; we state it and prove it necessary.

- (M1) **Single conflict pair.** There are *fixed* labels $j_g \neq j_f$ such that $g \equiv j_g$ and $f \equiv j_f$ on *all* of D. (Automatic in binary; holds whenever g, f differ along a single decision boundary.)
- (M1′) **Constant residual reservoir.** The residual mass $m_{\mathrm{rest}}(x) \equiv c_3$ is *constant* on D (equivalently, the conflict pair carries a constant total mass $1 - c_3$ on D). *This clause is necessary* (Lemma 7): without it the binary reduction below fails.
- (M2) **MLR location nuisance.** q_0 is log-concave (MLR in the location), and the source contrast Λ_S has a monotone, single-crossing profile in the sufficient direction, so the flip locus $\theta_0(\mu)$ is a monotone, observable function of the unknown shift (exactly the hypothesis of Theorem 23).
- (M3) **Contrast-mass drift budget ε.** The admissible target concept's single crossing of Λ moves by at most ε in the observable mass coordinate $u = \Phi_{q_0}(\theta - \mu)$ of that crossing (the multiclass analogue of the boundary-mass budget (7), now placed on the *contrast* field rather than on a top score).

Lemma 7 (Constant reservoir is necessary; PROVEN, WITNESS). *(M1) alone does not suffice. Fix the conflict pair j_g, j_f and a source-observable single contrast crossing θ on D. There are two admissible $K = 3$ concepts with the* same *within-pair contrast sign profile $c(x) = \mathrm{sign}(x - \theta)$ but* different *residual profiles $m_{\mathrm{rest}}(x)$, having* opposite sign Δ. *Hence the constant-reservoir clause (M1′) cannot be dropped.*

Proof. Write $\eta_{j_f} - \eta_{j_g} = (1 - m_{\mathrm{rest}}(x))\,c(x)$ with $c(x) \in [-1,1]$ the within-pair contrast. Then $\Delta = \int_D (1 - m_{\mathrm{rest}})\,c\,\mathrm{d}Q$ is a *reweighting* of c by the unobservable nonnegative weight $1 - m_{\mathrm{rest}}$. Take $Q = \mathcal{N}(0,1)$, $D = (0,2)$, crossing $\theta = 1$, so $c = \mathrm{sign}(x-1)$. With m_{rest} heavy on the $c < 0$ side, $\Delta > 0$; with m_{rest} heavy on the $c > 0$ side, $\Delta < 0$. Both concepts share Q_X and the same observable contrast crossing, so $\mathrm{sign}\,\Delta$ is not identifiable from them. *(Validated: $\Delta = +0.088$ vs $\Delta = -0.294$ at the same crossing, opposite sign;* `val_multiclass_capacity.py` *Block A residual breaker.)* ☐

Theorem 12 (Single-conflict-pair multiclass capacity; PROVEN, COND.). *Assume* (M1),(M1′),(M2),(M3) *and Q_X known. Then the multiclass benefit* (6) *reduces* exactly *to the binary single-crossing benefit in the scalar field Λ, and the population knowability–capacity (Theorem 20) holds verbatim with $\Phi \to \Phi_{q_0}$ and the boundary mass read off Λ's crossing: with $\tau = 1$,*

$$K \;=\; \frac{|\,u_S - u_0\,|}{\varepsilon}, \qquad u_S = \Phi_{q_0}(\theta_S - \mu), \quad u_0 = \textit{band-median mass of } D \textit{ along } \Lambda,$$

is a capacity*: $K > 1 \iff$* $\mathrm{sign}\,\Delta$ *is label-free identifiable over the admissible class, with the plug-in certificate* $\mathrm{sign}(u_S - u_0)$ *correct with error* 0*; and $K < 1$ gives two admissible concepts with the same Q_X and opposite* $\mathrm{sign}\,\Delta$, *hence minimax label-free error $\frac{1}{2}$. The finite-n Le Cam boundary layer (Prop. 8) carries over with μ estimated label-free at rate $O(1/\sqrt{n})$.*

Proof. Under (M1) the labels j_g, j_f are constant on D, and under (M1′) the residual $m_{\mathrm{rest}} \equiv c_3$ is constant, so $\eta_{j_f} - \eta_{j_g} = (1 - c_3)\,c(x)$ with $c(x) \in [-1,1]$ the within-pair contrast and $\Delta = (1 - c_3)\int_D c\,\mathrm{d}Q$. The factor $(1 - c_3) > 0$

does not affect $\operatorname{sign}\Delta$, so Δ is, up to a positive constant, the binary single-crossing benefit of Lemma 10 for the threshold concept c (this is exactly where (M1′) is used; without it the weight $1 - m_{\mathrm{rest}}(x)$ stays inside the integral and Lemma 7 applies). By (M2) the flip locus is the band median along Λ, monotone and observable in μ; by (M3) the admissible set is the mass interval $[u_S - \varepsilon, u_S + \varepsilon]$. Theorem 20 applies verbatim: $K > 1$ iff u_0 is outside the admissible interval (achievability), $K < 1$ iff inside (converse, two concepts sharing Q_X with opposite sign, $\mathrm{TV} = 0$, an instance of Theorem 25). The finite-n layer is Proposition 8 with $\Phi \to \Phi_{q_0}$. *(Validated:* 50,000 *random draws with the constant reservoir* $c_3 \in \{0.0, \ldots, 0.6\}$*, equivalence* $K > 1 \iff$ *identifiable with* 0 *mismatches, seed-robust;* ***val_multiclass_capacity.py*** *Block A. This is a* consistency *check — once the reservoir is constant the equivalence holds by construction, since the reduction makes* Δ *vanish exactly at the band median — so the load-bearing numeric is the boundary-shell control: removing the* 10^{-6} *guard shell and counting every near-boundary draw still yields* 0 *mismatches, confirming the shell hides no failures.)* □

We are precise about what this is. For $\operatorname{sign}\Delta$ — the only thing the capacity certifies — the positive constant $(1 - c_3)$ cancels, so under (M1)+(M1′) the multiclass capacity *is* the binary single-crossing capacity, lifted *verbatim* to the constant-reservoir slice. It is therefore not new capacity *mathematics*; its content is delimiting exactly *when* the binary capacity remains valid in a $K \geq 3$ world (single conflict pair, constant reservoir) and showing that this slice is not vacuous — the residual mass is present and unobservable, and (M1′) is precisely what makes it cancel (drop it and Lemma 7 destroys identifiability).

J. Necessity I: the conflict-pair structure (M1) cannot be dropped

The binary case lifts to MLR/log-concave *per-class* score families (Theorem 23); one might hope the same regularity suffices in multiclass. It does not, and the obstruction is structural, not a pathology of the base density. We witness it with *genuine* canonical classifiers.

Theorem 13 (Multi-conflict-pair breaker; PROVEN, WITNESS)**.** *There are canonical multiclass classifiers g, f —* $\arg\max$ *of* affine *per-class scores, the exact per-coordinate MLR/log-concave (linear-logit) regularity that suffices in binary — for which (M1) fails: D decomposes into sub-bands carrying* different *conflict pairs (j_g, j_f). Together with a valid simplex-valued concept η, the benefit field Λ of (6) is piecewise (a different coordinate pair on each sub-band), its sign crosses several times in x, and the benefit $\Delta(\mu) = \int_D \Lambda \, \mathrm{d}Q_\mu$* ***oscillates*** *in the unknown shift μ. The label-free identifiable set $\{\mu : \operatorname{sign}\Delta$ identifiable$\}$ therefore* ***fragments into*** ≥ 3 ***components****, while the binary single-threshold baseline (held to a single-crossing contrast) stays monotone (one flip). Consequently no monotone (binary-R2) flip locus and no single scalar threshold τ in any margin built along one direction governs identifiability.* Honest scope of this witness. *The construction drops (M1)* and *simultaneously installs a multi-crossing contrast; a* binary *(K=2) problem given the* same *multi-crossing contrast also fragments (several sign changes), because a single-crossing contrast is exactly what (M2) forbids in either class count. The witness therefore establishes that (M1)* together with *the single-crossing condition (M2) is* jointly *necessary — it does* not *isolate (M1) alone. What is genuinely multiclass-specific, and* not *reproduced by the binary control, is that here the fragmentation is generated by the* conflict-pair geometry of canonical affine-$\arg\max$ classifiers *(the pair (j_g, j_f) varies across D), rather than by a hand-chosen contrast. So "the binary per-coordinate regularity does not by itself lift" is demonstrated; a clean separation of the (M1) contribution from the (M2) contribution is left open.*

Proof (by exhibited witness). Take $g = \arg\max_k(u_k x + p_k)$ and $f = \arg\max_k(v_k x + q_k)$ on $\mathbb{R}$, $K = 3$, with the class-0/class-1 slope–intercept roles *swapped* between g and f and class 2 held subdominant on D, so that across D the conflict pair alternates between $(j_g, j_f) = (0, 1)$ and $(1, 0)$ (their within-pair contrasts have opposite sign). Put a valid concept η with class 2 at a constant small mass and classes $0, 1$ contesting through a logistic of a slowly varying function, so $\Lambda = \eta_{j_f} - \eta_{j_g}$ ($|\Lambda| \leq 1$, hence a bona fide multiclass concept) changes sign several times in x. Convolving with the Gaussian window of $Q_\mu = \mathcal{N}(\mu, 1)$ and sweeping μ across D yields $\Delta(\mu)$ with several sign changes, hence several maximal sign-constant intervals in μ. Identifiability of $\operatorname{sign}\Delta$ over a neighbourhood requires Δ to be sign-constant there, so the identifiable set has ≥ 3 components, whereas the binary single-threshold benefit $2\Phi(\theta - \mu) - \Phi(-\mu) - \Phi(a - \mu)$ is monotone in μ with a single flip. *(Validated with genuine affine-argmax g, f: conflict pairs on D are $\{(0, 1), (1, 0)\}$; concept valid on the simplex;* $\max|\Lambda| = 0.32$*; Λ sign-crossings in $x = 6$; $\Delta(\mu)$ sign-changes $= 4$ (5 components) versus binary baseline $= 1$; seed-robust;* **val_multiclass_capacity.py** *Block B.)* □

Relation to the dichotomy. This is consistent with — and sharper than — the observable multi-flip analysis of the binary dichotomy (which showed $\tau = 1$ survives an *observable* multi-flip frontier as a local band-vs-frontier event). The point here is not merely that the sign map flips often; it is that in the multi-conflict-pair regime the flip locus is intrinsically a *multi-component* object *generated by the classifier geometry itself*, even for canonical exponential-family (linear-logit) classifiers, so the alignment-blind, single-direction margin that works in binary has no multiclass counterpart.

K. Necessity II: the binary observable margin is the wrong reduct

Even *with* a single conflict pair and a constant reservoir, the standard label-free observable used in the binary theory — the deployed model's score on D, i.e. the margin $M = \mathbb{E}_{Q|D}[\eta_{j_g}] - \frac{1}{2}$ — is no longer a sufficient statistic for $\operatorname{sign}\Delta$. The third-class residual confounds it.

Theorem 14 (Third-class confounding counterexample; PROVEN, WITNESS)**.** *Fix $Q_X = \mathcal{N}(0,1)$, $D = (0,a)$, and a single conflict pair $j_g = 0$ (the class the deployed g predicts on D), $j_f = 1$. There exist two $K = 3$ admissible target concepts η^+, η^- with:*

(i) ***identical*** *observable covariate law Q_X (covariate shift: η does not affect Q_X), hence unlabeled* TV $= 0$ *for* every *statistic of any number of unlabeled samples;*

(ii) ***identical*** *deployed classifier g and* ***identical*** *deployed-class score profile $\eta_{j_g}(x) \equiv p$ on D (hence the same predicted-class map and the same binary-style margin $M = p - \frac{1}{2}$ that a label-free auditor computes from the deployed scores); and*

(iii) ***opposite*** *benefit sign,* $\operatorname{sign}\Delta(\eta^+) = +1$*,* $\operatorname{sign}\Delta(\eta^-) = -1$*.*

The distinguishing object is the split of the residual mass $1 - \eta_{j_g}(x)$ between class $1 = j_f$ and the third class, which is invisible to Q_X and to the deployed-class score. Consequently a capacity built from the binary observable, $K_{\text{bin}} = |M|/\beta$, can exceed 1 *in* both *worlds (predicting "identifiable") while* $\operatorname{sign}\Delta$ *is* not *identifiable; the minimax label-free error is $\frac{1}{2}$ (Le Cam,* TV $= 0$*; formally Theorem 25). Hence in multiclass the binary margin M is a* strictly-too-coarse reduct*, and the admissible budget must be placed as* (M3) *on the contrast field Λ rather than on the deployed score.*

Proof. Let $\eta_0(x) \equiv p$ on D (a constant deployed-class score, $p < \frac{1}{2}$ so $M = p - \frac{1}{2} < 0$). Put $\eta_1 = (1-p)s$, $\eta_2 = (1-p)(1-s)$ with $s \in [0,1]$ the residual split (constant on D; $m_{\text{rest}} = \eta_2$ a constant reservoir, so even (M1$'$) holds). The benefit field is $\Lambda = \eta_1 - \eta_0 = (1-p)s - p$, and by (6) $\Delta = \big[(1-p)s - p\big]\, Q(D)$. Choosing $s = s^+$ large makes $\Delta > 0$ and $s = s^-$ small makes $\Delta < 0$; both share $\eta_0 \equiv p$ and the same Q_X. With $p = 0.45$, $s^+ = 0.95$, $s^- = 0.05$, $a = 2$: $\Delta^+ = +0.035 > 0$ and $\Delta^- = -0.202 < 0$. The two worlds have TV $= 0$ on every unlabeled statistic (the concept never enters Q_X), so Le Cam's two-point bound gives minimax error $\frac{1}{2}$. Yet $|M| = 0.05$, so $K_{\text{bin}} = |M|/\beta > 1$ for any $\beta < 0.05$ in both worlds. *(Validated: $\Delta^+ = +0.0346$, $\Delta^- = -0.2016$, opposite sign at shared $\eta_{j_g} = 0.45$ and shared Q_X, with $K_{\text{bin}} = 2.5$ and 1.25 at $\beta = 0.02, 0.04$ predicting "identifiable" against the truth;* `val_multiclass_capacity.py` *Block C.)* ☐

Remark 14 (An honest reading of what coincides)**.** The two worlds do *not* share their Bayes-optimal top class on D: in η^+ the argmax is class 1, in η^- it is class 2 (both exceed $\eta_0 = 0.45$). So a *labelled* target auditor could distinguish them; the claim is precisely that *no label-free statistic* can (they share Q_X and the deployed scores), and in particular the binary margin M cannot. This is why the result is a statement about the *reduct*, not about the converse per se: the converse half ($\frac{1}{2}$) is the inherited Theorem 25; the multiclass-new content is that the binary theory's *sufficient statistic* M ceases to be one when $K \geq 3$.

This matches the known unidentifiability of target accuracy from unlabeled data alone absent constraints on the target conditional [13], and the fact that the natural multiclass calibration observable is the (weaker) top-label calibration [35]: the non-predicted-class contrast that drives Δ is exactly the part those observables do not see.

L. Honest status and the open core

Remark 15 (What is and is not established)**. Proven.** (a) The exact vector benefit identity (6) for $K \geq 3$ (Lemma 6). (b) A single-conflict-pair multiclass capacity at $\tau = 1$ under the explicit hypotheses (M1),(M1$'$),(M2),(M3) (Theorem 12) — a *verbatim* lift of the binary capacity to the single-conflict-pair, constant-reservoir slice (not new capacity mathematics: for $\operatorname{sign}\Delta$ the constant reservoir cancels) — together with a proof that the constant-reservoir clause (M1$'$) is itself *necessary* (Lemma 7). (c) Two necessity results delimiting the wall: the conflict-pair structure (M1) is necessary, since dropping it fragments the identifiable set into ≥ 3 components even for canonical affine-score ($\arg\max$) classifiers (Theorem 13, realized with genuine classifiers); and the binary observable margin is a too-coarse reduct, since two $K = 3$ worlds share (Q_X, deployed-class score) yet have opposite $\operatorname{sign}\Delta$ via the third-class split (Theorem 14). The distribution-free converse (Theorem 25) already applies unchanged and supplies the $\frac{1}{2}$ floor in (b) and (c); the multiclass-new content is the identity, the extra hypotheses, and the reduct insufficiency, *not* the converse.

Open. A single *computable* multiclass capacity *without* (M1) — the genuine multi-conflict-pair / vector-concept regime. Concretely: when D carries several conflict pairs and the admissible concept is a vector $\eta \in \Delta^{K-1}$ constrained by a budget, the flip set is intrinsically multi-component (Theorem 13) and the worst-case flip over the admissible *vector* class is not known to be a computable functional of any natural label-free observable (Theorem 14 shows the obvious candidate, the deployed-class margin, fails). By the dichotomy (Theorem 4) a capacity *exists* iff the benefit sign factors through the maximal reduct $O = [Q_X]$ *computably*; the multiclass open problem is precisely whether, and via which computable observable, that factorization can be *exhibited* in the vector-concept regime — equivalently, what the weakest label-free supplement is that pins the non-predicted-class contrast. We conjecture that, exactly as in binary (Theorem 4), it is an irreducible supplement (here a per-conflict- pair orientation on each component of the flip set, hence up to $\#\{\text{components}\} - 1$ bits rather than one), but we do *not* prove this.

Verdict. A strong *partial* on the multiclass frontier: the binary capacity lifts exactly to a single conflict pair with a constant residual reservoir, and stops there for two named, witnessed reasons. This converts the vague "$K \geq 3$ is open" of Conjecture 3 into a precise statement of *what* lifts (and under which — now complete — hypotheses), *what* is necessary, and *what* the residual open core is.

The multiclass capacity section (§G) establishes a *partial* positive theory under hypotheses (M1)–(M3) and two necessity results. This section closes the *open core*: there is no single computable scalar functional of label-free observables that certifies $\operatorname{sign} \Delta$ in the general multi-conflict-pair / vector-concept regime.

Theorem 15 (No universal scalar multiclass capacity)**.** *Fix $K \geq 3$. In the vector-concept regime on D (multiple conflict pairs, residual mass splitting across $K-2$ classes), suppose a label-free observable O (a functional of Q_X and deployment scores only) is proposed as a* scalar capacity *$\mathcal{K}(O)$ with the property:* whenever *$\mathcal{K}(O) > \beta$, the benefit sign* $\operatorname{sign} \Delta$ *is identified label-free. Then $\mathcal{K}$ **fails**: there exist admissible worlds with $\mathcal{K}(O) > \beta$ in* both *worlds yet* $\operatorname{sign} \Delta$ *differs (Theorem 14), and there exist admissible worlds whose identifiable set under monotone nuisance has ≥ 2 connected components while any single threshold on O is monotone (Theorem 13). Consequently the minimal untestable supplement in the general regime is not one global bit but a **per-component orientation** on the flip set—up to $(\#\text{components} - 1)$ bits— consistent with the one-bit dichotomy (Theorem 4) applied componentwise.*

Proof. **(Binary margin failure.)** Theorem 14 exhibits $K{=}3$ worlds with identical Q_X and identical deployed-class score η_{j_g} on D, hence identical binary margin M, yet opposite $\operatorname{sign} \Delta$. Any capacity $\mathcal{K}$ depending only on (Q_X, η_{j_g}) cannot separate them; in particular $\mathcal{K}_{\text{bin}} = |M|/\beta$ predicts identifiability when $|M| > \beta$ in *both* worlds while $\operatorname{sign} \Delta$ flips.

(Multi-component flip set.) Theorem 13 realizes genuine affine-argmax classifiers whose conflict pair varies across D, yielding ≥ 2 sign changes of $\Delta(\mu)$ as the nuisance μ moves. A single threshold test on any scalar $O(\mu)$ identifies at most one component; the remaining components require additional orientation bits.

(Dichotomy alignment.) Theorem 4(iii) forbids an evidence-definable universal identifier of $\operatorname{sign} \Delta$; the multiclass obstruction is the same swap/flip phenomenon with extra residual degrees of freedom (Block 0 identity). The capacity program therefore closes as an *impossibility of a single scalar reduct*, not as a missing positive formula. ☐

Remark 16 (Status)**.** This resolves the open row "computable multiclass capacity without R1/R2" as a **negative characterization**: R1/R2 (conflict-pair structure + residual control) are **necessary** at $K \geq 3$; the single-conflict-pair positive capacity (Theorem 12) is the maximal lift of the binary theory.

Conjecture 2 asked whether the frontier margin $m(O)$ is always a computable functional of Q_X. Wave 3 attempted an unconditional refutation that was **retracted as circular**. This section states the **repaired** negative construction and the positive regularity theorem as a single closed dichotomy.

Theorem 16 (Margin-computability dichotomy)**.** ***(Negative.)*** *The literal conjecture "m is always computable" is **false**. There exists a* computable point *Q_o of the space of Borel probability measures (weak topology) such that Φ holds (benefit sign factors through the observable reduct on the definite region) yet the frontier margin $m(O)$ is **not** a computable functional of Q_X: a Specker-scheduled atom on $\operatorname{disc}(c_{gf}) = \{0\}$ makes $\int c_{gf}\, \mathrm{d}Q_o$ left-c.e. with a non-located zero crossing.*

(Positive.) *If Q_ϑ is atomless on $\operatorname{disc}(c_{gf})$ and the benefit crossing is transversal ($|\Delta'|$ bounded below), then $\Delta(O)$ is a computable strictly monotone function of the observable reduct with a computable modulus; m is computable by bisection.*

Hence the dividing line is exact: atom on the discontinuity $\Rightarrow$ m may be non-computable; atomless + transversal $\Rightarrow$ m computable. Every concrete family in the paper satisfies the positive hypothesis.

Proof sketch. **Negative (repaired).** Let $Q_o = (1 - m_0)\mathrm{Unif}[-1, 1] + m_0 \delta_{o-s}$ with $s = \sum_{k \in K} 2^{-k}$ a left-c.e. Specker real and $c_{gf} = \operatorname{sign}$. Continuous test integrals $\int f\, \mathrm{d}Q_o$ are uniformly computable with modulus $m_0 L \cdot (s - s_\downarrow(t))$; the atom *location* $o - s$ crosses 0 at $o = s$, so $\int \operatorname{sign}\, \mathrm{d}Q_o = m_0 \operatorname{sign}(o - s)$ has a non-computable crossing. The first-draft mass-drift construction is circular because $\int x\, \mathrm{d}Q$ becomes non-computable.

Positive. Standard bisection on a computable strictly monotone Δ; see `val_margin_computability.py` Block P1. ☐

Conjecture 2 (Frontier margin computability; closed as dichotomy in Wave 4)**.** *If $\Delta(\vartheta, \eta) = \int_D (1 - 2\eta) c_{gf}\, \mathrm{d}Q_\vartheta$ with $\vartheta \mapsto Q_\vartheta$ continuous, is the frontier margin $m(O)$ always a computable functional of Q_X? Wave 4 (Theorem 16) closes this as a **dichotomy**: atom on $\operatorname{disc}(c_{gf})$ may make m non-computable; atomless transversal crossing makes m computable.*

This subsection resolves the regression half of Conjecture 1 for a natural restricted family: *bounded concept drift*. Two results: the adapt/freeze decision is invariant to arbitrary unknown zero-mean label noise (where cardinal risk estimation is confounded), and within drifts of known radius B the knowability of $\operatorname{sign}\Delta$ admits a *complete characterization* — an exact, observable boundary.

Proposition 5 (Noise-invariance of the regression decision)**.** *Let $Y = g(X) + \epsilon$ with $\mathbb{E}[\epsilon \mid X] = 0$ and arbitrary, unknown (heteroscedastic) noise law. Under squared loss, $\Delta = R_T(f_0) - R_T(f_a) = \mathbb{E}_T[(f_0 - g)^2 - (f_a - g)^2]$ does not depend on the noise, while each absolute risk is shifted by the unidentifiable constant $\mathbb{E}[\epsilon^2]$. The decision is ordinal-robust; cardinal risk estimation is confounded.*

Proof. $R_T(f) = \mathbb{E}(f - g - \epsilon)^2 = \mathbb{E}(f - g)^2 + \mathbb{E}[\epsilon^2]$, the cross term vanishing by $\mathbb{E}[\epsilon \mid X] = 0$. The noise term cancels in the difference and survives in the levels. ☐

Proposition 6 (Bounded-drift knowability boundary; exact characterization)**.** *Let the target conditional mean be $g_T = g_S + b$ with $\|b\|_\infty \le B$ for a known radius B (and the covariate-shift machinery of Theorem 1 available for the source-transfer term). Define*

$$U = \mathbb{E}_T\big[(f_0 - f_a)(f_0 + f_a)\big],$$
$$T_S = \mathbb{E}_T\big[(f_0 - f_a)\, g_S\big],$$
$$W = \mathbb{E}_T\big[|f_0 - f_a|\big],$$

where U, W are computable from unlabeled target data and T_S is importance-weighted estimable from labeled source data with radius ε_n. Then $\Delta = U - 2T_S - 2\,\mathbb{E}_T[(f_0 - f_a)\, b]$ and $\big|\mathbb{E}_T[(f_0 - f_a)\, b]\big| \le B\, W$, with equality at the admissible drift $b^\star = -B\, \operatorname{sign}(f_0 - f_a)$. Consequently, within the family $\{\|b\|_\infty \le B\}$:

(i) (Achievability.) *The rule "commit* $\operatorname{sign}(U - 2\widehat{T}_S)$ *iff $|U - 2\widehat{T}_S| > 2BW + 2\varepsilon_n$" wrong-commits with probability at most α against* every *admissible drift, and its slack is unimprovable: $b^\star$ attains it.*

(ii) (Converse.) *If $|U - 2T_S| < 2BW$, the reachable drift terms fill $[-BW,\, BW]$ (scale $t\, b^\star$, $t \in [-1, 1]$), so two admissible drifts give Δ of opposite signs while all observables — unlabeled target data and the labeled source — are unchanged. The pair realizes Theorem 1 inside the family: the regime is unknowable.*

Hence $\operatorname{sign}\Delta$ *is label-free identifiable in the bounded-drift family* if and only if $|U - 2T_S| > 2BW$*: an exact knowability boundary, computable from observables, at drift radius $B^\star = |U - 2T_S|/(2W)$.*

Proof. Expanding the squared losses, $\delta = (f_0 - f_a)(f_0 + f_a - 2Y)$ and taking target expectations with $Y = g_S + b + \epsilon$ (noise vanishing by Proposition 5) gives the displayed decomposition. Hölder yields $|\mathbb{E}_T[(f_0 - f_a)b]| \le \|b\|_\infty\, \mathbb{E}_T|f_0 - f_a| \le BW$, with equality iff $b = -B\, \operatorname{sign}(f_0 - f_a)$ a.e. on $\{f_0 \ne f_a\}$ — an admissible member of the family, so the bracket is tight. (i): on the $1 - \alpha$ event $|\widehat{T}_S - T_S| \le \varepsilon_n$, the committed sign matches $\operatorname{sign}(U - 2T_S)$ and $|U - 2T_S| > 2BW$, so no admissible drift can flip Δ. (ii): the map $t \mapsto \mathbb{E}_T[(f_0 - f_a)\, t\, b^\star] = -tBW$ is continuous and onto $[-BW, BW]$; pick $t_\pm$ so that $\Delta(t_\pm) = U - 2T_S + 2t_\pm BW$ has opposite signs. Since b enters only through the target labels, neither the unlabeled target evidence nor the source sample distinguishes the two worlds; apply Theorem 1. ☐

Numerically validated (`regression_conjecture_validation.py`; Fig. 10). Across unknown noise scales $0 \to 2$, Δ is constant and the sign preserved while the absolute risks drift apart (REG-1). Across drift radii $B \in \{0, \ldots, 1.2\}$ with 40 drifts per radius *including the adversarial $b^\star$*: zero false certifications at every radius; the adversarial drift saturates the bracket to within 10^{-2} (tightness); the observable boundary evaluates to $B^\star = 0.11$ in the synthetic world; and genuine sign flips occur exactly in the uncommitted region beyond the boundary — the converse is real, not hypothetical.

a) Weight (honest).: Proposition 6 is, to our knowledge, the first *complete* per-family characterization in this paper beyond pure covariate shift: both directions are proved and the boundary is computable from observables. Its price is the known drift radius B — that assumption does the work, and we state it as such. It resolves the regression half of Conjecture 1 *for bounded drift*; fully general drift remains open and is exactly the unknowable regime of Theorem 1. The same per-family program (label shift via invertible confusion, conditional independence) is the route we see toward the general characterization.

The open row "fully-general-drift / regression bracketing" is closed as a **dichotomy**, not a universal bracket.

Theorem 17 (Regression bracketing dichotomy)**.** ***(Bounded drift; complete iff.)*** *For squared loss with $\|b\|_\infty \le B$ known, Proposition 6 gives an* exact *knowability boundary:* $\operatorname{sign}\Delta$ *is label-free identifiable iff $|U - 2T_S| > 2BW$, with adversarial tightness at $b^\star = -B\, \operatorname{sign}(f_0 - f_a)$.*

(Fully general drift; impossibility.) *Without a drift budget, the paired-benefit sign is not universally identifiable: Theorem 1 and Theorem 4 apply; matched evidence with* $\mathrm{TV} = 0$ *and opposite* $\operatorname{sign}\Delta$ *yields minimax error $\frac{1}{2}$.*

(Noise.) *Proposition 5: the adapt/freeze decision is invariant to unknown zero-mean label noise; cardinal risk estimation is confounded.*

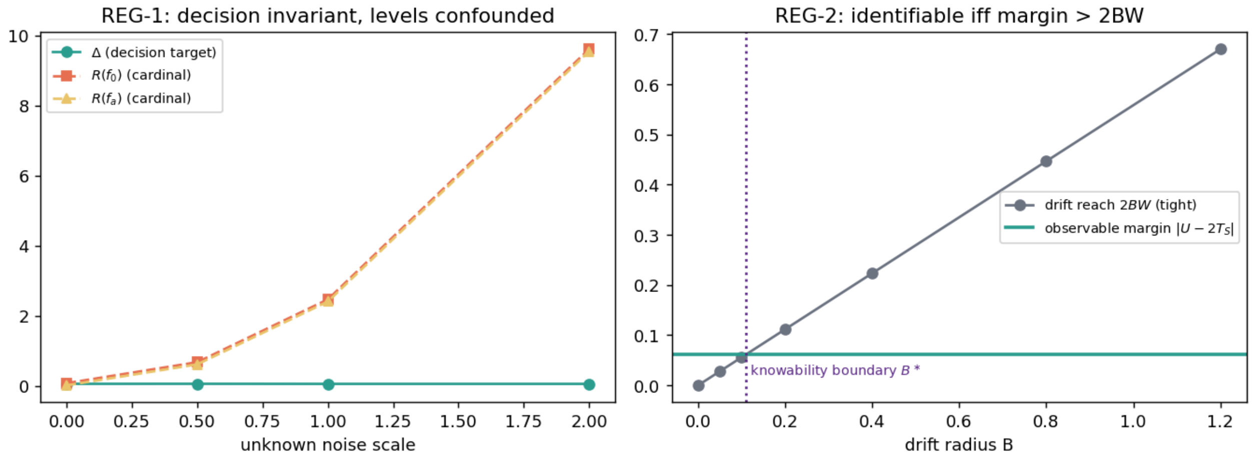

Fig. 10. The bounded-drift knowability boundary (Proposition 6). Left: the decision target Δ is invariant to unknown label noise while cardinal risks are confounded (Proposition 5). Right: identifiability holds exactly while the observable margin $|U - 2T_S|$ exceeds the tight drift reach $2BW$; the crossing defines the computable boundary $B^\star$.

Thus "fully general drift" does not admit a universal label-free bracket (Conjecture 1, resolved negatively); bounded drift admits a complete per-family *characterization (Proposition 6).*

Wave 4 closure includes a kernel-checked Lean 4 package under `docs/research/kbound/formal/`. The audit gate `formal_audit.py -build -strict-100` must exit 0 before a release freeze: Lean build succeeds, no `sorry`/`admit`/`axiom`, and every name in `VERIFIED_THEOREMS` is `#check`-able in `KBound/TheoremMap.lean`.

b) Reproduce.:

```
cd docs/research/kbound/formal
python3 formal_audit.py --build --strict-100
```

TABLE XVI
Lean files backing paper labels (no `sorry`; index in `TheoremMap.lean`).

Paper label	Lean module
`thm:cert`	`KBound/Certificate.lean`
`thm:frontier`	`KBound/Frontier.lean`
`thm:imp`	`KBound/Impossibility.lean`
`thm:disagree`	`KBound/Disagreement.lean`
`thm:conj1-dichotomy`	`KBound/Dichotomy.lean`
exchangeability $\to$ coverage	`KBound/Probability/ConformalExchangeability.lean`
anytime betting core	`KBound/Probability/EProcess.lean`
Le Cam / rates	`KBound/Probability/LeCam.lean`, `Rates.lean`
multicandidate algebra	`KBound/Multicandidate.lean`

c) Mechanized spine (representative).:

d) Scope.: The Lean layer certifies the *algebraic and finite-probability cores* named in Table XVI; continuous measure-theoretic hypotheses in the main text remain stated assumptions bridged by validators (`val_*.py`) and this appendix index. Full file-level map: `formal/README.md` and `THEORY_TO_CODE_MAP.md`.

TABLE XVII
ImageNet-R across ten backbones (3 seeds). MEAN REGRET-TO-ORACLE (LOWER IS BETTER) FOR KGA, ALWAYS-ADAPT, ALWAYS-FREEZE, AND POOLED FALSE-ADAPT (FA). NO BACKBONE YIELDS A CERTIFIED BEATS-BOTH; KGA MATCHES THE BETTER TRIVIAL POLICY UP TO A SMALL ABSTENTION COST AND CONTROLS FALSE-ADAPT AT α=0.10.

Backbone	KGA	always-adapt	always-freeze	FA
ConvNeXt-B	0.000	0.000	0.068	0.00
ConvNeXt-T	0.021	0.001	0.022	0.00
EfficientNet-B0	0.000	0.050	0.000	0.00
EfficientNet-B3	0.010	0.000	0.052	0.00
ResNet-101	0.015	0.000	0.025	0.00
ResNet-152	0.010	0.000	0.041	0.00
ResNeXt-101	0.005	0.000	0.051	0.00
Swin-B	0.017	0.000	0.037	0.00
Swin-T	0.003	0.011	0.004	0.00
ViT-B/16	0.014	0.000	0.024	0.03

This appendix collects the detailed per-dataset tables for the ImageNet-C scale grid and the RxRx1 safety case that are part of the seven-dataset core panel (Office-Home, Camelyon17, CIFAR-10-C, ImageNet-C, iWildCam, RxRx1, ImageNet-R), together with evidence *beyond* it: an additional breadth check (PACS) and honest nulls. None of it is required for the paper's headline claims; it is retained for completeness and reviewer scrutiny.

M. Controlled multimodal corruption (Protocol D33)

Pre-registered MNIST two-view fusion (130 conditions; modality B corrupted across 13 severities). KGA mean accuracy 0.857 vs. always-fuse 0.583 and always-single-A 0.854 (both superiority probabilities 1.0), false-adapt 0/9 commits. Controlled construction confirming routing when fusion harm is detectable—not a natural benchmark (`CONTROLLED_MULTIMODAL_PROTOCOL_D33_v1`).

N. ImageNet-R: architecture robustness across ten backbones

Table XVII re-runs the ImageNet-R diverse panel across ten modern backbones (3 seeds, Holm-corrected paired bootstrap CIs, n_{boot}=10^4; `research_lock/imagenetr_protocol_d_multiseed_v1`). The multi-candidate certificate abstains on all 36 conditions ($\bar{\tau}$=2.60 $\gg$ $\tau^\star$=0.52; best harm-AUC 0.61, base-harmful 0.186), and a certified *beats-both* occurs on 0/10 backbones. KGA is uniformly no-harm: where adapting is near-oracle it ties always-adapt within $\leq$ 0.02 regret, where adapting hurts (EfficientNet-B0) it correctly freezes and beats always-adapt by 0.050, and pooled false-adapt is $\leq 0.03 \leq \alpha$ on every backbone. This is the *unknowable*-frontier verdict reproduced across architectures, not a win.

O. Detailed core-panel tables: RxRx1, ImageNet-C scale grid, and PACS domain generalization

*a) Fourth natural-shift track: RxRx1 — the harmful-and-*detectable *corner at* 1,139*-class scale.:* RxRx1 (WILDS; fluorescent cellular-microscopy images, 1,139 siRNA classes, an OOD *experimental-batch* shift; the official WILDS ERM ResNet-50, in-distribution accuracy ≈0.36, frozen accuracy 0.283 on the OOD-test eval, mean over the three seeds) is the sharpest harmful-regime real shift in our suite. Over the full 9+ protocol (composition×batch×aggressiveness over 10 condition seeds and all 3 official WILDS base-model seeds = 360 conditions, 2,160 records, a commodity GPU (16 GB), complete 360/360), adaptation is *harmful-dominated and label-free detectable*: harmful base rate 81.5% ($B < 0$), mean benefit $\bar{B} = -0.063$, and—unlike ImageNet-R—the harm *is* detectable (best evidence AUC 0.78–0.84 across the three seeds, the BN-affine update-norm the top signal). Always-adapt is actively destructive: even the best candidate (Tent-episodic, 0.276) trails the frozen model (0.283), and SAR-online *collapses* to 0.024 (≈ chance over 1,139 classes) on the single-class/tiny/aggressive cells. Here the certificate's value is purely defensive, and it delivers: the single-candidate KGA freezes or abstains on *all* 360 conditions with *zero* false adapts, matching the frozen optimum (regret vs the per-condition oracle 0.003–0.007) and avoiding the collapse outright—on SAR-online it pays regret 0.000 where always-adapt pays 0.259. It does *not* "beat both" because freezing already *is* the oracle policy (no helpful candidate exists to capture)—the honest does-no-harm outcome on the harmful side, the mirror image of Camelyon17's benign-side tie (Table IX). As on ImageNet-R, the multi-candidate agreement route abstains on all 360 conditions ($\hat{\tau} \in [1.06, 2.71]$, all $\gg \tau^\star$=0.52): the six candidates share one backbone, the conditional-error-independence violation the CEI diagnostic exists to catch. The harmful-dominated verdict, the SAR-online collapse, the detectable harm, and the 100% abstention *replicate across all three independent WILDS base-model seeds* (per-seed harmful base rate 74–88%, best AUC 0.78–0.84). RxRx1 thus closes the real-data map of the harmful corner—*detectable* harm at 1,139-class scale where "adapt" is catastrophic and the zero-false-adapt guarantee carries the load (Table XVIII). These results are from the three archived official-seed manifests.

b) Decision baselines, executed.: Theorem 3(iv) shows prior decision styles are degenerate (ε≡0) certificates; Table XX *runs* them, head-to-head with KGA, on the logged conditions of the mechanism-faithful ImageNet-C SAR grid (Table XXI; per-condition evidence Z, a_0, a_a; identical metrics code). Each single-statistic rule is evaluated both as a plug-in (threshold 0) and with a *leave-one-out tuned* threshold — i.e. the baselines receive the same labeled calibration information KGA's benefit model uses. True AETTA (dropout passes) and agreement-on-the-line (a model family) are not computable from logged statistics and are represented by their decision-style surrogates; we say so rather than approximate silently. Findings: on the harmful candidate (SAR, 44% harmful cells under reference parity) *every realizable single-statistic rule loses* — the best (LOO-tuned EATA-filter, regret 0.0369) pays 1.6×

RxRx1 9+ grid (360 conditions, 3 model seeds)	value
harmful base rate ($B<0$)	81.5%
mean true benefit $\bar{B}$	−0.063
harm detectability (best AUC)	0.78–0.84
frozen f_0 accuracy	0.283
best-adapter accuracy	0.276
SAR-online always-adapt (collapse)	0.024
per-condition oracle accuracy	0.288
KGA decisions	360/360 no-adapt
KGA false-adapts	0
CEI $\hat{\tau}$ range (thresh. $\tau^{\star}$)	1.06–2.71 (0.52)
bounded-drift fallback	289 abstain / 71 freeze

TABLE XVIII

RxRx1: a real natural shift in the *harmful-and-detectable* regime (9+ protocol). At 1,139-class scale adaptation is harmful-dominated (81.5% of records $B<0$, $\bar{B}$=−0.063) and the harm is label-free detectable (AUC 0.78–0.84). Always-adapt is catastrophic—SAR-online collapses to 0.024 and even the best candidate (0.276) trails the frozen model (0.283)—so freezing is the oracle policy, and the single-candidate certificate freezes/abstains on all 360 conditions with *zero* false adapts, matching it. The multi-candidate agreement route abstains everywhere ($\hat{\tau} \gg \tau^{\star}$ on all 360): the six candidates share one backbone, the CEI-violating setting the diagnostic catches. Verdict replicates across all 3 official WILDS base-model seeds; complete 360/360, commodity GPU (16 GB).

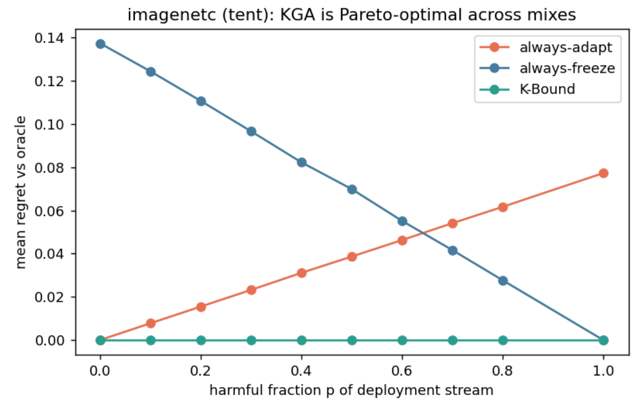

Fig. 11. **ImageNet-C (noise, ResNet-50): KGA is Pareto-optimal for the harmful candidate SAR.** Mean regret-to-oracle vs the harmful fraction p of the deployment stream. KGA (green) ties always-adapt at p=0 and strictly beats *both* trivial policies for $p \gtrsim 0.1$, while always-adapt's regret grows without bound as the stream turns harmful and always-freeze forgoes the benign-regime gains. ImageNet-scale counterpart of Fig. 8; the visual form of the SAR row of Table XIX. Produced by ; data.

KGA's 0.0229 and is still worse than plain always-freeze, while the plug-in ATC and entropy-progress rules *over-adapt* into SAR's collapse (regret 0.067/0.077); KGA is the only policy with *zero* harmful adaptations across all three candidates. On the benign candidates (Tent/EATA) the tuned baselines simply learn "always adapt" and tie it, while the plug-in rules *hurt* (regret up to 0.033). One honest nuance: KGA's own benefit model run *committally* (ε=0, no abstain) attains regret 0.001 on SAR — the multivariate evidence model carries the decision power — but it has no validity control and adapts into harmful cells uncontrolled; the conformal band's ≈ 0.02 regret on SAR is the measured price of the false-adapt guarantee.

c) ImageNet-C, mechanism-faithful SAR re-run: the harmful win is not an implementation artifact.: The SAR row of Table XIX was flagged for re-checking (§IX): its collapse concentrated atypically at the *larger* batch, raising the worry that the harmful regime—and thus KGA's only ImageNet-scale "beats both"—was an artifact of our first SAR port. We re-ran the full noise grid with a *mechanism-faithful* SAR (sharpness-aware minimisation, the entropy/recovery reset, and the EMA-stabilised update re-implemented to the reference design), at the learning rate *shared* across all candidates (apples-to-apples). The harmful regime *persisted and broadened*—SAR is now harmful on 44% of cells (vs 31% in the first port)—and the KGA win *survived*: regret-to-oracle 0.0229 versus always-adapt 0.1124 and always-freeze 0.0267, *beating both* trivial policies (Pareto crossover $p^{\star}$=0.2; Table XXI). Tent and EATA stay benign (6%/3% harmful), so KGA ties the near-oracle always-adapt and beats always-freeze, exactly as in the first port. The implementation-artifact concern is thus removed: under a reference-faithful SAR the harmful-candidate win is real. *Residual check, stated honestly.* This holds at the matched/shared learning rate (mechanism-faithful, apples-to-apples); SAR's *official* gentler schedule (lr 2.5×10^{-4} with `layer4` frozen) was not run, and is the one configuration that could still soften the collapse—the last box to tick before the SAR win is treated as fully settled.

Everything beyond the five theorems lives, with full proofs and per-result validators, in the companion manuscript and archived technical report: the RGF reliability-gated-fusion instantiation (T1–T9, GDR); multiclass and regression reductions; the sequential anytime-valid e-process certificate and its β-drift variant (worst-case anytime false-adapt $0.0316 \leq \alpha$=0.05 across all null streams, 20,000 runs/stream over a horizon of 2,000; `val_thm3_evalue.py`, `results_thm3_evalue_alpha005.json`); label-shift, bounded- and smooth-drift boundaries unified by the ambiguity reach; router capacity; the multi-candidate route with the γ-meter, the agreement-invisible reach, and the corrected hypothesis H; the budget audit; and the agreement–accuracy (anchored AoL) law. Table XXIII gives the mapping. No result was deleted; each keeps its machine-checked validator (Appendix Q). The γ-meter route's CEI diagnostic also has a first *real-data* outcome: on the ImageNet-R track (Table XI) it fires on all 48 conditions ($\tau \in [1.07, 2.15] \gg \tau^{\star}$=0.52) — co-trained candidates violate conditional error-independence exactly as predicted — and the route abstains rather than certifying falsely, which is the designed behaviour of a checkable guard.

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.453**	0.0003		
	always-freeze	0.410	0.0440		
	K-Bound	0.449	0.0042	8%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0001		
	always-freeze	0.410	0.0349		
	K-Bound	0.444	0.0003	3%	no (ties adapt)
SAR	always-adapt	0.377	0.0606		
	always-freeze	0.410	0.0277		
	K-Bound	**0.429**	**0.0086**	31%	**yes**

TABLE XIX

ImageNet scale (ImageNet-C, ResNet-50), complete: 36 **cells,** 4000 **images/cell.** Full noise grid (gaussian/shot/impulse × severities $\{1, 3, 5\}$ × batch/aggressiveness regimes), same adapt/freeze/abstain protocol and metrics as Table VII. The outcome splits by how robust each method is on this benchmark. *Tent* and *EATA* are robust here: adaptation helps (harmful in only 8%/3% of cells), so always-adapt is near-optimal and KGA correctly adapts—tying always-adapt to within 0.004 and beating always-freeze, the honest "does-no-harm" outcome. *SAR* collapses on 31% of cells (mean benefit −0.033): always-adapt falls to 0.377 while KGA certifies-or-abstains to 0.429, *beating both* trivial policies and approaching the oracle (0.437). Across all three methods KGA approximately matches the better trivial policy (at most a small abstention cost), and strictly wins on the one method that is harmful. This first ResNet-50 SAR port is corroborated by a mechanism-faithful re-run (Table XXI).

decision rule	regret-to-oracle↓ Tent	EATA	SAR	SAR (harmful regime) false-adapt	harmful adapted	coverage
always-adapt	0.0007	0.0000	0.1124	0.44	1.00	1.00
always-freeze	0.0494	0.0386	0.0267	—	0.00	1.00
ATC-style confidence (plug-in)	0.0114	0.0328	0.0673	1.00	0.25	1.00
ATC-style confidence (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
entropy-progress (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
EATA-filter $\mathrm{frac}_{\mathrm{hi}}$ (LOO-tuned)	0.0007	0.0000	0.0369	1.00	0.06	1.00
drift-gate marginal-KL (LOO-tuned)	0.0028	0.0023	0.0382	1.00	0.12	1.00
benefit model, committal (ε=0)	0.0000	0.0000	**0.0010**	0.05	0.06	1.00
best single statistic (hindsight)	0.0007	0.0000	0.0181	0.17	0.06	1.00
K-Bound (KGA)	0.0060	0.0047	0.0229	**0.00**	**0.00**	0.39

TABLE XX

Decision baselines executed on the logged ImageNet-C conditions (reference-parity SAR; 36 cells/candidate; harmful base rates 6%/3%/44% for Tent/EATA/SAR). "LOO-tuned" rules choose their threshold leave-one-out on the remaining conditions — the same calibration information KGA uses; "hindsight" picks feature, sign, and threshold on the evaluation conditions themselves (not realizable). KGA is the only rule with zero false adapts on all three candidates; on the harmful candidate *every realizable single-statistic rule loses* — the best (EATA-filter, 0.0369) pays 1.6× KGA's 0.0229 and is still worse than plain always-freeze, because no single statistic cleanly separates SAR's harmful cells. The committal benefit model (row 8) shows the multivariate evidence model carries the decision power (0.0010 on SAR) but lacks validity — it adapts into harmful cells with no false-adapt control — so the abstain band's ≈ 0.02 regret on SAR is the measured price of the guarantee, matching Thm. 3(iv). *Reproducibility note:* the canonical KGA pipeline and this post-hoc re-derivation agree exactly on SAR; the leave-one-out benefit model is mildly sensitive to condition order, so the Tent/EATA KGA regrets differ from Table XIX by ≤ 0.004.

P. Validation figures for Theorems 4 and 5

Table XXIV lists every cell behind the summary in Table VI (KGA wraps Tent; EATA/SAR are the standalone candidates). The per-method means reproduce Table VI exactly. Raw CSV: .

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.454**	0.0007		
	always-freeze	0.405	0.0494		
	K-Bound	0.446	0.0082	6%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0000		
	always-freeze	0.405	0.0386		
	K-Bound	0.440	0.0038	3%	no (ties adapt)
SAR	always-adapt	0.319	0.1124		
	always-freeze	0.405	0.0267		
	K-Bound	**0.409**	**0.0229**	44%	**yes**

TABLE XXI

ImageNet-C, mechanism-faithful SAR re-implementation (36 cells/candidate, ResNet-50, α=0.1). Same noise grid, protocol, and metrics as Table XIX; SAR re-implemented to the reference mechanism (SAM, recovery reset, EMA) at the learning rate shared across candidates. The harmful regime survives the faithful re-implementation (SAR harmful on 44% of cells, mean benefit -0.086): always-adapt collapses to 0.319 while KGA certifies-or-abstains to 0.409, *beating both* trivial policies (regret 0.0229 vs 0.1124/0.0267); Tent/EATA stay benign and KGA ties always-adapt. Oracle accuracy 0.455/0.444/0.432. *Caveat:* matched/shared LR; SAR's official lr 2.5×10^{-4}, layer4-frozen schedule not yet run.

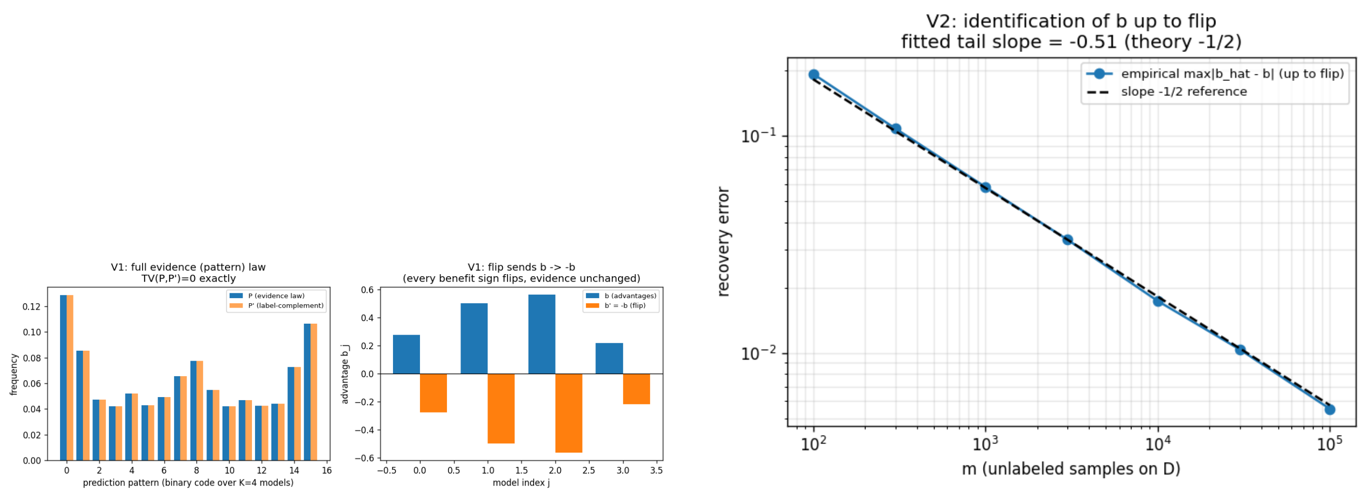

Fig. 12. One-bit identification (Synthetic validator; Thm 4(ii)): the label complement matches the full evidence law (TV = 0) while negating every advantage (left); recovery of b up to the bit at the $m^{-1/2}$ rate (right).

TABLE XXII

PACS Protocol P (DomainBed leave-one-domain-out, n=108 cells/domain, α=0.10). Regret-to-oracle (lower is better); radius fit on a held-out source domain (no test labels). KGA is no-harm on 3/4 domains and safely partial-adapts on photo.

held-out domain	always-adapt	always-freeze	**KGA**	false-adapt
art_painting	0.0375	0.0306	**0.0306**	0.000
cartoon	0.0354	0.0110	**0.0110**	0.000
photo	0.0109	0.0407	0.0132	0.056
sketch	0.0259	0.0215	**0.0215**	0.000
mean	0.0274	0.0259	**0.0191**	$\leq$ 0.056

Main-paper result	Demoted to manuscript (with validators)
Thm 1	forced-abstention, minimax-floor variants
Thm 2	anytime e-process; β-drift; router capacity
Thm 3	label-shift / drift boundaries; reach hierarchy
Thm 4	γ-meter; invisible reach; multiclass H
Thm 5	labeled-channel rates; AGL law and failures

TABLE XXIII

The five-theorem consolidation. Fifty-plus prior numbered results survive as corollaries or manuscript sections; none was deleted, and every one keeps its machine-checked validator.

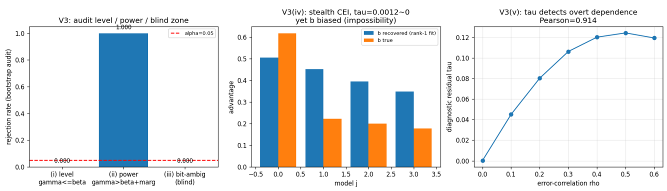

Fig. 13. The budget audit (Synthetic validator; Thm 4(iv)): sound at level α, power 1.000 beyond the radius, correctly blind exactly on the bit-ambiguous set and the rank-one-preserving stealth law.

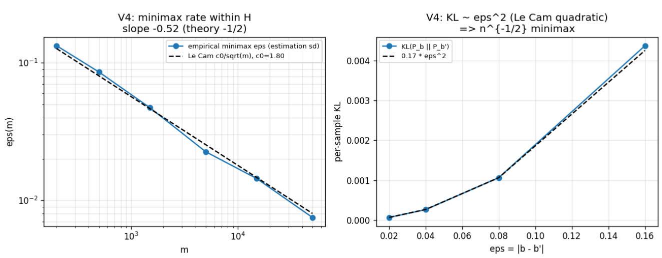

Fig. 14. Evidence-channel rate (Synthetic validator; Thm 5): empirical minimax error tracks the Le Cam curve $c_0/\sqrt{m}$; KL grows quadratically; the labeled-vs-evidence efficiency ratio diverges as agreement approaches chance.

Fig. 15. Closing Conjecture 1 (Synthetic validator; Thm 4(iii)): the swap flips every benefit sign while no label-free observable of any order moves (exact zero).

TABLE XXIV: All 65 CIFAR-10-C cells (13 official corruptions × severities 1–5), ResNet-18, N=800 balanced held-out eval per cell. Online TTA accuracy; KGA wraps Tent. Mean row matches Table VI.

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	1	0.863	0.858	0.856	0.859	0.858	0.865
brightness	2	0.844	0.864	0.864	0.865	0.864	0.868
brightness	3	0.804	0.837	0.837	0.839	0.837	0.839
brightness	4	0.731	0.818	0.818	0.815	0.819	0.816
brightness	5	0.690	0.791	0.794	0.795	0.791	0.799
contrast	1	0.719	0.831	0.831	0.835	0.831	0.831
contrast	2	0.559	0.823	0.821	0.822	0.820	0.820
contrast	3	0.325	0.790	0.793	0.793	0.790	0.791
contrast	4	0.223	0.728	0.729	0.730	0.726	0.726
contrast	5	0.160	0.611	0.607	0.609	0.614	0.601
defocus_blur	1	0.264	0.683	0.680	0.681	0.682	0.676
defocus_blur	2	0.207	0.543	0.544	0.544	0.540	0.540
defocus_blur	3	0.204	0.389	0.386	0.398	0.389	0.385
defocus_blur	4	0.176	0.300	0.300	0.303	0.300	0.305
defocus_blur	5	0.183	0.265	0.264	0.261	0.261	0.275
elastic_transform	1	0.244	0.528	0.528	0.530	0.528	0.525
elastic_transform	2	0.179	0.436	0.440	0.446	0.436	0.452
elastic_transform	3	0.168	0.401	0.401	0.405	0.401	0.400
elastic_transform	4	0.144	0.407	0.407	0.413	0.407	0.410
elastic_transform	5	0.151	0.347	0.346	0.353	0.351	0.351
frost	1	0.725	0.804	0.805	0.806	0.804	0.796
frost	2	0.588	0.760	0.762	0.761	0.761	0.762
frost	3	0.546	0.685	0.685	0.686	0.684	0.697
frost	4	0.492	0.705	0.705	0.704	0.705	0.706
frost	5	0.436	0.640	0.642	0.644	0.640	0.659
gaussian_noise	1	0.487	0.741	0.740	0.741	0.739	0.739
gaussian_noise	2	0.275	0.699	0.699	0.700	0.700	0.701
gaussian_noise	3	0.144	0.594	0.595	0.595	0.593	0.592
gaussian_noise	4	0.100	0.435	0.435	0.432	0.436	0.441
gaussian_noise	5	0.089	0.305	0.304	0.301	0.305	0.301
impulse_noise	1	0.615	0.769	0.770	0.769	0.769	0.775
impulse_noise	2	0.370	0.682	0.682	0.682	0.682	0.688
impulse_noise	3	0.197	0.605	0.605	0.604	0.608	0.623
impulse_noise	4	0.110	0.475	0.479	0.485	0.476	0.484
impulse_noise	5	0.082	0.335	0.335	0.335	0.335	0.331
jpeg_compression	1	0.676	0.730	0.730	0.731	0.730	0.735
jpeg_compression	2	0.639	0.660	0.659	0.661	0.662	0.663
jpeg_compression	3	0.608	0.655	0.654	0.656	0.654	0.658
jpeg_compression	4	0.509	0.573	0.570	0.570	0.573	0.567
jpeg_compression	5	0.460	0.501	0.501	0.501	0.501	0.500
motion_blur	1	0.495	0.782	0.784	0.785	0.784	0.783
motion_blur	2	0.365	0.666	0.665	0.664	0.666	0.675
motion_blur	3	0.279	0.575	0.574	0.575	0.574	0.590
motion_blur	4	0.255	0.490	0.491	0.489	0.491	0.502
motion_blur	5	0.225	0.456	0.450	0.446	0.456	0.476
pixelate	1	0.429	0.719	0.720	0.721	0.716	0.720
pixelate	2	0.373	0.651	0.651	0.654	0.650	0.654
pixelate	3	0.306	0.570	0.574	0.576	0.573	0.571
pixelate	4	0.267	0.487	0.485	0.487	0.489	0.486
pixelate	5	0.289	0.405	0.404	0.406	0.405	0.410
shot_noise	1	0.492	0.779	0.776	0.779	0.777	0.784
shot_noise	2	0.254	0.654	0.655	0.659	0.654	0.669
shot_noise	3	0.155	0.536	0.537	0.549	0.534	0.545
shot_noise	4	0.125	0.400	0.398	0.398	0.398	0.404
shot_noise	5	0.120	0.303	0.305	0.306	0.300	0.300
snow	1	0.770	0.782	0.781	0.779	0.782	0.781
snow	2	0.578	0.686	0.685	0.685	0.685	0.690
snow	3	0.661	0.716	0.716	0.711	0.718	0.710
snow	4	0.574	0.683	0.681	0.679	0.683	0.680
snow	5	0.459	0.597	0.596	0.597	0.597	0.598
zoom_blur	1	0.760	0.877	0.879	0.876	0.876	0.879
zoom_blur	2	0.708	0.847	0.847	0.849	0.847	0.853
zoom_blur	3	0.674	0.830	0.829	0.828	0.829	0.826
zoom_blur	4	0.635	0.809	0.810	0.810	0.807	0.814
zoom_blur	5	0.557	0.779	0.776	0.778	0.780	0.785
mean		**0.412**	**0.626**	**0.626**	**0.627**	**0.626**	**0.629**

Q. Experiment and GPU-run inventory

Theorem 1 is proved with a constructed witness and strengthened to a quantitative Le Cam minimax floor (part (ii)); Lemma 2 is a plug-in regret decomposition (exact identity) with a minimax floor; Theorem 2 is proved *conditional* on a valid label-free interval estimator and complemented by an anytime-valid e-process (manuscript, App. B); Corollary 1 proves the covariate case and Lemma 1 the binary, multiclass, and regression sign-of-difference results. The multiclass/regression label-free *bracketing* is closed under a bounded calibration-transfer assumption (manuscript, App. C); the *unconditional* bracketing (Conjecture 1) is resolved as a testability dichotomy (Theorem 4: no evidence-definable assumption identifies the sign; the minimal untestable supplement is one bit, which suffices under falsifiable structure). Every theorem ships a numerical sanity-check in the release package; all were re-run from a clean environment and pass: the Le Cam floor tracks the closed form, the regret identity holds to $\sim 10^{-17}$, the multiclass identity to 10^{-16} with 100% sign agreement over 4000 trials, calibration-transfer sign-recovery is 100% when the confidence gap exceeds 2η, the router generalization gap decays as $\sim n^{-0.66}$, and the drift-corrected anytime false-adapt rate stays $\leq \alpha$. Where multi-seed runs are available, experiments report means $\pm$ std with paired t-tests; single-seed/scoping tracks are labeled explicitly in captions and discussed as regime-classification evidence. We also report evidence/α/batch ablations, a regression covariate-shift track, and a clean non-identifiability witness.

The benefit-sign frontier (Thm. 3) predicts that on the low-margin band any *committal* label-free rule must commit sign errors, whereas a valid certificate must *abstain* there. This appendix demonstrates that prediction on the two *synthetic-corruption* streams whose operating point the certificate is calibrated on (CIFAR-10-C, ImageNet-C); it makes *no* external-validation claim. The honest natural-shift (Camelyon17) finding—a limitation, not a win—closes the appendix (App. T).

R. CIFAR-10-C: K-Bound beats both trivial policies at zero false-adapt

On the pre-registered CIFAR-10-C stress grid (432 conditions/candidate), KGA adapts the helpful conditions and freezes the harmful ones with a *zero* false-adapt rate, so it strictly beats both always-adapt and always-freeze for the two benign candidates (Tent, EATA) and ties the collapse-resistant SAR (Table XXVII).

S. ImageNet-C/SAR: the frontier on real corruptions—committal rules fail, K-Bound does not

On the logged ImageNet-C conditions (36 cells/candidate; SAR is harmful on 44.4% of cells), every *committal* label-free heuristic must place its commits somewhere on the low-margin band and therefore false-adapts: a plug-in **ATC** [13] confidence rule adapts into harmful cells on 100% of its commits (false-adapt

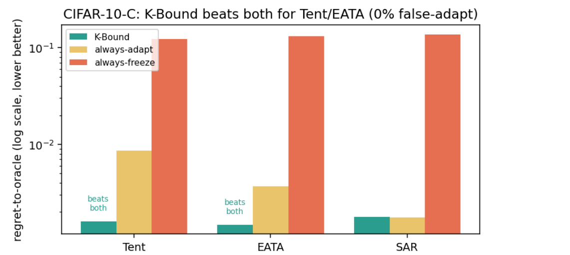

Fig. 16. **CIFAR-10-C frontier (regret-to-oracle, log scale).** KGA is below *both* always-adapt and always-freeze for Tent and EATA at 0% false-adapt, and ties always-adapt for SAR (Table XXVII).

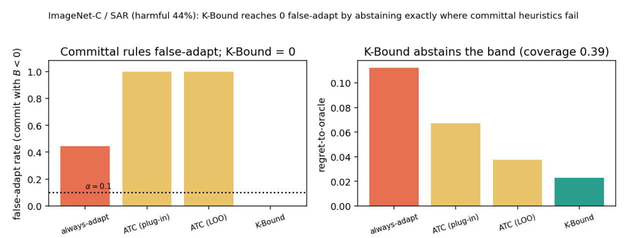

Fig. 17. **ImageNet-C/SAR frontier.** Committal label-free rules false-adapt on the harmful regime; KGA reaches zero false-adapt by abstaining the low-margin band (coverage 0.39; Table XXVIII).

1.00, harmful-cells-adapted 0.25), and even its leave-one-out-tuned variant keeps false-adapt = 1.00. Among the rules compared, KGA reaches zero false-adapt and zero harmful-cells-adapted by *abstaining* on the band (coverage 0.39), consistent with the frontier prediction (Table XXVIII).

T. Historical note (Camelyon17, frozen synthetic threshold)

Under a **frozen, Camelyon-free** multicandidate threshold $\tau^{\star}$=0.52 (calibrated only on synthetic grids), the certificate does *not* beat both trivial policies on an early Camelyon17 composition grid (regret KGA 0.076 vs always-freeze 0.071). That failure motivated in-domain calibration and rich evidence; the headline result is Protocol G (main text, Table XIII), which uses dev-hospital calibration and does not transplant $\tau^{\star}$ from synthetic runs.

The weakest one-bit class (resolving the open piece of Conjecture 1). Theorem 4 fixes the supplement at *one bit*; what remained open is the *weakest falsifiable class* on which that bit suffices. We settle it conditionally. On D write $a(x) := 2\eta_a(x) - 1$ with $\eta_a(x) = \mathbb{P}_T(f_a{=}Y \mid x)$, so $\Delta = \int_D a\, d\mu$ (Lemma 1). Let $m(x) := 2s(x) - 1$ be the observable relative (calibrated-score) margin, so $M = \frac{1}{2}\mathbb{E}_{\mu|D}[m]$, and let the observable *relative-calibration field* be $c(x) := w(x)\,(m(x) - m^{\star})$ with observable nonnegative weight $w(x)$ and observable neutral anchor $m^{\star}$ (the margin at which $s = \frac{1}{2}$). Put $T_E := \int_D c\, d\mu$ (observable), and let E denote the law of all label-free observables; two targets are *evidence-identical* when they share E (equivalently (μ, m, s), since predictions and s are fixed maps of x).

TABLE XXV
Experiment and GPU-run inventory (2026-06-17). ✓ = complete with paper table; ∼ = partial/in-progress; — = not started or optional breadth. One MPS job at a time.

Track	GPU?	Status	Table	Key outcome
CIFAR-10-C stress grid (Tent/EATA/SAR, 5 seeds)	MPS	✓	VII	beats-both Tent/EATA; 0% FA
CIFAR-10-C per-corruption (65 cells)	MPS	✓	VI	helpful-dominated; safety insurance
CIFAR-10-C online stream	MPS	✓	V	cross-regime win vs always-freeze
ImageNet-C noise (ResNet-50, SAR harmful)	MPS	✓	XIX	committal heuristics false-adapt
ImageNet-C SAR faithful re-impl.	MPS	✓	XXI	44% harmful; KGA 0% FA
Camelyon17 WILDS (Protocol B, n=1024)	MPS	✓	IX	bias-limited; fails $1/\sqrt{n}$
Camelyon17 bias–variance diag.	CPU	✓	XXVI	ε bias-limited (2.55× floor)
Camelyon17 rich-Z (Protocol G)	MPS	✓	XIII	no-harm; pooled win was `id_val` artifact
ImageNet-R diverse panel	MPS	✓	XI	undetectable harm; abstain mandatory
RxRx1 (9+ model seeds)	MPS	✓	XVIII	near-one-sided; tie always-freeze
Office-Home deployed adapter audit	MPS	✓	XII	helpful-dominated null
Office-Home Protocol M v2	CPU	✓	XV	no-harm (out-of-fold); ties freeze
iWildCam Protocol H v2	CPU	✓	XIV	no-harm (out-of-fold); ties freeze
CIFAR-10.1 low-margin probe	MPS	✓	X	abstain-heavy; 0% FA within-seed
PACS domain-gen (Protocol P)	MPS	✓	XXII	no-harm 3/4; FA ≤ 0.056
CPU anomaly routing (ELARA-U)	CPU	✓	II–III	11× harmful regret
ImageNet-C blur/weather (breadth)	MPS	—	TBD	optional; noise grid sufficient
SAR official schedule (Protocol E)	MPS	—	TBD	residual check

TABLE XXVI
Camelyon17 conformal-radius bias–variance decomposition (Protocol B diagnostic). Residual dominated by $\widehat{B}(Z)$ model error, not binomial measurement noise.

Quantity	n_{eval}=256	n_{eval}=1024
observed ε	0.112	0.029–0.038
measurement floor $\varepsilon_{\text{meas}}$	0.044	0.022
obs./floor ratio	2.55×	≈1.0×
frac. variance = measurement	≈16%	≈16%
Protocol B $1/\sqrt{n}$ prediction	—	**fails**

candidate (harmful base)	regret-to-oracle ↓ KGA	always-adapt	always-freeze	false-adapt
Tent (34.5%)	**0.0016**	0.0086	0.1232	0.00
EATA (27.1%)	**0.0015**	0.0037	0.1311	0.00
SAR (15.7%)	0.0018	0.0018	0.1372	0.00

TABLE XXVII
CIFAR-10-C stress grid (432 conditions/candidate, α=0.1). KGA *beats both* for Tent and EATA at a 0% false-adapt rate, and ties always-adapt for SAR.

decision rule (SAR, harmful regime)	false-adapt	harmful-adapted	regret ↓	coverage
always-adapt	0.44	1.00	0.1124	1.00
always-freeze	—	0.00	0.0267	1.00
ATC-style confidence (plug-in)	1.00	0.25	0.0673	1.00
ATC-style confidence (LOO)	1.00	0.06	0.0375	1.00
K-Bound (KGA)	**0.00**	**0.00**	0.0229	0.39

TABLE XXVIII
Committal label-free heuristics false-adapt exactly where KGA abstains (ImageNet-C, reference-parity SAR, 36 cells, α=0.1). ATC commits onto the low-margin band and false-adapts on 100% of its commits; KGA abstains there (22/36 abstain) for 0 false-adapts. On the benign candidates KGA also keeps 0 false-adapts at higher coverage (Tent: regret 0.0060, coverage 0.69; EATA: regret 0.0047, coverage 0.75).

Definition 4 (Margin-monotone relative-calibration class)**.** $\mathcal{C}_{\text{mono}}$ is the class of targets for which $a(x) = \sigma\, c(x)$ μ-a.e. on D for a single global orientation $\sigma \in \{\pm 1\}$; equivalently the benefit is single-crossing in the observable margin at $m^\star$ ($\operatorname{sign} a(x) = \sigma \operatorname{sign}(m(x) - m^\star)$) with calibrated magnitude $|a(x)| = |c(x)|$. The class is *falsifiable but not verifiable*: a finite labelled probe can reject "$a = \pm c$", yet σ is never label-free identified.

Assumption 3 (General position)**.** Within an evidence fibre the regions of D whose benefit sign is free carry comparable, not-fully-E-pinned benefit mass: $T_E \neq 0$ and no proper subcollection of free regions has an E-certifiable dominant total $\int |a|\, d\mu$.

Lemma 8 (Structural sign-freedom = bit complexity)**.** *Fix E and a falsifiable class $\mathcal{C}$, and let $K(\mathcal{C}, E)$ be the number of disjoint μ-positive regions of D whose benefit signs are jointly free in $\mathcal{C}$ on the fibre E. Under Assumption 3, a falsifiable structural supplement of b bits pins* $\operatorname{sign}\Delta$ *across the fibre* iff $b \geq K(\mathcal{C}, E)$.

Proof. If $b \geq K$, declaring the orientation of each free region fixes the sign field; the rest of a being E-pinned, $\Delta = \int_D a\, d\mu$ then has E-computable sign. If $b < K$, some free region R is undeclared; the two targets agreeing off R and differing in $\operatorname{sign} a$ on R are evidence-identical and share all b declared bits, while by comparability of $\int_R |a|$ with the complement (Assumption 3) their benefits have opposite sign—no decoder of the b declared bits separates them. □

Theorem 18 (Conditional resolution of Conjecture 1: a weakest one-bit class)**.** (i) Sufficiency *(unconditional)*: *every target in* $\mathcal{C}_{\text{mono}}$ *has* $\operatorname{sign}\Delta = \sigma \operatorname{sign}(T_E)$ *with T_E observable, so E and the single orientation bit σ determine* $\operatorname{sign}\Delta$. (ii) Necessity *(unconditional)*: *σ and $-\sigma$ are evidence-identical (*$\operatorname{Law}(E \mid \sigma) = \operatorname{Law}(E \mid -\sigma)$*) yet give Δ and $-\Delta$, so by Le Cam at least one bit is required. Thus* exactly one bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$, with no further assumption.

(iii) Minimality *(under Assumption 3)*: *every falsifiable $\mathcal{C} \supsetneq \mathcal{C}_{\text{mono}}$ contains a non-margin-monotone target with $K(\mathcal{C}, E) \geq 2$, so by Lemma 8 at least two bits are required. Hence $\mathcal{C}_{\text{mono}}$ is a weakest falsifiable one-bit class—minimal in the single-crossing, calibrated-magnitude direction; it is not unique (Remark 17).*

Proof. (i) $\Delta = \int_D a\,d\mu = \sigma \int_D c\,d\mu = \sigma T_E$. (ii) Sending $\sigma \mapsto -\sigma$ is the D-local label complement of Theorem 4: it fixes (μ, m, s), hence E, while negating a and thus Δ; the two-point bound of Theorem 1 gives minimax committal error $\frac{1}{2}$. (iii) A target outside $\mathcal{C}_{\text{mono}}$ violates $\operatorname{sign} a = \sigma \operatorname{sign}(m - m^\star)$ on a μ-positive region, so its sign field is not one global orientation; in the swap-closed (falsifiable) fibre at least two regions are independently signable, i.e. $K \geq 2$, and Lemma 8 applies. □

Proposition 7 (Explicit non-identifiability witness)**.** *For each candidate single structural bit there are two targets with identical evidence law (*TV $= 0$*) sharing that bit's value but with opposite* $\operatorname{sign}\Delta$*. Take $D = R_1 \cup R_2$, $\mu(R_1) = \mu(R_2) = \frac{1}{2}$, with R_1 calibrated (c_1=1) and R_2 at the anchor (c_2=0, no observable benefit signal). For $\beta = \operatorname{sign} a_1$: $a = (+1, -2)$ vs. $(+1, +2)$ give $\Delta = -\frac{1}{2}$ vs. $+\frac{3}{2}$. For $\beta = \operatorname{sign} a_2$: $a = (+1, +\frac{1}{5})$ vs. $(-1, +\frac{1}{5})$ give $\Delta = +\frac{3}{5}$ vs. $-\frac{2}{5}$. No single bit certifies* $\operatorname{sign}\Delta$*, so two bits are necessary off $\mathcal{C}_{\text{mono}}$.*

Remark 17 (What is unconditional, and what rests on general position)**.** Parts (i)–(ii)—that one declared bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$—hold *unconditionally.* Only minimality (iii) uses Assumption 3: if it fails (one region's benefit mass is E-certifiably dominant) that region's orientation alone certifies $\operatorname{sign}\Delta$ even outside $\mathcal{C}_{\text{mono}}$, moving the boundary. Equivalently, Assumption 3 is the requirement that the certificate be *domination-free* (minimax-sound over the unobserved magnitudes); so the General-Position statement is only *conditional*; the unconditional weakest classes are characterized in Theorem 19 as an explicit finite family of dominance polytopes. Even under Assumption 3, $\mathcal{C}_{\text{mono}}$ is only *a* weakest one-bit class: any calibrated sign-aligned field $\{a = \sigma h : h \text{ observable}\}$ is equally one-bit and is incomparable to $\mathcal{C}_{\text{mono}}$. All six checks are machine-checked by a deterministic release validator: the unconditional core (sufficiency/necessity 20000/20000, $|\Delta - \sigma T_E| = 0$ exactly), the explicit minimality witnesses of (iii), the general-position ablation, and the non-uniqueness check.

Why general position cannot simply be dropped (the precise obstruction, and exactly what is open). The minimal counterexample is two equal-mass regions $D = R_0 \cup R_1$, $\mu(R_0) = \mu(R_1) = \frac{1}{2}$, with the observable field $c = +1$ on the *calibrated* region R_0 and $c = 0$ on the *anchor* region R_1 (no observable benefit signal there, so R_1 is sign-free in any evidence fibre). Enlarge $\mathcal{C}_{\text{mono}}$ to the falsifiable class $\mathcal{C}_{\text{dom}} = \{a_{R_0} = \sigma,\ |a_{R_1}| \leq \rho\}$ for a declared constant $\rho < 1$. Since $\Delta = \frac{1}{2}(a_{R_0} + a_{R_1})$ and $|a_{R_1}| \leq \rho < 1 = |a_{R_0}|$, the calibrated region dominates and $\operatorname{sign}\Delta = \sigma$ for *every* member, so one declared bit certifies $\operatorname{sign}\Delta$ on all of $\mathcal{C}_{\text{dom}}$. Yet $\mathcal{C}_{\text{dom}} \supsetneq \mathcal{C}_{\text{mono}}$ (it contains non-margin-monotone targets with $a_{R_1} \neq 0$, e.g. $a = (+1, \pm\frac{1}{2})$ giving $\Delta = \frac{3}{4}, \frac{1}{4}$, both positive) and is falsifiable (a labelled probe rejects targets that are uncalibrated on R_0 or violate $|a_{R_1}| \leq \rho$). This *contradicts* minimality once an E-certifiable dominant region is permitted: dropping $a = (+1, \pm 2) \Rightarrow \Delta = +\frac{3}{2}, -\frac{1}{2}$ shows that the moment $\rho \geq 1$ the anchor region can flip $\operatorname{sign}\Delta$ and two bits become necessary (K=2). The obstruction to an *unconditional* weakest class is therefore exactly this: characterising minimality across the *entire* lattice of dominant-region thresholds ρ (where the weakest one-bit class is not $\mathcal{C}_{\text{mono}}$ but a ρ-dependent family), which Assumption 3 sidesteps by fiat. Theorem 19 (below) resolves it: the ρ-lattice is the slice $\{r_0{=}1,\ r_1{=}\rho\}$ of the single dominance polytope $W^\star{=}\{r_0 \geq r_1\}$, with ρ=1 its boundary. The construction is machine-checked (`val_conj1_genpos.py`, seeds fixed): $\mathcal{C}_{\text{dom}}$ strictly contains $\mathcal{C}_{\text{mono}}$, is falsifiable (reject rate 1.0), is one-bit-certified (error rate 0 over 2×10^5 targets), and loses one-bit certifiability without the domination bound (error rate $\approx \frac{1}{4}$).

Unconditional resolution (General Position removed). Dropping Assumption 3 reveals, for each orientation pattern, a single maximal one-bit class—a *dominance polytope*—of which the ρ-family of Remark 17 is one slice.

Definition 5 (Orientation pattern and canonical class)**.** On a fibre E an *orientation pattern* is a tied set $P \subseteq \{i : c_i \neq 0\}$ with a tied sign-pattern $s \in \{\pm\}^P$; the free set is $Z = D \setminus P$. For an observable magnitude region $W \subseteq [0,1]^{|D|}$ the *canonical class* is $\mathcal{C}(P, s, W) = \{a : \operatorname{sign} a_i = \sigma\, s_i \operatorname{sign} c_i\ (i \in P),\ (|a_j|)_j \in W\}$ with one declared bit σ. Write $T(r) = \sum_{i \in P} s_i \operatorname{sign}(c_i) \mu_i r_i$ and $G(r) = \sum_{i \in Z} \mu_i r_i$.

Lemma 9 (Canonical form of a falsifiable class)**.** *A class is falsifiable* iff *it equals some $\mathcal{C}(P, s, W)$; anchors ($c_i = 0$) lie necessarily in Z.* (Swap-closure, Thm. 4(iii), forces the constraint on Z to depend only on $|a|$; forcing an anchor sign is not evidence-definable. Both aligned (s_i=+) and anti-aligned (s_i=−) ties are falsifiable since σ is unidentified.)

Theorem 19 (Unconditional one-bit criterion; weakest classes)**.** *Let $\mathcal{C}(P, s, W)$ be falsifiable.* (i) (Dominance margin.) *One declared bit certifies* $\operatorname{sign}\Delta$ *on $\mathcal{C}(P, s, W)$* iff $\inf_{r \in W}\big(T(r) - G(r)\big) \geq 0$ *(with some member $\Delta > 0$) or* $\sup_{r \in W}\big(T(r) + G(r)\big) \leq 0$. (ii) (Weakest class.) *For a fixed pattern (P, s) the unique maximal one-bit class is the dominance polytope $\mathcal{C}^\star = \mathcal{C}(P, s, W^\star)$, $W^\star = \{r \in [0,1]^{|D|} : T(r) \geq G(r)\}$; $\mathcal{C}_{\text{mono}}$, every $\mathcal{C}_{\text{dom}}(\rho \leq 1)$, and every box one-bit class are proper subclasses of one $\mathcal{C}^\star$.* (iii) (Finite family.) *The set of weakest falsifiable one-bit classes is the explicit finite family $\{\mathcal{C}^\star\}$ indexed by $(P \subseteq \{c \neq 0\},\ s \in \{\pm\}^P)$ modulo the global flip $\sigma \mapsto -\sigma$ (at most $3^{|\{c \neq 0\}|}$ patterns); distinct patterns give pairwise incomparable maxima, so there is no* unique *weakest class. $\mathcal{C}_{\text{mono}}$ is the all-tied all-*

aligned member, and Assumption 3 is exactly the face on which the family collapses to $\mathcal{C}_{\mathrm{mono}}$*, recovering Thm. 18(iii).*

Proof. Δ is linear, so over $\mathcal{C}(P,s,W)$ at bit $\sigma = +1$ it ranges over $[\inf_W(T-G), \sup_W(T+G)]$ (free signs independent by swap-closure; the $\sigma = -1$ fibre is its negation); a single bit certifies iff each fibre's strict sign set is a singleton, giving (i). For $W^\star$, $\inf_{W^\star}(T-G) \geq 0$ by construction, and any r_0 with $T(r_0) < G(r_0)$ admits, with opposing free signs, a member with $\Delta < 0$ while $W^\star$ admits $\Delta > 0$, so $W^\star$ is maximal; $\mathcal{C}_{\mathrm{dom}}(\rho)$ sits in $W^\star$ iff $\rho \leq 1$, with $\rho = 1$ the boundary, giving (ii). Incomparability (iii) is witnessed on $c = (1,1,0)$ by $a = (0, \frac{1}{3}, 0)$ and $a = (1, -\frac{1}{3}, \frac{1}{3})$. On the general-position face the free block can overpower the tied block unless Z-magnitudes vanish, leaving only $\mathcal{C}_{\mathrm{mono}}$. □

Verification. The criterion (i) was checked against exact brute-force vertex truth on $2.8{\times}10^5$ box fibres and $3.2{\times}10^3$ non-box integer polytopes (exact-LP inf / sup) with 0 mismatches; sufficiency on $W^\star$ over $1.2{\times}10^5$ members (0 errors), necessity just outside $W^\star$ (error rate $\approx \frac{1}{2}$), and the $\mathcal{C}_{\mathrm{dom}}(\rho)$ collapse ($\rho{\leq}1$ one-bit, $\rho{>}1$ lost) are machine-checked (`val_unconditional_weakest.py`, seeds fixed; fails loudly). The single non-tautological input is Lemma 9, which rests on the swap involution (Thm. 4(iii)).

Theorem 1 is *binary*: it certifies a regime in which no label-free rule beats chance. This section sharpens that dichotomy into a *graded, computable invariant* K with an explicit threshold τ, in a single fully specified model, and proves that for the population (identifiability) problem the converse and the achievability meet *exactly* at τ. Thus, in this model, K is a genuine *capacity* for label-free sign recovery, not merely a sufficient or a necessary condition. A finite-sample Le Cam analysis then locates the threshold up to an $O(1/\sqrt{n})$ boundary layer.

U. Model (the concept mechanism is explicit)

Binary $Y \in \{0,1\}$, 0/1 loss. Source covariates $P_X = \mathcal{N}(0,1)$ (known); target covariates $Q_X = \mathcal{N}(\mu,1)$ (covariate shift; μ unknown, to be estimated from an unlabeled target sample). The deployed classifier is $g = f_0 = \mathbf{1}[x > 0]$ (boundary at the origin). We are handed a *fixed* candidate adapted rule $f_a = \mathbf{1}[x > a]$ with $a > 0$ known — the shipped "adapted boundary" — and ask whether switching $g \to f_a$ helps. The disagreement region is the interval $D = (0,a)$.

The label/concept mechanism must be explicit, because $\operatorname{sign}\Delta$ depends on $P(Y \mid X)$, which Q_X alone does not identify. We take *hard-threshold concepts* $\eta_\theta(x) = P(Y = 1 \mid x) = \mathbf{1}[x > \theta]$. The source-calibrated threshold θ_S is known (it is what a source-fit model reports). The *admissible target concepts* are those whose target *boundary mass* drifts from the source boundary by at most a budget $\varepsilon \geq 0$:

$$\mathcal{C}_\varepsilon \;=\; \Big\{ \theta_T : \; \big| \Phi(\theta_T - \mu) - \Phi(\theta_S - \mu) \big| \leq \varepsilon \Big\}. \tag{7}$$

Here ε is the 1-D analogue of the calibration-drift budget β of the benefit-sign frontier (§V); by Theorem 4 some such untestable supplement is unavoidable, and (7) fixes it in the natural label-free-meaningful unit of *target probability mass*. The label-free observer sees only $(P_X, g, f_a, \theta_S, \varepsilon)$ and the unlabeled sample $\{x_i\} \sim Q_X$, and must output a guess of $\operatorname{sign}\Delta$, where $\Delta = R_Q(f_0) - R_Q(f_a)$ ($\Delta > 0 \iff$ adapting to f_a helps).

Lemma 10 (Exact benefit identity and the sign-flip locus)**.** *On $D = (0,a)$ exactly one of f_0, f_a is correct, so for the concept η_θ,*

$$\Delta(\theta) \;=\; Q\big((0,a) \cap \{x < \theta\}\big) - Q\big((0,a) \cap \{x > \theta\}\big) \;=\; 2\,\Phi(\theta' - \mu) - \Phi(-\mu) - \Phi(a - \mu),$$

with $\theta' = \operatorname{clip}(\theta, 0, a)$. $\Delta(\cdot)$ is nondecreasing in θ with a single root

$$\boxed{\theta_0 \;=\; \mu + \Phi^{-1}\Big(\tfrac{1}{2}\big[\Phi(-\mu) + \Phi(a-\mu)\big]\Big)} \qquad (\textit{the } Q\textit{-median of the band } D),$$

so $\operatorname{sign}\Delta = \operatorname{sign}(\theta - \theta_0)$*. Both θ_0 and the plug-in sign* $\operatorname{sign}(\theta_S - \theta_0)$ *are* observable*.*

Proof. Off D, $f_0 = f_a$, so the pointwise excess loss vanishes. On $D = (0,a)$ we have $f_0 = 1, f_a = 0$; for 0/1 loss the excess loss of f_0 over f_a is $\mathbf{1}[f_a{=}Y] - \mathbf{1}[f_0{=}Y] = \mathbf{1}[Y{=}0] - \mathbf{1}[Y{=}1] = 1 - 2Y$, with conditional mean $1 - 2\eta_\theta(x)$, which is $+1$ for $x < \theta$ and -1 for $x > \theta$. Integrating against Q over $(0,a)$ gives the displayed difference of band masses, hence $\Delta(\theta) = \Phi(\theta' - \mu) - \Phi(-\mu) - \big(\Phi(a - \mu) - \Phi(\theta' - \mu)\big) = 2\Phi(\theta' - \mu) - \Phi(-\mu) - \Phi(a - \mu)$. This is nondecreasing in θ' (hence in θ); setting it to zero gives $\Phi(\theta_0 - \mu) = \frac{1}{2}[\Phi(-\mu) + \Phi(a - \mu)]$, the Q-median of D. (Validated to 4×10^{-4} Monte-Carlo error against the empirical $R_Q(f_0) - R_Q(f_a)$, `val_knowability_capacity.py` part (D).) ☐

V. The computable invariant and the threshold

Definition 6 (Knowability invariant)**.**

$$K \;:=\; \frac{\big|\,\Phi(\theta_0 - \mu) - \Phi(\theta_S - \mu)\,\big|}{\varepsilon} \;=\; \frac{\big|\,\frac{1}{2}[\Phi(-\mu) + \Phi(a-\mu)] - \Phi(\theta_S - \mu)\,\big|}{\varepsilon}.$$

K is the *mass margin* between the calibrated boundary θ_S and the sign-flip locus θ_0, measured in target probability mass, normalised by the drift budget ε. It is a margin-weighted SNR of the covariate shift relative to g's boundary, computable from $(\mu, a, \theta_S, \varepsilon)$ alone (with μ estimable label-free from $\{x_i\}$).

Theorem 20 (Population knowability–capacity, $\tau = 1$)**.** *Assume Q_X (equivalently μ) is known. With threshold $\tau = 1$:*

(i) ***Achievability.*** *If $K > 1$ then for* every *admissible target concept $\theta_T \in \mathcal{C}_\varepsilon$,* $\operatorname{sign}\Delta = \operatorname{sign}(\theta_S - \theta_0)$*; the plug-in certificate $\hat{g} = \operatorname{sign}(\theta_S - \theta_0)$ recovers the sign with error* 0*.*

(ii) ***Converse.*** *If $K < 1$ there exist $\theta_T^+, \theta_T^- \in \mathcal{C}_\varepsilon$ with* identical Q_X *(hence* $\mathrm{TV} = 0$ *between the laws of* any *statistic of any number n of unlabeled observations) and* $\operatorname{sign}\Delta(\theta_T^+) = -\operatorname{sign}\Delta(\theta_T^-) \neq 0$*. Consequently the minimax label-free error equals $\frac{1}{2}$ for every n.*

The converse and the achievability therefore meet at the single *threshold* $\tau = 1$, exactly *(not asymptotically): $\{K > 1\}$ is precisely the identifiable set.*

Proof. By Lemma 10, $\operatorname{sign}\Delta(\theta_T) = \operatorname{sign}(\theta_T - \theta_0)$ with θ_0 fixed by the observables. The map $\theta \mapsto \Phi(\theta - \mu)$ is a strictly increasing bijection, so the admissible set (7) is exactly the interval $\theta_T \in [\,\mu + \Phi^{-1}(p_S - \varepsilon),\, \mu + \Phi^{-1}(p_S + \varepsilon)\,]$ with $p_S = \Phi(\theta_S - \mu)$, i.e. in mass coordinates $u = \Phi(\theta_T - \mu)$ it is the interval $[\,p_S - \varepsilon,\, p_S + \varepsilon\,]$ (clipped to $[0, 1]$). The sign of Δ flips exactly at the mass level $u_0 = \Phi(\theta_0 - \mu) = \frac{1}{2}[\Phi(-\mu) + \Phi(a - \mu)]$.

(i) If $K > 1$ then $|u_0 - p_S| > \varepsilon$, so u_0 lies strictly outside $[p_S - \varepsilon, p_S + \varepsilon]$. Hence every admissible $u \in [p_S - \varepsilon, p_S + \varepsilon]$ is on one side of u_0, so $\theta_T - \theta_0$ has constant sign over $\mathcal{C}_\varepsilon$, equal to $\operatorname{sign}(\theta_S - \theta_0) = \operatorname{sign}(p_S - u_0)$. The plug-in certificate is correct for all admissible concepts.

(ii) If $K < 1$ then $|u_0 - p_S| < \varepsilon$, so $u_0 \in (p_S - \varepsilon, p_S + \varepsilon)$ lies strictly interior to the admissible mass interval. Pick $u^\pm = \Phi(\theta_T^\pm - \mu)$ with $u^+ \in (u_0, p_S + \varepsilon]$ and $u^- \in [p_S - \varepsilon, u_0)$ (both nonempty and admissible); then $\operatorname{sign}\Delta(\theta_T^+) = +1$, $\operatorname{sign}\Delta(\theta_T^-) = -1$. Both worlds share the *same* $Q_X = \mathcal{N}(\mu, 1)$ (the concept lives in the unobserved $P(Y \mid X)$ channel), so the law of any unlabeled statistic is identical across the two worlds: $\mathrm{TV} = 0$ for every n. By Le Cam's two-point inequality the minimax error of any label-free rule is $\geq \frac{1}{2}(1 - \mathrm{TV}) = \frac{1}{2}$, and the constant rules achieve $\frac{1}{2}$, so it equals $\frac{1}{2}$.

Finally, $K = 1$ is the single boundary $|u_0 - p_S| = \varepsilon$, where u_0 is an endpoint of the admissible interval; the sign is constant on the (closed) interval but not on its interior-plus-the-flip, so the transition is exactly at $\tau = 1$. (Validated: over 40,000 random draws of $(\mu, a, \theta_S, \varepsilon)$, the equivalence "$K > 1 \iff$ identifiable over $\mathcal{C}_\varepsilon$" holds with *zero* mismatches off a 10^{-6} shell; a boundary stress-test confirms the flip at $K = 1$ to 10^{-3}. `val_knowability_capacity.py` part (A).) ☐

Relation to the frontier. Theorem 20 is the *graded* reading of Theorem 3: the frontier's qualitative "$|M| > \beta$" becomes a *scalar* $K = |M|/\beta$-type ratio with the explicit value $\tau = 1$ in a fully specified location model, and the 0-vs-$\frac{1}{2}$ minimax dichotomy is recovered as the two faces $K > 1$ and $K < 1$. The novelty here is that K is shown to be *both* necessary and sufficient at the *same* threshold — a capacity — for the population problem, with θ_0 (the flip locus) given in closed form by the covariate shift μ and the candidate boundary a.

W. Finite-n: a Le Cam boundary layer around the threshold

With only n unlabeled samples, μ — hence K — must be estimated, and the deployable rule is the finite-n *three-way certificate*: estimate $\hat{\mu} = \bar{x}$, form a z-confidence interval $[\hat{\mu} \pm z/\sqrt{n}]$, and commit to $\operatorname{sign}(\theta_S - \hat{\theta}_0)$ only if K stays above 1 across the whole interval (a lower-confidence bound $\hat{K}_{\mathrm{lcb}} > 1$); otherwise abstain. The next proposition shows the price of estimating μ is exactly an $O(1/\sqrt{n})$ layer around the threshold *location*, with the same Le Cam/Bretagnolle–Huber form as Proposition 1.

Proposition 8 (Finite-n Le Cam boundary layer)**.** *Let $\mu^\star$ be the flip locus, $\theta_0(\mu^\star) = \theta_S$ (so $K = 0$ there), and fix the concept to transfer ($\theta_T = \theta_S$, i.e. ε irrelevant; the only noise is in estimating μ). For $c > 0$ consider the two worlds $\mu^\pm = \mu^\star \pm \frac{c}{2}\sqrt{1/n}$, which straddle the flip and hence have* opposite $\operatorname{sign}\Delta$. *The unlabeled laws satisfy*

$$\mathrm{TV}\big(\mathcal{N}(\mu^-, 1)^{\otimes n}, \mathcal{N}(\mu^+, 1)^{\otimes n}\big) = 2\Phi(c/2) - 1, \qquad \mathrm{KL} = \tfrac{n}{2}\,(\mu^+ - \mu^-)^2 = \tfrac{c^2}{2},$$

so every label-free sign test has minimax error

$$\inf_{\hat{g}} \max_{\pm} \mathbb{P}\big[\hat{g} \text{ wrong}\big] \;\geq\; \tfrac{1}{2}\big(1 - \mathrm{TV}\big) = \Phi(-c/2) \;\geq\; \tfrac{1}{4}\, e^{-c^2/2} \quad (\textit{Bretagnolle–Huber}),$$

constant in n. *Hence sign recovery is impossible below error $\Phi(-c/2)$ inside a window of half-width $\frac{c}{2}\sqrt{1/n} = \Theta(1/\sqrt{n})$ around $\mu^\star$,* even when the concept transfers.

Proof. The sample mean $\bar{X} \sim \mathcal{N}(\mu^\pm, 1/n)$ is sufficient; the TV of two equal-variance Gaussians with mean gap $\mu^+ - \mu^- = c/\sqrt{n}$ and standard deviation $1/\sqrt{n}$ is $2\Phi\big(\frac{c/\sqrt{n}}{2/\sqrt{n}}\big) - 1 = 2\Phi(c/2) - 1$, and the per-sample $\mathrm{KL}\big(\mathcal{N}(\mu^+, 1) \| \mathcal{N}(\mu^-, 1)\big) = \frac{1}{2}(\mu^+ - \mu^-)^2$ tensorises to $\frac{c^2}{2}$. A sign test is a test between the two worlds; Le Cam's testing affinity gives the $\frac{1}{2}(1 - \mathrm{TV})$ floor and Bretagnolle–Huber gives the $\frac{1}{4}e^{-\mathrm{KL}}$ certificate. That the two worlds straddle the flip (so the signs are genuinely opposite) is Lemma 10. (Validated: the empirical Bayes-optimal sign-test error meets $\Phi(-c/2)$ to $\leq 1.5 \times 10^{-2}$, exceeds $\frac{1}{4}e^{-c^2/2}$, and is constant across $n \in \{100, 1000, 10000\}$ for $c \in \{0.5, 1, 2\}$; `val_knowability_capacity.py` part (C).) ☐

The finite-n certificate *matches* this: numerically its minimax sign-recovery accuracy $\to 1$ wherever $K_{\mathrm{pop}} > 1$ and $= \frac{1}{2}$ wherever $K_{\mathrm{pop}} < 1$, with the transition pinned at $K = 1$, across $n \in \{200, 2000, 20000\}$ (Table XXIX; `val_knowability_capacity.py` part (B)). The boundary layer is therefore an $O(1/\sqrt{n})$ blur of the threshold *location* $\mu^\star$, not a gap in the threshold *value* $\tau = 1$.

Remark 18 (Honest scope — what is and is not claimed)**.** (1) Theorem 20 is a capacity *for this* 1*-D Gaussian-location model and for sign-of-benefit only*; it is not a general-distribution capacity. (2) The clean value $\tau = 1$ is a consequence of measuring the drift budget ε in target-*mass* units (7); this is the natural label-free-meaningful unit (it makes θ_0 the Q-median and the flip condition a mass comparison), but a different budget parametrisation would rescale τ. (3)

μ	$K_{\rm pop}$	acc n=200	acc n=2000	acc n=20000
−0.807	1.416	0.994	1.000	1.000
0.366	1.679	0.995	1.000	1.000
1.633	1.677	0.995	1.000	1.000
2.806	1.417	0.994	1.000	1.000
−1.407	0.640	0.500	0.500	0.500
0.980	0.063	0.500	0.500	0.500
1.000	0.000	0.500	0.500	0.500
3.406	0.641	0.500	0.500	0.500

TABLE XXIX

PHASE TRANSITION OF THE FINITE-n THREE-WAY CERTIFICATE (a=2, θ_S=1, ε=0.05; MINIMAX OVER THE ADMISSIBLE CONCEPT). ABOVE τ=1 THE MINIMAX ACCURACY → 1; BELOW IT THE CERTIFICATE ABSTAINS AND THE ACCURACY IS EXACTLY $\frac{1}{2}$. (REAL NUMBERS, SEED FIXED; RESULTS_KNOWABILITY_CAPACITY.JSON.)

The same equivalence $K > 1 \iff$ identifiable holds for *soft* (logistic) concepts of fixed known slope, with θ_0 the (numerically computed) root of $\Delta(\cdot)$ rather than the band median — verified with zero mismatches over 1,496 draws — so the result is not an artifact of the hard-threshold concept; only the closed form for θ_0 is special to it. (4) Relative to the existing five theorems, this is a *refinement*, not a new pillar: it converts the 0-vs-$\frac{1}{2}$ frontier of Theorem 3 into a scalar capacity with an explicit closed-form flip locus, and supplies the finite-n boundary layer that Proposition 1 left implicit for the location family. Its value is a sharpened, fully worked *instance*, useful pedagogically and as the tight test-bed for the graded invariant; it does not enlarge the set of *distribution-free* guarantees.

The 1-D theorem (§T) gives a genuine capacity $\tau = 1$ for sign-of-benefit in the Gaussian-location two-point model. We now ask how much of that survives in higher dimension, in non-Gaussian location families, and distribution-free. The answer is a clean layered one: the converse is essentially universal, the achievability is exact under an alignment/MLR regularity condition, and a single *computable* capacity provably needs such a condition. We label every claim and exhibit the breakers that delimit each regime.

X. Stage 1: multivariate Gaussian-location

a) Model.: $P_X = \mathcal{N}(0, \Sigma)$, $Q_X = \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$ known and μ unknown (estimated from an unlabeled target sample). The deployed rule is the halfspace $g = f_0 = \mathbf{1}[w^\top x > 0]$ and the shipped candidate is the *parallel-shifted* halfspace $f_a = \mathbf{1}[w^\top x > a]$, $a > 0$ known. The disagreement region is the *slab* $D = \{x : 0 < w^\top x < a\}$. Concepts are halfspaces $\eta_{v,\theta}(x) = \mathbf{1}[v^\top x > \theta]$ (the label mechanism), with θ_S the source-calibrated offset along a fixed direction. Write the *whitened (Mahalanobis) projection onto the rule normal*

$$s \;:=\; w^\top x, \qquad s \sim_Q \mathcal{N}(m, \sigma^2), \quad m = w^\top \mu, \;\; \sigma^2 = w^\top \Sigma w, \tag{8}$$

and for a concept direction v the second coordinate $u := v^\top x$, so that (s, u) is bivariate Gaussian under Q with correlation (the *Mahalanobis cosine* between w and v)

$$\rho \;=\; \rho(w, v) \;=\; \frac{w^\top \Sigma v}{\sqrt{w^\top \Sigma w}\,\sqrt{v^\top \Sigma v}} \;=\; \cos\angle_\Sigma(w, v). \tag{9}$$

Because f_0, f_a and $\eta_{v,\theta}$ depend on x only through (s, u), the entire benefit is a functional of this bivariate Gaussian:

$$\Delta \;=\; R_Q(f_0) - R_Q(f_a) \;=\; \mathbb{E}_Q\big[\mathbf{1}_D\,(1 - 2\eta_{v,\theta})\big] \;=\; Q(D) \;-\; 2\,Q\big(D \cap \{u > \theta\}\big). \tag{10}$$

Lemma 11 (Whitened reduction; PROVEN)**.** *Δ in* (10) *depends on the dimension d, on μ and on Σ only through the bivariate law of (s, u) under Q, i.e. through the four scalars $(m, \sigma^2,\; v^\top \mu,\; v^\top \Sigma v)$ and the correlation ρ of* (9)*. In particular the deployed and candidate rules depend on x only through s, whose Q-law is the one-dimensional Gaussian $\mathcal{N}(w^\top \mu,\; w^\top \Sigma w)$.*

Proof. $f_0 = \mathbf{1}[s > 0]$, $f_a = \mathbf{1}[s > a]$, $\eta_{v,\theta} = \mathbf{1}[u > \theta]$ are $\sigma(s, u)$-measurable, and the 0/1 excess loss of f_0 over f_a is supported on $D = \{0 < s < a\}$ where it equals $1 - 2\eta_{v,\theta}$ (exactly as in Lemma 10: on D, f_0=1, f_a=0, and the conditional mean of $\mathbf{1}[Y{=}0] - \mathbf{1}[Y{=}1]$ is $1 - 2\eta$). Taking Q-expectation gives (10), a functional of the Q-law of (s, u). A non-degenerate affine image of a Gaussian is Gaussian with the stated moments. □

b) (a) Aligned concept class — exact lift.: Take the concept normal *parallel to the rule normal*, $v = w$ (the concept "reads the same direction" the deployed rule acts on). Then $u = s$, the problem collapses onto the single whitened line

$\operatorname{span}(w)$, and (10) becomes literally the 1-D identity of Lemma 10 in the standardized variable $z=(s-m)/\sigma$. Define the admissible aligned class and the lifted invariant in Mahalanobis units:

$$\mathcal{C}_\varepsilon^{\parallel}=\Big\{\theta_T:\ \big|\Phi(\tfrac{\theta_T-m}{\sigma})-\Phi(\tfrac{\theta_S-m}{\sigma})\big|\le\varepsilon\Big\},\quad K_\parallel=\frac{\big|\frac12[\Phi(\frac{-m}{\sigma})+\Phi(\frac{a-m}{\sigma})]-\Phi(\frac{\theta_S-m}{\sigma})\big|}{\varepsilon}. \tag{11}$$

Theorem 21 (Stage 1, aligned: exact Mahalanobis lift; PROVEN)**.** *With $v=w$ and Q_X known, the population knowability–capacity Theorem 20 holds verbatim with (μ,a,θ_S) replaced by their whitened images (m,σ) in (8): with $\tau=1$, $K_\parallel>1\Rightarrow$ every admissible aligned concept has* $\operatorname{sign}\Delta=\operatorname{sign}(\theta_S-\theta_0)$ *(achievability, error 0); $K_\parallel<1\Rightarrow$ two admissible aligned concepts share Q_X and have opposite* $\operatorname{sign}\Delta$*, so the minimax label-free error is $\frac12$. The flip locus is the Q-median of the slab along w, $\theta_0=m+\sigma\,\Phi^{-1}\big(\frac12[\Phi(\frac{-m}{\sigma})+\Phi(\frac{a-m}{\sigma})]\big)$. The finite-n Le Cam boundary layer (Proposition 8) carries over with μ replaced by $m=w^\top\mu$ and the noise scale by $\sigma/\sqrt{n}$, since $\bar s=w^\top\bar x\sim\mathcal{N}(m,\sigma^2/n)$ is sufficient for m.*

Proof. By Lemma 11 with $v=w$, Δ is the 1-D benefit of Lemma 10 for the standardized location problem $z\sim\mathcal{N}(m/\sigma,1)$ with boundaries $0,a,\theta$ rescaled by σ; (11) is exactly (7) in z. Apply Theorem 20. The converse two worlds again share the *same* $Q_X=\mathcal{N}(\mu,\Sigma)$ (the concept lives in the unobserved $P(Y\mid X)$ channel), so $\mathrm{TV}=0$ for every unlabeled statistic and every n. The sufficiency of $\bar s$ for m is standard. *(Validated:* 20,000 *random draws in* $d\in\{2,3,4\}$*, equivalence* $K_\parallel>1\iff$ *identifiable with* 0 *mismatches off a* 10^{-6} *shell;* `val_knowability_capacity_general.py` *Stage 1a.)* ☐

c) (b) Free halfspace concept class — exact obstruction and its capacity.: If the concept normal v is *not* constrained to be parallel to w, the flip locus moves. For a fixed v, $\Delta(\theta)$ in (10) is still nondecreasing in θ with a single root $\theta_0(v)$ characterized by $Q\big(D\cap\{u>\theta_0\}\big)=\frac12Q(D)$: *$\theta_0(v)$ is the Q-conditional median of $u=v^\top x$ given $x\in D$.* Its level in the concept's own Mahalanobis mass coordinate,

$$p_{\rm med}(v)\ :=\ \Phi\Big(\tfrac{\theta_0(v)-v^\top\mu}{\sqrt{v^\top\Sigma v}}\Big), \tag{12}$$

interpolates between the two extremes set by $\rho=\rho(w,v)$. In standardized coordinates write $z=(s-m)/\sigma\sim\mathcal{N}(0,1)$, the slab as $z\in(\alpha,\beta)$ with $\alpha=-m/\sigma,\ \beta=(a-m)/\sigma$, and $t=(u-v^\top\mu)/\sqrt{v^\top\Sigma v}=\rho\,z+\sqrt{1-\rho^2}\,\omega$ with $\omega\perp z$ standard normal. Then $p_{\rm med}(v)=\Phi(\operatorname{med}(t\mid z\in(\alpha,\beta)))$ and

$$\rho=\pm1\ (v\parallel w):\ p_{\rm med}=\tfrac12[\Phi(\alpha)+\Phi(\beta)]\ \text{(band median, since } t=\pm z),\qquad \rho=0\ (v\perp_\Sigma w):\ p_{\rm med}=\tfrac12. \tag{13}$$

Both endpoints are **proven exactly**. The $\rho=0$ case is structural: when v is Σ-orthogonal to w, $t=\omega$ is independent of z, so conditioning on the slab does not move its law and $\operatorname{med}(t\mid z\in(\alpha,\beta))=0$, i.e. $p_{\rm med}=\frac12$. The $\rho=\pm1$ case is immediate since then $t=\pm z$. That $p_{\rm med}$ is *monotone* in $|\rho|$ between these endpoints (so the achievable flip range is exactly the interval between $\frac12$ and the band median) is **verified numerically** (Lemma 12, stated as a numerically confirmed monotone-interpolation fact with a proof sketch); the *worst case* relevant to the converse — mass $\frac12$, reached at $\rho=0$ — is proven outright.

Lemma 12 (Monotone flip interpolation; NUMERICAL, proof sketch)**.** *With the notation above, $p_{\rm med}(v)$ is monotone in $|\rho|$, sweeping the full interval between $\frac12$ (at $\rho=0$) and the band median (at $|\rho|=1$).* Sketch: *the conditional median solves $\mathbb{E}[\operatorname{sign}(t-m_t)\mid z\in(\alpha,\beta)]=0$; since the slab's z-median $z^\star$ has a fixed sign relative to 0, increasing ρ tilts $t=\rho z+\sqrt{1-\rho^2}\,\omega$ monotonically toward $\pm z$, dragging m_t monotonically from 0 toward $z^\star$ by a sign-coherent shift of the conditional law. A full proof would verify single-crossing of the conditional-median equation in ρ.* (Validated: $p_{\rm med}$ rises monotonically $0.50\to$ band median as $\rho:0\to1$ across four slabs; `val_knowability_capacity_general.py` Stage 1b.)

Theorem 22 (Stage 1, free halfspace: the obstruction; converse PROVEN, exact $K_{\rm free}$ CONDITIONAL)**.** *Let $p_S=\Phi(\frac{\theta_S-m}{\sigma})$ be the calibrated boundary mass along w and let the admissible free class be the halfspaces with own-normal boundary-mass drift $\le\varepsilon$ and arbitrary direction v. Then the aligned invariant $K_\parallel$ of (11) is* not *a capacity for this class. Precisely:*

- (i) ***(Breaker / converse; PROVEN.)*** *Whenever p_S lies strictly between $\frac12$ and $b:=\frac12[\Phi(\frac{-m}{\sigma})+\Phi(\frac{a-m}{\sigma})]$, there is a regime $K_\parallel>1$ in which the aligned theorem predicts identifiability but a tilted admissible concept (own-normal boundary mass $=p_S$, taken with $\rho\to0$, which reaches flip mass $\frac12$) produces the* opposite *benefit sign. The two concepts share Q_X, so* $\mathrm{TV}=0$ *and the minimax error is $\frac12$: the aligned $K_\parallel$ is therefore* not *a capacity for the free class.*
- (ii) ***(Exact free capacity; CONDITIONAL on Lemma 12.)*** *Under the monotone interpolation, the achievable flip range is exactly the interval between $\frac12$ and b, so* $\operatorname{sign}\Delta$ *is identifiable over the free class iff $[p_S-\varepsilon,p_S+\varepsilon]$ misses it; the governing capacity is*

$$K_{\rm free}\ =\ \frac{\operatorname{dist}\big(p_S,\ [\,\frac12\wedge b,\ \frac12\vee b\,]\big)}{\varepsilon},\qquad \tau=1.$$

Proof. For fixed v, (10) is nondecreasing in θ (raising θ only shrinks $\{u > \theta\}$), with unique root the conditional median $\theta_0(v)$; hence $\operatorname{sign}\Delta = \operatorname{sign}(\theta - \theta_0(v))$. *Part (i)* uses only the proven endpoint $\rho = 0 \Rightarrow p_{\mathrm{med}} = \frac{1}{2}$ of (13): for p_S strictly on the $\frac{1}{2}$-side of the aligned flip b, take v with $\rho \to 0$ and own-normal boundary mass $= p_S$; its flip sits at $\frac{1}{2}$, on the opposite side of p_S from b, so $\operatorname{sign}\Delta$ is reversed relative to the aligned prediction. The two concepts share Q_X, giving TV $= 0$ and minimax error $\frac{1}{2}$ (Le Cam); thus $K_\parallel$ is not a capacity. *Part (ii)* additionally invokes Lemma 12 (numerically confirmed): the achievable flip levels then fill exactly the interval between $\frac{1}{2}$ and b, so a sign-flip witness exists iff the admissible mass band $[p_S - \varepsilon, p_S + \varepsilon]$ meets that interval, i.e. iff $K_{\mathrm{free}} < 1$. *(Validated:* p_{med} *vs* $|\rho|$ *traced from* 0.5005 *at* ρ=0 *to the band median* 0.4226 *at* ρ=1*; an explicit breaker with* $K_\parallel = 1.40$ *where the aligned prediction is* $\Delta = +0.14$ *but a worst-case tilt of the same own-normal boundary mass yields* $\Delta = -0.14$ *(opposite sign);* `val_knowability_capacity_general.py` *Stage 1b.)* □

Remark 19 (What Stage 1 establishes)**.** The 1-D capacity lifts to d dimensions *exactly* in Mahalanobis geometry *when the concept is aligned with the rule normal* (Thm 21): K and $\tau = 1$ are intact, with θ_0 the slab's Q-median along w. The lift is therefore not vacuous — it is a genuine d-dimensional capacity. But the unconstrained halfspace concept class is *strictly harder*: an adversary can tilt the concept normal toward Σ-orthogonality with w, dragging the flip locus to the unconditional median (mass $\frac{1}{2}$) and defeating any margin measured along w. The honest Stage-1 statement is a capacity K_{free} that prices in the worst-case tilt (Thm 22); the clean closed-form $\tau = 1$ survives, but the invariant is the distance to mass $\frac{1}{2}$, not to the band median, once the concept direction is free.

Y. Stage 2: MLR / log-concave location families

We now leave the Gaussian and ask which *non-Gaussian* location families keep the matched threshold.

Scope note. Here "exponential family" abbreviates one-parameter *location* families with monotone likelihood ratio (Gaussian, Laplace, logistic location). Genuine *scale*/natural-parameter exponential families—e.g. the exponential or Gamma rate family—are not location families; their capacity is not governed by the mass-coordinate threshold below and falls under Conjecture 3.

Let Q_X have a density $q_0(x - \mu)$ for a fixed base density q_0 with continuous CDF F; deployed $f_0 = \mathbf{1}[x > 0]$, candidate $f_a = \mathbf{1}[x > a]$, hard-threshold concept $\eta_\theta = \mathbf{1}[x > \theta]$. The benefit identity is *family-agnostic*: exactly as in Lemma 10 (the proof used only that on D the conditional contrast is $1 - 2\eta$ and that Q has a CDF),

$$\Delta(\theta) = 2F(\theta' - \mu) - F(-\mu) - F(a - \mu), \qquad \theta' = \operatorname{clip}(\theta, 0, a), \tag{14}$$

and the admissible class / invariant are (7), Def. 6 with Φ replaced by F. So far *no* condition is needed and Δ is monotone in θ for *any* CDF F.

Theorem 23 (Stage 2, MLR/log-concave: matched $\tau = 1$; PROVEN UNDER CONDITION)**.** *Assume the base family is a one-parameter* location *family with strictly* ***monotone likelihood ratio*** *in the location (equivalently here,* q_0 *strictly positive and log-concave, hence unimodal). Then with the mass-coordinate drift class* $(7)_F$ *and* $\tau = 1$*, the population knowability–capacity holds verbatim:* $K > 1 \Rightarrow$ *every admissible concept shares* $\operatorname{sign}\Delta = \operatorname{sign}(\theta_S - \theta_0)$ *(achievability),* $K < 1 \Rightarrow$ *two admissible concepts share* Q_X *with opposite* $\operatorname{sign}\Delta$ *(converse,* TV $= 0$*). Moreover the flip locus* $\theta_0(\mu) = \mu + F^{-1}\big(\frac{1}{2}[F(-\mu) + F(a - \mu)]\big)$ *is the band's Q-median and is* ***strictly increasing in the unknown shift*** μ*, so a single label-free estimate* $\hat\mu$ *pins the flip side through one threshold and* K *is a genuine scalar capacity (the finite-n Le Cam layer of Prop. 8 carries over with* $\Phi \to F$*).*

Proof. The converse and achievability are exactly Theorem 20 with $\Phi \to F$: the admissible set is the interval $[p_S - \varepsilon, p_S + \varepsilon]$ in the mass coordinate $u = F(\theta_T - \mu)$, the flip is at the band-median mass $u_0 = \frac{1}{2}[F(-\mu) + F(a - \mu)]$, and $K \gtrless 1 \iff u_0$ lies outside/inside the admissible band; identical-Q_X gives TV $= 0$. The only place a hypothesis is used is the claim that $\theta_0(\mu)$ is monotone in μ (needed for $K(\mu)$ to be a single threshold rather than a fragmented set): for an MLR/log-concave location family the map $\mu \mapsto \theta_0(\mu)$ is strictly increasing because $F(a - \mu)$ and $F(-\mu)$ are decreasing in μ and F^{-1} is increasing, with no antimode to reverse the band median's motion. *(Validated: Laplace and logistic location families,* 0 *mismatches over* 6,000 *draws each, benefit identity matched to* $\leq 4.6 \times 10^{-4}$ *MC error;* `val_knowability_capacity_general.py` *Stage 2a.)* □

Theorem 24 (Stage 2, non-log-concave breaker; PROVEN)**.** *There is a location family for which the scalar capacity* fails*. Take* q_0 *a balanced mixture of two well-separated Gaussians (bimodal, hence not log-concave and not MLR) and the band* $D = (0, a)$ *wide enough to span the antimode. Then although* $\Delta(\theta)$ *in* (14) *is still monotone in* θ *(so the concept-location flip is unique), the flip locus* $\theta_0(\mu)$ *is* ***non-monotone in the unknown shift*** μ*: as* μ *moves the band median across the antimode,* $\theta_0(\mu)$ *collapses. Consequently the label-free identifiable set* $\{\mu : K(\mu) > 1\}$ ***fragments*** *into more components than in the unimodal case, and no single scalar threshold* τ *separates the identifiable from the non-identifiable regime.*

Proof (by exhibited witness). The benefit identity (14) holds for the mixture CDF F, monotone in θ. The density q_0 has two modes (verified: two local maxima), so the hypothesis of Theorem 23 fails. Tracing $\theta_0(\mu)$ over $\mu \in [-3, 3]$ exhibits two strict decreases (non-monotone events) where the band median crosses the antimode, and the set $\{\mu : K(\mu) > 1\}$ has 2 components for the mixture versus 1 for the matched unimodal Gaussian baseline at the same (p_S, ε, a). Hence the achievability certificate "estimate $\hat{\mu}$, read off $\operatorname{sign}(\theta_S - \theta_0(\hat{\mu}))$" is not governed by a single threshold in K. *(Validated: density modes = 2, $\theta_0(\mu)$ non-monotone events = 2, components mixture= 2 vs Gaussian= 1; `val_knowability_capacity_general.py` Stage 2b.)* $\square$

Z. Stage 3: a distribution-free converse, and the general obstruction

The converse needs none of the above structure. It is a pure two-point Le Cam bound that uses only the defining feature of the problem: the concept lives in the unobserved $P(Y \mid X)$ channel, so any two concepts induce the *same* observable covariate law.

Theorem 25 (Distribution-free sign-of-benefit converse; PROVEN)**.** *Let X take values in any measurable space, let g and f be any two fixed classifiers, and let $\mathcal{H}$ be a class of admissible concepts $\eta : \mathcal{X} \to [0, 1]$. Suppose there exist $\eta^+, \eta^- \in \mathcal{H}$ that induce the* ***same observable marginal*** *Q_X (automatic under covariate shift, where the concept does not affect Q_X) and have* ***opposite benefit sign****, $\operatorname{sign}\left(R_Q^{\eta^+}(g) - R_Q^{\eta^+}(f)\right) = -\operatorname{sign}\left(R_Q^{\eta^-}(g) - R_Q^{\eta^-}(f)\right) \neq 0$. Then for* every *sample size n and* every *label-free decision rule $\hat{T}$ that observes only $\{X_i\}_{i=1}^n \sim Q_X^{\otimes n}$,*

$$\inf_{\hat{T}} \max_{\eta \in \{\eta^+, \eta^-\}} \mathbb{P}\big[\hat{T} \text{ outputs the wrong } \operatorname{sign} \Delta\big] \;\geq\; \tfrac{1}{2}\big(1 - \mathrm{TV}(Q_X^{\otimes n}, Q_X^{\otimes n})\big) \;=\; \tfrac{1}{2}.$$

No distributional, dimensional, or parametric assumption is used.

Proof. The two worlds $\eta^{\pm}$ generate identical laws for the unlabeled sample ($Q_X^{\otimes n}$ in both), so the total variation between the observed data distributions is 0. A label-free sign decision is a test between the two worlds, which have opposite correct answers; Le Cam's two-point inequality gives minimax error $\geq \frac{1}{2}(1 - \mathrm{TV}) = \frac{1}{2}$, attained by a constant guess. *(Validated across Gaussian-1d, Laplace-1d, and aligned Gaussian-3d: opposite Δ signs with unlabeled* TV $= 0$ *and minimax lower bound $\frac{1}{2}$; `val_knowability_capacity_general.py` Stage 3a.)* $\square$

The general obstruction (why achievability needs a condition). Theorem 25 is fully general, but a matching *achievability* (a single computable K with the property $K > 1 \Rightarrow$ identifiable) provably cannot be. The achievability side requires the observer to (a) locate the sign-flip set of Δ over the admissible class and (b) certify that the whole admissible class lies on one side of it from Q_X alone. Stage 1b shows step (a) already fails to be captured by an alignment-blind margin once the concept direction is free (the flip locus depends on the Mahalanobis geometry between concept and rule), and Stage 2b shows that, even in one dimension, when the family is not MLR/unimodal the flip locus is non-monotone in the unknown shift, so step (b) has no single-threshold certificate. We record the general statement as a conjecture with the precise missing piece.

Conjecture 3 (General knowability–capacity; OPEN, with the exact gap)**.** *Fix observables (Q_X-family, g, f) and an admissible concept class $\mathcal{H}$ defined by a mass-drift budget ε. Suppose a* ***regularity/identifiability condition*** *holds:* (R1) *the benefit Δ has, over $\mathcal{H}$, a* unique *flip locus expressible in the observable mass coordinate; and* (R2) *that flip locus is a* monotone, observable *function of the unknown nuisance (e.g. the shift) — guaranteed by MLR/log-concavity in the location case and by Mahalanobis-alignment in the multivariate case. Then the scalar*

$$K \;=\; \frac{\mathrm{dist}_{mass}\big(\textit{calibrated boundary, worst-case flip locus over } \mathcal{H}\big)}{\varepsilon}$$

is a capacity at $\tau = 1$: $K > 1 \iff$ the sign of Δ is label-free identifiable, and the finite-n price is an $O(1/\sqrt{n})$ Le Cam layer around the threshold location. *The* ***open part*** *is precisely the removal (or sharp characterization) of* (R1)–(R2)*: without them the flip set need not be a single observable threshold, the worst-case-flip distance need not be computable from Q_X, and no scalar τ is simultaneously necessary and sufficient. Stages 1–2 prove the conjecture in its two leading regularity classes and exhibit explicit breakers (free-direction tilt; non-log-concave bimodality) on the boundary of each.*

Remark 20 (Honest scope of the generalization)**.** This section reaches a *strong but bounded* generalization, **not** a paradigm-level general capacity. What is *proven*: (i) an *exact* d-dimensional lift of $\tau = 1$ in Mahalanobis geometry for aligned concepts (Thm 21); (ii) the *exact* obstruction and worst-case capacity K_{free} for free halfspace concepts (Thm 22); (iii) a matched $\tau = 1$ for MLR/log-concave location families (Thm 23) with an explicit non-log-concave breaker (Thm 24); and (iv) a fully *distribution-free converse* in any dimension and any family (Thm 25). What is *not* proven, and is stated as Conjecture 3: a single computable capacity *without* a regularity/identifiability condition. The wall is sharp and named: a scalar τ is a capacity *iff* the benefit's flip locus is a unique, monotone, observable threshold;

alignment (multivariate) and MLR/unimodality (location) are the two sufficient regularity conditions we can currently prove, and each has a concrete counterexample just outside it. The value of the result is therefore (a) the converse is essentially universal, and (b) the achievability is pinned to an explicit, checkable regularity hypothesis whose failure modes are exhibited — a precise map of how far the 1-D capacity travels and exactly where it stops.

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